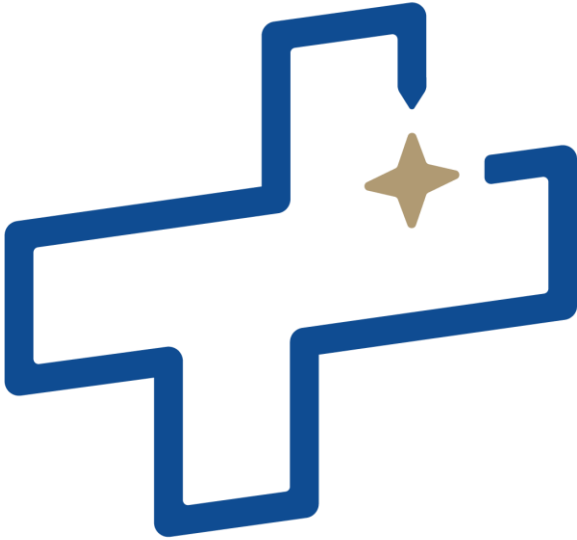




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Reference Documents

- [1] N/Acube-computer-5/control-program-8.xml (distributed with software package)
- [2] N/Acube-computer-5/cube-computer-common-3.xml (distributed with software package)
- [3] N/Acommon-framework/common-framework-1.xml (distributed with software package)
- [4] CS-DEV.UM.CA-01 CubeADCS User Manual Ver.1.04 or later
- [5] CS-DEV.FRM.CA-01 CubeProduct Firmware Reference Manual Ver 6.0 or later
- [6] CS-DEV.AMNL.BL-01 Bootloader Application Manual Ver.2.01 or later
- [7] CS-DEV.UM.CU-01 CubeSupport HMI
- [8] CS-DEV.GD.TPL-02 ADCS HW config and options Ver.1.00 or later (Note Document number will be client specific)
- [9] CD-DEV.ICD.CA-01 CubeADCS Standard ICD Ver.1.00 or later



List of Acronyms/Abbreviations

ACP	ADCS Control Program
ADCS	Attitude Determination and Control System
CAN	Controller Area Network
COTS	Commercial Off The Shelf
CSS	Coarse Sun Sensor
CVCM	Collected Volatile Condensable Materials
DUT	Device Under Test
EDAC	Error Detection and Correction
EHS	Earth Horizon Sensor
EM	Engineering Model
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
ESD	Electrostatic Discharge
FDIR	Fault Detection, Isolation, and Recovery
FM	Flight Model
FSS	Fine Sun Sensor
GID	Global Identification
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GYR	Gyroscope
I2C	Inter-Integrated Circuit
ID	Identification
IGRF	International Geomagnetic Reference Field (model)
LTDN	Local Time of Descending Node
LEO	Low Earth Orbit
MCU	Microcontroller Unit
MEMS	Microelectromechanical System
MTM	Magnetometer
MTQ	Magnetorquer
NDA	Non-Disclosure Agreement
OBC	On-board Computer
ORC	Orbit Reference Coordinates
PCB	Printed Circuit Board



RTC	Real-Time Clock
RWA	Reaction Wheel Assembly
RW	Reaction Wheel
SBC	Satellite Body Coordinate
SOFIA	Software Framework for Integrated ADCS
SPI	Serial Peripheral Interface
SRAM	Static Random-Access Memory
SSP	Sub-Satellite Point
STR	Star Tracker
TC	Telecommand
TCTLM	Telecommand and Telemetry (protocol)
TID	Total Ionizing Dose
TLM	Telemetry
TML	Total Mass Loss
UART	Universal Asynchronous Receiver/Transmitter



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1. Introduction

This document describes the operation of a CubeADCS in flight. Throughout this manual, a distinction is made between commissioning, and post-commissioning operations. During commissioning additional steps are required to verify the ADCS configuration and establish safe operation.

This manual focuses on the different control and estimation modes that apply to each part of the operations sequence, as well as telemetry logging requirements and key decision points.



2. ADCS modes reference

This chapter serves as a reference for the various modes that apply to the CubeADCS.

Throughout this document various references are made to commands, telemetry, modes and parameters used by the CubeADCS software. The API references include detail for such definitions in [1], [2], [3] and [4]. Wherever telecommand and telemetry, or parameters in telecommand messages, or fields in telemetry messages are referred to, it will be displayed using [this style](#), for example “[TlmController](#)”. In such cases, the code name from [1], [2] and [3] will be used to identify the telecommand, telemetry, parameter or field. This also applies to mode names in this section, such as [ConBdot](#).

2.1 Operational States

The future “Automatic” operational state of the ADCS is a setting which will allow the ADCS to perform automatic control mode transitions. This functionality has not been implemented yet. It will be described in future versions of this document after implementation.

Currently the ADCS can only be operated in Manual operations state – the [OpState](#) setting in the [AdcsOpState](#) telecommand is ignored, as well as the [OpStateDefault](#) setting in the [ConfigDefaultModes](#) configuration TCTLM.

Table 1: Operational States

OPERATIONAL STATE	DESCRIPTION
OpStateManual	Manual operational state
OpStateAuto	Autonomous operational state
OpStateSafe	Safe operational state
OpStateFault	Fault operational state

2.2 Run Modes

The CubeComputer control program will by default run in a processing loop. For debugging or HIL simulation reasons it might be required to change the run mode, however for normal operational use the default run mode (Enabled) should be kept. The possible run modes are detailed in Table 2.

Table 2: ADCS Run Modes

RUN MODE	DESCRIPTION
AdcsOff	ADCS loop is inactive
AdcsEnabled	ADCS loop is active (power-on default)
AdcsSimulation	ADCS is in simulation mode (used only for HIL simulations)

2.3 Estimation Modes

The ADCS requires knowledge of the current attitude in order to control the spacecraft. There are different options to estimate (measure / determine) the attitude, available as estimation mode selections. Not all modes provide the full attitude state, as some modes only measure angular rates. The combination of sensors required also varies by estimation mode. The various estimation modes are elaborated next.



2.3.1 Estimation Mode 0 (EstNone)

Description:

What:

No active ADCS estimator selection. Typical after an ADCS reset and before any ADCS controllers are enabled.

2.3.2 Estimation Mode 1 (EstGyro)

Description:

What:

Estimation of the angular rate vector only. Note that this estimation mode will use the inertially referenced angular rates measured by the gyro as the orbit reference rates without correction. There will thus be an error in the (orbit reference) estimated rate corresponding with the orbital rate of the satellite.

How:

Not actually an estimator of the angular rates in the true sense of the word but using the direct rate sensor (gyro) measurements as the estimated rates, without any dynamic bias corrections.

When:

This “estimator” is typically used during detumbling to measure the 3-axis body rates that may be too high for a typical 1 Hz state estimator. It will also be immune against state estimator divergence due to unmodelled disturbances and will be the best source of body rate knowledge during the initial phases of ADCS commissioning.

Sensor Dependency: Rate Sensors (Gyr0 or Gyr1)

Convergence time: Immediate

Configurable values: None

2.3.3 Estimation Mode 2 (EstMagRkf)

Description:

What:

Estimate the 3-axis body orbit referenced angular rate vector

How:

by using successive measurements of the magnetic field vector measurements. The estimated inertial referenced angular rates will be the same as the orbit referenced rates. As the geomagnetic field vector rotates twice per orbit with respect to the inertial coordinate frame and once per orbit with respect to the orbit coordinate frame, the estimated rates will have absolute errors in the order of the orbital rate.

When:

No orbit position knowledge (TLEs for SGP4) is required for this estimator making it ideal for use during the initial ADCS commissioning steps to verify the rate sensor measurements at absolute spin rates less than 30 °/sec.



Convergence time:

The convergence time for this estimator is approximately 10 samples or 10 seconds at 1 Hz sample period.

Sensor Dependency: Magnetometer (Mag0 or Mag1)

Configurable values:

The system noise covariance and sensor noise used in the filter can be changed by the [ConfigAdcsEstimator](#) (ADCS estimator configuration) message. See 3.6 and [1] for details.

2.3.4 Estimation Mode 3 (EstPitchRkf)

Description:

What:

Estimates the 3-axis body orbit referenced angular rate vector, and pitch angle. Roll and yaw angles will be reported as zero. The EstPitchRkf estimator will only estimate the pitch accurately while in Y-Thomson spin.

How:

Estimation based on magnetic field vector measurements. Orbit position knowledge is required to use an IGRF model of the geomagnetic field.

This estimation makes use of modelled vectors for the magnetic field. To calculate these modelled vectors, the orbital position and velocity of the satellite must be known. The latter information is obtained from an SGP4 orbit propagator. ***The mean orbital elements used by the propagator must be updated regularly to ensure that the estimation errors remain low. It is recommended to use orbital elements that are no more than 2 days old.***

When:

This estimator is at least required by the [ConYwheelInit](#) controller to estimate the pitch angle and determine whether automatic transition to the Y-momentum stabilisation [ConYwheel1](#) controller is possible.

Convergence time:

The convergence time is like the [EstMagRkf](#) estimator.

Sensor Dependency:

Magnetometer (Mag0 or Mag1) along with orbital position and velocity knowledge.

Configurable values:

The system noise covariance and sensor noise used in the filter can be changed by the [ConfigAdcsEstimator](#) (ADCS estimator configuration) message. Orbital elements are updated by sending a [ConfigOrbitSatParams](#) (Satellite orbit parameters configuration) telecommand.

2.3.5 Estimation Mode 4 (EstTriad)

Description:

What:

Estimates attitude quaternion, attitude rotation matrix (transformation from orbit reference coordinates ORC to satellite body coordinates SBC) and Euler 2-1-3 RPY attitude angles, as well as inertial referenced body rate vector.



How:

This estimator requires two different vector pairs of sensor SBC (Satellite Body Coordinate frame) vector measurements and ORC (Orbit Reference Coordinate frame) modelled vectors. The first vector pair can be configured from, a) a valid nadir sensor SBC unit vector measurement and the modelled ORC unit vector (+Z₀ unit vector), b) the best fine sun sensor SBC unit vector measurement and the sun modelled ORC unit vector (from the sun and satellite orbit models) or c) the best magnetometer calibrated SBC unit vector and the IGRF (International Geomagnetic Reference Field) ORC modelled unit vector. The second vector pair can be configured from, a) to c) above or d) the coarse sun sensor SBC unit vector measurement and the sun-modelled ORC unit vector (from the sun and satellite orbit models).

When both vectors from a vector pair are not available, the unit quaternion, unit rotation matrix and zero RPY attitude angles are returned. This estimation mode will not return measurements in eclipse if the sun sensor is used.

The inertially referenced angular rate vector is directly obtained from the angular rate sensor (gyro).

Convergence time:

The estimator's convergence is instantaneous (deterministic).

Sensor Dependency:

Magnetometer (Mag0 or Mag1) and/or Coarse (CSS) and/or Fine sun sensor (Fss0 to Fss3) and/or Horizon (Hss0 or Hss1) sensor

Configurable values:

Orbital elements are updated by sending a [ConfigOrbitSatParams](#) (Satellite orbit parameters configuration) telecommand.

2.3.6 Estimation Mode 5 (*EstFullEkf*)

Description:

What:

Estimate attitude quaternion and angular rates. From the estimated quaternion vector, the attitude rotation matrix and Euler 2-1-3 RPY attitude angles are calculated. The estimated rates include both the inertial-referenced IRC and orbit-referenced ORC vectors.

How:

Using an extended Kalman filter to estimate the attitude quaternion and angular rates from a selectable combination of several attitude sensor SBC unit vector measurements and their equivalent modelled ORC unit vectors. The body angular rates are also estimated from these vectors and no direct rate sensor measurements are required.

Sensor Dependency:

The selectable attitude sensor measurements include these from the following sensors:

1. The best valid calibrated magnetometer (use Mag0 if valid, else Mag1)
2. A valid coarse sun sensor unit vector (the absolute CSS cell intensity must be valid)
3. The best valid fine sun sensor unit vector (Fss0 to Fss3) (the measurement closest to a sun sensor's boresight will be the best)
4. All valid horizon sensor's unit vectors (Hss0 and Hss1)
5. All the star tracker matched unit vectors (Str0 and Str1) (the best 3 vectors from each star tracker are used)

Details on which sensor measurements are used when multiple of the same type are available, or when none are available, are detailed in 3.8.



This estimator depends on the accuracy of a model of the satellite's dynamics, therefore unmodelled disturbance torques can reduce the estimation accuracy.

Convergency:

The convergence time for this estimator is approximately 30 samples or 6 seconds for a 5 Hz sample period.

Configurable Values:

The system and measurement noise covariance used in the filter can be changed by [ConfigAdcsEstimator](#) (ADCS estimator configuration) telecommand. The Kalman filter estimators make use of the satellite moment of inertia tensor, and estimated control torque to propagate the attitude state (the moment of inertia and control torque is used in the Euler dynamics equation). **The satellite moment of inertia tensor should ideally be measured before flight and is part of the configuration block.** The [ConfigAdcsSatellite](#) (ADCS satellite configuration) telecommand can be sent to update it. The control torque used in the propagation step is calculated from the actuator commands and model parameters for the actuators. The latter are also configuration parameters modifiable via [ConfigActuatorMtq](#) (Magnetorquer configuration) telecommand and [ConfigActuatorRwl](#) (Wheel configuration) telecommand. Orbital elements are updated by sending a [ConfigOrbitSatParams](#) (Satellite orbit parameters configuration) telecommand.

2.3.7 Estimation Mode 6 (EstGyroEkf)

Description:

What:

Estimation of the attitude quaternion and gyro bias vector from a selectable combination of several attitude sensor SBC unit vector measurements and their equivalent modelled ORC unit vectors. The body angular rates are measured from the rate sensor measurements and the estimated rate sensor bias subtracted to obtain the best body rates. From the estimated quaternion vector, the attitude rotation matrix and Euler 2-1-3 RPY attitude angles are calculated. The estimated rates include both the inertial referenced IRC and orbit referenced ORC vectors.

How:

A gyro based estimator using an extended Kalman filter to estimate the attitude quaternion and gyro bias vector from a selectable combination of several attitude sensor SBC unit vector measurements and their equivalent modelled ORC unit vectors.

When:

This estimator does not rely on a model of the satellite's dynamics and unmodelled disturbance torques does not reduce the estimation accuracy as the body rates are directly measured. The rate sensor bias drift can negatively impact the estimation accuracy especially when no attitude sensor measurements are available, e.g., during eclipse for satellites with no star trackers or IR nadir sensors, when only magnetometer measurements are available. This EKF is therefore more robust against unmodelled external disturbance torques, actuator modelling errors and satellite mass property uncertainties (moments and products of inertia).

Sensor Dependency:

The selectable attitude sensor measurements are like the [EstFullEkf](#) estimator.

Convergence:

The convergence time for this estimator is approximately 30 samples or 6 seconds for a 5 Hz sample period.

Configurable values:



The system and measurement noise covariance used in the filter can be changed by [ConfigAdcsEstimator](#) (ADCS estimator configuration) telecommand.

Orbital elements are updated by sending a [ConfigOrbitSatParams](#) (Satellite orbit parameters configuration) telecommand.

2.4 Control Modes

The selected control mode will determine which algorithm is used when computing actuator commands. It will determine the attitude outcome of the satellite, and the selection is fully dependent on the operational need of the mission at the time (detumbling vs. sun-tracking, vs. ground-tracking, for instance).

The control modes are grouped as in Table 3, based on commonality in function and mission phase.

Table 3: Available control modes

CONTROL MODE GROUP	ENUMERATION VALUE	CONTROL MODE
-	0: ConNone	No control
Detumbling and safe-mode magnetic spin control	1: ConBdot	Detumbling single axis Bdot
	2: ConYspin	Y-Thomson spin
	3: ConBdot3	3-axis Bdot detumbling
	4: ConDetumble	3-axis low power fast (8 Hz) detumbling
	5: ConSunYspin	Sun Y-axis spin detumbling
	6: ConZspin	Z-Thomson spin
	7: ConSunZspin	Sun Z-axis spin detumbling
Gravity-gradient	8: ConGGboom	MTQ control of GG boom & yaw zero control
	9: ConGGsun	MTQ control of GG boom & Z-RW yaw sun pointing control
Y-Momentum	10: ConYwheelInit	Y-Momentum wheel (Initial pitch acquisition)
	11: ConYwheel	Y-Momentum wheel (Steady state)
3-Axis Wheel control	12: ConXYZwheel	3-axis reaction wheel control
	13: ConSunTrack	3-axis reaction wheel sun tracking control
	14: ConTgtTrack	3-axis reaction wheel earth target tracking control (Nadir +Z facet)
	15: ConTgtSteer	3-axis reaction wheel earth target steering control (Nadir +Z facet)
	16: ConGndTrack	3-axis reaction wheel ground station target tracking control
	17: ConIrcTrack	3-axis reaction wheel inertial target tracking control
	18: ConMoonTrack	3-axis reaction wheel moon tracking control
	19: ConSatTrack	3-axis reaction wheel satellite tracking control
	20: ConYawSun	Nadir pointing and yaw towards the sun
	21: ConYawTarget	Nadir pointing and yaw towards a ground target
	22: ConRollSun	Nadir pointing and roll towards the sun



CONTROL MODE GROUP	ENUMERATION VALUE	CONTROL MODE
	23: ConRollTarget	Nadir pointing and roll towards a ground target
	24: ConYawTargetBest	Nadir pointing and yaw best +/-Y body facet towards a ground station
	25: ConFmcTarget	Forward Motion Compensation Target Scan
	26: ConYawSunSpin	RW sun yaw-spin control
	29: ConSunPayload	RW sun tracking and rotation for payload pointing to Zenith
	30: ConIrcSunTrack	RW sun tracking and rotation for payload pointing to Zenith
Safe-mode sun (with reaction wheels)	27: ConSunDetumble	RW detumbling to sun-point control
	31: ConSunAvoid	3-axis reaction wheel inertial target tracking control with optimal sun angle rotation
Astro-steering	28: ConAstroSteer	
Wheel management	50: ConStopRW	Controlled stop of reaction wheels
	51: ConHxyzRW	Open loop XYZ wheel momentum command control

The control modes make use of gain parameters and other settings that influence their behaviour. These settings may be changed using the [ConfigAdcsController](#) (ADCS controller configuration) message.

Control behaviour may also be affected by satellite configuration settings, such as chosen SBC referenced directions for sun-tracking and target tracking. These may be changed using the [ConfigAdcsSatellite](#) (ADCS satellite configuration) message.

Some controllers make use of “reference commands” to affect their behaviour. This includes ground target locations, and roll-, pitch- and yaw reference angles. Reference commands can be set using the [ReferenceRpy](#), [ReferenceTarget](#), [ReferenceIrcVector](#) and [ReferenceFmcParams](#) (Reference commands for controllers) messages.

2.4.1 Detumbling and safe-mode magnetic spin control

Detumbling and safe-mode spin control modes control the angular rates of the satellite (but not the attitude angles directly). These modes are deemed “safe” modes as they only make use of magnetorquer control (they do not require reaction wheels), have minimal dependencies on sensor inputs, are not dependent on orbit position information, and are robust against modelling errors and disturbances.

These modes can be used to recover from any initial tumble condition and place the satellite in a slow, stable and known tumbling motion – e.g., Y-Thomson or Z-Thomson spin, Sun Y-axis or Sun Z-axis spin.

The *Detumbling* and *spin* control mode group encompasses 7 control modes:



2.4.1.1 Control Mode 1: (ConBdot)

Description:

What:

1 Hz magnetic controller using only the single Y_B -axis magnetorquer to damp the X_B - and Z_B -axis angular rates.

When:

This controller is used during initial detumbling of the satellite if the body angular rates are less than ± 20 °/sec as a precursor to Y-Thomson detumbling.

How:

This single axis Bdot controller uses a gain K_D (typical value = 0.1). No orbit position, attitude or angular rate measurements/estimations are required, only the ground calibrated 3-axis magnetometer measurements are used. However, it is a good idea to use the 3-axis rate sensor measurements to monitor the angular rates during active control to ensure the correct polarity of the Y_B -axis magnetorquer. With a wrong polarity the X_B - and Z_B -axis angular rates will increase. This controller can be activated with a time-out period to ensure that a wrong magnetorquer polarity will not increase the angular rates to very high rates.

Sensor and Estimator dependence:

Magnetometer measurements are required.

Actuator Dependence:

Magnetorquers

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- gain K_D (typical value = 0.1).

Other Requirements and Restrictions:

Control Mode 1 can only detumble the satellite from angular rates below 20 deg/s. For higher initial rates, Control Mode 3 or 4 can be used.

2.4.1.2 Control Mode 2: (ConYSpin)

Description:

What:

1 Hz magnetic Y-Thomson controller. Control Mode 2 adds Y-Thomson Spin to Mode 1 using the X- and Z-axis magnetorquers

When:

This controller is used to place the satellite in a known tumbling motion, and can be used from any initial condition where the body angular rates are less than ± 20 °/s.

How:

using Y_B -axis magnetorquer to damp the X_B - and Z_B -axis angular rates (like the Bdot controller above) and the X_B - and Z_B -axis magnetorquer to control the Y_B -axis angular rate to a reference value specified by the configuration parameter $wRef$ (typical values = -1, -2 °/sec). For orbits with inclination $> 20^\circ$ and satellites with dominant Y_B -axis MOI (I_{YY} at least 10% larger than I_{XX} and I_{ZZ}) the satellite will align its Y_B spin axis to the orbit anti-normal direction (negative $wRef$ value) or orbit normal direction (positive $wRef$ values), resulting in small roll and yaw attitude angles with only pitch rotations.

Configurable and Reference Values:



- **ConfigAdcsController** configuration values
- The Bdot controller gain K_D (typical value = 0.1)
- The Y-spin controller gain K_S (typical value = 0.02 to 0.04)
- w_{Ref} , the reference spin rate (typical values = -1, -2 °/sec).

Sensor/Estimator Dependency:

Like the **ConBdot** controller, the ground calibrated 3-axis magnetometer measurements are used. In this control mode, it is also necessary to estimate or measure the Y_B -axis angular rate, so that the controller can control it to the w_{Ref} reference rate. Estimation is normally performed in this mode by using the 3-axis rate sensor measurements (estimation mode 1). The X_B - and Z_B -axis magnetorquer polarities can also be verified, as the incorrect polarities will not track the Y_B -axis reference rate. This controller can also be activated with a time-out period to ensure that a wrong magnetorquer polarity will not increase the angular rates to very high rates.

Actuator dependency: Magnetorquers

Other Requirements and Restrictions:

Control Mode 2 can only detumble the satellite from angular rates below 20 deg/s. For higher initial rates, Control Mode 3 or 4 can be used.

2.4.1.3 Control Mode 3: (ConBdot3)

Description:

What:

1 Hz magnetic controller to do 3-axis B-dot detumbling, by activating the magnetorquers. The BDot3 control mode has an option to select individual magnetorquer axes, instead of using all three magnetorquers at once. This is useful in situations where torquer polarity needs to be confirmed in-flight. A configuration parameter (**bdot3Axis**) is used to select which axis' magnetorquer(s) may be used. The default is 0 (**Bdot3AxisAll**) which results in all torquers being used. Setting a value of 1 (**Bdot3AxisX**) will use only the X-axis torquer, etc.

When:

This controller is used to recover from relatively high angular rates ($> \pm 20$ °/sec), either due to unintentional spin-up or after initial release from the launch vehicle.

How:

This controller will reduce the angular rates of all body axes to low values, using only the ground calibrated 3-axis magnetometer measurements. However, it is a good idea to use the 3-axis rate sensor measurements to monitor the angular rates during active control to ensure the correct polarity of the magnetorquers. The controller can detumble from initial angular rates less than 100 °/sec. (For higher angular rates, control mode 4 should be used).

This 3-axis Bdot controller uses a single gain K_D for all axes (typical value = 0.1). This controller can also be activated with a time-out period to ensure that a wrong magnetorquer polarities will not increase the angular rates to very high rates.

Sensor and Estimator dependence:

Magnetometer measurements are required.

Actuator Dependence:

Magnetorquers



Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
 - gain K_D (typical value = 0.1)
 - [bdot3Axis](#) axis selection

Other Requirements and Restrictions:

The controller can detumble from initial angular rates less than 100 °/sec. For higher rates, control mode 4 should be used.

2.4.1.4 Control Mode 4: (ConDetumble)

Description:

What:

8 Hz magnetic controller to do 3-axis fast detumbling to low angular rates using only a single axis magnetorquer at a time.

When:

This controller is used to recover from relatively high angular rates ($> \pm 20$ °/sec), either due to unintentional spin-up or after initial release from the launch vehicle.

How:

This controller will reduce the angular rates of all body axes to low values, using only the ground calibrated 3-axis magnetometer measurements. It is more power efficient than ConBdot3 since it only activates one torquer at a time. This 3-axis controller uses a single gain K_{DF} for all axes (typical value = 1.0). This controller can also be activated with a time-out period to ensure that a wrong magnetorquer polarities will not increase the angular rates to very high rates.

Sensor and Estimator dependence:

Magnetometer measurements are required.

Actuator Dependence:

Magnetorquers

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
 - gain K_{DF} (typical value = 1.0).

Other Requirements and Restrictions:

The controller can detumble from high initial angular rates up to 1000 °/sec.

2.4.1.5 Control Mode 5: (ConSunYspin)

Description:

What:

1 Hz magnetic spin controller like [ConYspin](#), but to point the Y_B spin-axis towards the sun direction.



When:

After the satellite has been placed in a stable Y-Thomson spin, this control mode can be used to achieve optimal power generation from solar panels that are mounted or deployed with normal vector along the Y_B axis, without required the use of reaction wheels.

How:

This controller uses the ground calibrated magnetometer measurements and the sun vector measurement from the coarse sun sensors (CSS), or the fine sun sensors (FSS) when available. During the sunlit part of each orbit the controller will slew the spin axis towards the sun and during eclipse the normal Y-Thomson controller will be active to damp any X_B - and Z_B -axis angular rates and to control the Y_B -axis angular rate to a reference value specified by the configuration parameter $wRef$ (typical values = -1, -2 °/sec). Negative $wRef$ values will result in the satellite's $+Y_B$ -axis pointing to the sun on the orbit anti-normal side of the orbit plane and positive $wRef$ values will point the satellite's $+Y_B$ -axis to the sun on the orbit normal side of the orbit plane. The 3-axis rate sensor measurements are used to control the Y_B -axis angular rate towards the $wRef$ reference rate. The X_B - and Z_B -axis magnetorquer polarities can also be verified, as the incorrect polarities will not track the Y_B -axis reference rate. This controller can also be activated with a time-out period to ensure that a wrong magnetorquer polarity will not increase the angular rates to extremely high rates.

Sensor and Estimator dependence:

Magnetometer measurements are required. In this control mode, it is also necessary to estimate or measure the Y_B -axis angular rate, so that the controller can control it to the $wRef$ reference rate. Estimation is normally performed in this mode by using the 3-axis rate sensor measurements (estimation mode 1).

CSS or Fine sun sensor (FSS) measurements must be available to track the sun.

Actuator Dependence:

Magnetorquers

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The Bdot controller gains K_{D-sun} and $K_{D-eclipse}$
- The Y-spin controller gain K_S (typical value = 0.02 to 0.04)
- $wRef$, the reference spin rate (typical values = -1, -2 °/sec).

Other Requirements and Restrictions:

The normal Y-spin controller ([ConYSpin](#)) must have been active long enough to result in a stable Y-Thomson spin before transitioning to this control mode.

2.4.1.6 Control Mode 6: (ConZspin)

This is the same controller as [ConYSpin](#), except that the dominant moment of inertia axis must be I_{ZZ} , and the Z_B axis will be aligned with the orbit normal, or anti-normal (depending on the sign of the $wRef$ parameter). All other parameters and requirements are the same.

2.4.1.7 Control Mode 7: (ConSunZspin)

This is the same controller as for [ConSunYspin](#), except that the dominant moment of inertia axis must be I_{ZZ} , and the Z_B axis will track the sun vector on either the orbit normal or orbit anti-normal side of the orbit plane, depending on the sign of the $wRef$ parameter. All other parameters and requirements are the same.



2.4.2 Gravity Gradient

The **Gravity-Gradient** group of control modes is intended to be used on satellites with small Z_B -axis moment of inertia, compared to the X_B and Y_B axes, where the satellite can make use of gravity gradient torque for passive attitude stability. This is typical for a satellite that deploys a boom towards the $+Z_B$ or $-Z_B$ direction.

This control mode group encompasses 2 control modes:

2.4.2.1 Control Mode 8: (ConGGboom)

Description:

What:

1 Hz magnetic controller for a satellite with a deployed boom along the Z_B -axis to align this axis along the nadir/zenith direction

When:

To be used on satellites with boom deployed along the Z_B direction, where gravity gradient can be used to aid in passive attitude stability.

How:

This controller will do libration damping of roll and pitch angles and control the yaw angle magnetically towards zero.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The libration rate damping gain K_N (typical value = 0.4) and
- The quaternion error gain K_Q (typical value = 5.0).

Sensor/Estimator Dependency:

The controller requires not only knowledge of the measured/estimated angular rates, but also the estimated attitude quaternion. This means a Full State EKF or Gyro EKF must be selected.

Actuator dependency: Magnetorquers

Other Requirements and Restrictions:

For this controller to perform best, the ratio between the transverse I_{xx} and I_{yy} MOI and boom axis I_{zz} MOI must be maximised, typically I_{xx}/I_{zz} and $I_{yy}/I_{zz} > 20$.

2.4.2.2 Control Mode 9: (ConGGsun)

Description:

What:

1 Hz magnetic controller for a satellite with a deployed boom along the Z_B -axis to align this axis along the nadir/zenith direction, while controlling the yaw angle using a reaction wheel, to orient a chosen body axis direction to track the sun vector in the ORC X/Y plane.

When:

To be used on satellites with boom deployed along the Z_B direction, where gravity gradient can be used to aid in passive attitude stability.



How:

This controller will do libration damping of roll and pitch angles. The Z-wheel is used to yaw control a specified body axis direction in the X_B/Y_B plane to optimally track the sun vector direction. The Z-wheel angular momentum is also dumped by magnetic control towards zero

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The libration rate damping gain K_N (typical value = 0.4) and
- The quaternion error gain K_Q (typical value = 5.0).
- The proportional gain K_{P1} and
- The derivative gain K_{D1}

Sensor/Estimator Dependency:

The controller requires not only knowledge of the measured/estimated angular rates, but also the estimated attitude quaternion. This means a Full State EKF or Gyro EKF must be selected.

Actuator dependency: Magnetorquers

Other Requirements and Restrictions:

For this controller to perform best, the ratio between the transverse I_{xx} and I_{yy} MOI and boom axis I_{zz} MOI must be maximised, typically I_{xx}/I_{zz} and $I_{yy}/I_{zz} > 20$.

2.4.3 Y-Momentum

The Y-momentum control mode group can only be activated once the satellite is in a stable Y-Thomson tumbling state. In *Y-momentum* control a single Y-axis aligned wheel is used to absorb the angular momentum of the satellite, causing it to stop spinning and stabilize to the nominal orientation (zero roll, pitch, and yaw angles).

The Y-Momentum control mode is only used on satellites stabilised with a single Y-Momentum wheel. Although there are two control modes in the group, they are essentially different stages of the same mode.

2.4.3.1 Control Mode 10: (ConYwheelInit)

Description:

What:

This is not a feedback Y-wheel controller, but an initialisation step to the next actual Y-wheel controller.

When:

Once the satellite is in a stable Y-Thomson spin and it is required to make use of a single Y-momentum wheel.

How:

This controller will open-loop ramp the Y-wheel momentum to a reference value $hRef$ to transfer the Y-Thomson Y_B -axis body spin rate to the wheel disc. The controller will then wait for the satellite's pitch angle to become less than $\pm 45^\circ$ before it automatically enables the Y-wheel controller [ConYwheel](#).

Configurable and Reference Values:

None (open-loop ramp)



Sensor/Estimator Dependency:

For this controller to work, the pitch angle must be estimated along with the angular rate vector, i.e., any pitch attitude estimator must be active.

Actuator dependency: Magnetorquers, Y-axis wheel

Other Requirements and Restrictions:

For this controller to perform best the Y_B moment of inertia must dominate. (I_{YY} at least 10% larger than I_{XX} and I_{ZZ})

2.4.3.2 Control Mode 11: (ConYwheel)

Description:

What:

A magnetic and Y-wheel controller for a satellite that is Y-momentum stabilised, with optional control of the pitch angle (to reference value)

When:

An automatic transition to this mode will take place from [ConYwheelInit](#) when the pitch angle is below a threshold.

How:

A magnetic nutation damping controller is used to damp the roll (X_B) and yaw (Z_B) angular rates and to maintain the Y-wheel angular rate at the reference value of $hRef$. The Y-wheel is used to control the pitch angle to any reference value, the default is zero pitch or nadir pointing.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The nutation rate damping gain K_N (typical value = 0.4) and
- The wheel momentum dumping gain K_H (typical value = 0.5).
- The proportional gain K_{P1} and
- The derivative gain K_{D1}
- The reference Y-wheel angular momentum, $hRef$

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and pitch angle must be estimated, i.e., any angular rate and pitch attitude estimator must be active, e.g., the Magnetometer estimated rate and pitch estimator (EstPitchRkf), the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, Y-axis wheel

Other Requirements and Restrictions:

For this controller to perform best the Y_B moment of inertia must dominate. (I_{YY} at least 10% larger than I_{XX} and I_{ZZ})

2.4.4 XYZ wheel

The **XYZ wheel control** group uses 3-axis reaction wheels with zero momentum bias or a pyramid configuration of four reaction wheels with individual wheel speed bias, but zero external momentum bias. This group has a control mode to track the nadir vector (zero roll, pitch, and yaw angles) or constant attitude reference angles which are set using the [Reference commands for controllers](#) telecommand. It also has control modes to track the Sun vector with the satellite's solar panels for maximum power or to track a reference target at a specific location on Earth or to track an inertially referenced target or another target satellite in the same or different orbit. The ground target must first be selected with the [Reference commands for controllers](#) telecommand.



The **XYZ wheel group** is only used on satellites with 3-axis reaction wheel capability and can make use of 13 control modes:

2.4.4.1 Control Mode 12: (ConXYZwheel)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised.

When:

Stable attitude, nadir pointing, or specific roll, pitch and yaw offset angles are required.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PD controller to control the full 3-axis attitude to reference roll, pitch, and yaw values according to the Euler 213 definition: the first rotation angle is pitch to $\pm 180^\circ$ range, then roll to $\pm 90^\circ$ range and finally yaw to $\pm 180^\circ$ range.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, KH (typical value = 0.5)
- The PD wheel controller proportional gain K_{p2} and
- The PD wheel controller derivative gain K_{D2}
- [ReferenceRpy](#) reference values
- Roll, pitch and yaw angles

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.4.2 Control Mode 13: (ConSunTrack)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, that will point the satellite body referenced (SBC) vector specified by the [ConfigAdcsSatellite.SunPointBodyVec](#) [XYZ] configuration setting to the sun. Depending on the [sunTrackEclEn](#) flag, the controller will either automatically transition to nadir pointing (zero roll, pitch and yaw) in eclipse ([sunTrackEclEn](#) set to FALSE), or continue to track the sun, even though it may fall below the horizon ([sunTrackEclEn](#) set to TRUE).



When:

Optimal solar power generation is required.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PD controller to control the full 3-axis attitude to track the sun vector direction with a specified body vector direction. In eclipse the satellite will be controlled to a nadir pointing attitude, i.e., zero roll, pitch and yaw angles.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- The eclipse sun-track flag, [sunTrackEclEn](#)
- [ConfigAdcsSatellite](#) configuration values
- Sun-pointing SBC vector ([SunPointBodyVec](#) [XYZ])

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEKF) or the Gyro EKF (EstGyroEKF).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.4.3 Control Mode 14: (ConTgtTrack)

Description:

What:

A reaction wheel controller for a satellite that is 3-axis stabilised to track a specified ground target by pointing the $+Z_B$ body axis.

When:

Required to track ground target, such as a ground station, with the $+Z_B$ axis.

How:

The reaction wheels are controlled with a quaternion feedback PID controller to control the full 3-axis attitude while tracking a specified ground target direction with the $+Z_B$ body axis (the nominal nadir pointing facet). The ground target will be specified beforehand by setting the geocentric latitude, longitude and altitude. The pitch and roll reference angles are automatically adjusted to point towards the ground target while the yaw angle will be controlled towards the commanded (reference) yaw angle, as specified using the [ReferenceRpy](#) command.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The PID wheel controller proportional gain K_{P3} and
- The PID wheel controller derivative gain K_{D3}
- The PID wheel controller integral gain K_{I3}
- [ReferenceRpy](#) reference values
- Reference yaw angle, [CmdRpyYaw](#)



- [ReferenceTarget](#) reference values
- Target geocentric latitude, [CmdTargetLatitude](#)
- Target longitude, [CmdTargetLongitude](#)
- Target geocentric altitude, [CmdTargetAltitude](#)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.4.4 Control Mode 15: (ConTgtSteer)

Description:

What:

A reaction wheel controller for a satellite that is 3-axis stabilised to track a specified ground target by pointing the $+Z_B$ body axis, with the ability to steer the roll and pitch angles from an external feedback instrument, e.g., a laser reflection terminal. Magnetic control will be inactive during this control mode to minimise magnetic disturbances.

When:

Required to track ground target, such as a ground station, with the $+Z_B$ axis with external feedback for finer control.

How:

The reaction wheels are controlled with a quaternion feedback PID controller to control the full 3-axis attitude while tracking a specified ground target direction with the $+Z_B$ body axis (the nominal nadir pointing facet). The ground target will be specified beforehand by setting the geocentric latitude, longitude and altitude. The pitch and roll reference angles are automatically adjusted to point towards the ground target while the yaw angle will be controlled towards zero or towards a reference yaw angle. The quaternion feedback reaction wheel PID controller uses 3 gain parameters derived from the ConTgtTrack gains above but having a much faster settling time.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The PID wheel controller proportional gain K_{P3} and
- The PID wheel controller derivative gain K_{D3}
- The PID wheel controller integral gain K_{I3}
- Yaw compensation flag, [YawCompensate](#)
- [ReferenceTarget](#) reference values
- Target geocentric latitude, [CmdTargetLatitude](#)
- Target longitude, [CmdTargetLongitude](#)
- Target geocentric altitude, [CmdTargetAltitude](#)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels



Other Requirements and Restrictions: This controller can only be entered from the ConTgtTrack controller while the satellite's $+Z_B$ body axis is already pointing to the preselected ground target.

2.4.4.5 Control Mode 16: (ConGndTrack)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised to track a specified ground target by pointing a user configured SBC vector to the target.

When:

Required to track ground target, such as a ground station, with arbitrary body axis.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PID controller to control the full 3-axis attitude to track the ground target vector direction with a specified body vector direction. The ground target must be specified beforehand by setting the geocentric latitude, longitude and altitude. The attitude reference angles are automatically adjusted to point towards a ground target, while the rotation angle around the pointing vector will be optimised for minimum Euler angle rotation.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
 - The wheel momentum management gain, K_H (typical value = 0.5)
 - The PID wheel controller proportional gain K_{P3} and
 - The PID wheel controller derivative gain K_{D3}
 - The PID wheel controller integral gain K_{I3}
- [ConfigAdcsSatellite](#) configuration values
 - Target tracking SBC vector ([TgtTrackBodyVec \[XYZ\]](#))
- [ReferenceTarget](#) reference values
 - Target geocentric latitude, [CmdTargetLatitude](#)
 - Target longitude, [CmdTargetLongitude](#)
 - Target geocentric altitude, [CmdTargetAltitude](#)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels



Other Requirements and Restrictions: N/A.

2.4.4.6 Control Mode 17: (*ConIrcTrack*)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised to track an inertially referenced target, e.g., a specific star vector direction.

When:

Required to track inertial vector, such as a star vector direction, with arbitrary body axis.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PID controller to control the full 3-axis attitude to track the inertial target with a specified body vector direction. The target will be specified beforehand by setting the inertial referenced unit vector direction. The attitude reference angles are automatically adjusted to point towards the inertial target, while the rotation angle around the pointing vector will be optimised for minimum Euler angle rotation.

Configurable and Reference Values:

- **ConfigAdcsController** configuration values
 - The wheel momentum management gain, K_H (typical value = 0.5)
 - The PID wheel controller proportional gain K_{P3} and
 - The PID wheel controller derivative gain K_{D3}
 - The PID wheel controller integral gain K_{I3}
- **ConfigAdcsSatellite** configuration values
 - IRC reference vector (**IrcTrackBodyVec** [XYZ])
- **ReferenceIrcVector** reference values
 - ECI referenced vector to track, **CmdEciVec** [XYZ]

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEKF) or the Gyro EKF (EstGyroEKF).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A.

2.4.4.7 Control Mode 18: (*ConMoonTrack*)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised to track the moon.

When:

Required to track the moon with arbitrary body axis.



How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PID controller to control the full 3-axis attitude to track the moon with a specified body vector direction. The moon position and reference unit vector direction are automatically updated from a lunar orbit model. The attitude reference angles are automatically adjusted to point towards the moon, while the rotation angle around the pointing vector will be optimised for minimum Euler angle rotation.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
 - The wheel momentum management gain, K_H (typical value = 0.5)
 - The PID wheel controller proportional gain K_{P3} and
 - The PID wheel controller derivative gain K_{D3}
 - The PID wheel controller integral gain K_{I3}
- [ConfigAdcsSatellite](#) configuration values
 - Moon tracking SBC vector ([MoonTrackBodyVec \[XYZ\]](#))

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEKF) or the Gyro EKF (EstGyroEKF).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A.

2.4.4.8 Control Mode 19: (ConSatTrack)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised to track another satellite.

When:

Required to track a satellite with arbitrary body axis.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PID controller to control the full 3-axis attitude to track the target satellite with a specified body vector direction. The target satellite position and unit vector direction are automatically updated from a SGP4 orbit propagator using the target satellite's TLEs. The attitude reference angles are automatically adjusted to point towards the satellite, while the rotation angle around the pointing vector will be optimised for minimum Euler angle rotation.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
 - The wheel momentum management gain, K_H (typical value = 0.5)
 - The PID wheel controller proportional gain K_{P3} and
 - The PID wheel controller derivative gain K_{D3}
 - The PID wheel controller integral gain K_{I3}
- [ConfigOrbitTargetParams](#) configuration values
 - Orbit elements (SGP4) for target satellite



- [ConfigAdcsSatellite](#) configuration values
 - Satellite tracking SBC vector ([SatTrackBodyVec \[XYZ\]](#))

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEKF) or the Gyro EKF (EstGyroEKF).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A.

2.4.4.9 Control Mode 20: (ConYawSun)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, to point the $+Z_B$ axis towards nadir while rotating around the yaw axis to track the sun. Depending on the [sunTrackEclEn](#) flag, the controller will either automatically transition to nadir pointing (zero roll, pitch and yaw) in eclipse ([sunTrackEclEn](#) set to FALSE), or continue to track the sun using yaw changes only, even though it may fall below the horizon ([sunTrackEclEn](#) set to TRUE).

When:

Required to point the $+Z_B$ axis towards nadir while still optimizing power generation from solar panels.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value h_{Bias} aligned to the orbit normal direction (default h_{Bias} is zero). The reaction wheels are controlled with a quaternion feedback PD controller to control the full 3-axis attitude to point towards nadir and optimise the yaw attitude angle to track the sun vector direction with a specified body vector direction. When the sun vector is aligned to the zenith vector direction the yaw tracking will be momentarily stopped as the required yaw angle becomes undefined

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- The eclipse sun-track flag, [sunTrackEclEn](#)
- [ConfigAdcsSatellite](#) configuration values
- Sun-tracking SBC vector ([SunPointBodyVecX \[XYZ\]](#))

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEKF) or the Gyro EKF (EstGyroEKF).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A.

2.4.4.10 Control Mode 21: (ConYawTarget)

Description:

What:



A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, to point the $+Z_B$ axis towards nadir while rotating around the yaw axis to optimally track a ground target.

When:

Required to point the $+Z_B$ axis towards nadir while rotating the yaw axis to minimize a specific body direction to a target, for example wide-beam antenna pointing to a ground station that does not require direct pointing.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PD controller to control the full 3-axis attitude to point towards nadir and optimise the yaw attitude angle to track the ground target with a specified body vector direction. When the ground target vector is aligned to the nadir vector direction the yaw tracking will be momentarily stopped as the required yaw angle becomes undefined.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- [ConfigAdcsSatellite](#) configuration values
- Target-tracking SBC vector ([TgtTrackBodyVecX \[XYZ\]](#))
- [ReferenceTarget](#) reference values
- Target geocentric latitude, [CmdTargetLatitude](#)
- Target longitude, [CmdTargetLongitude](#)
- Target geocentric altitude, [CmdTargetAltitude](#)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A.

2.4.4.11 Control Mode 22: (ConRollSun)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, to keep the $+X_B$ axis in the velocity direction while controlling the roll axis to optimally track the sun. Depending on the [sunTrackEc1En](#) flag, the controller will either automatically transition to nadir pointing (zero roll, pitch and yaw) in eclipse ([sunTrackEc1En](#) set to FALSE), or continue to track the sun using roll rotation only, even though it may fall below the horizon ([sunTrackEc1En](#) set to TRUE).

When:

Required to point the $+X_B$ axis towards velocity direction while still optimizing power generation from solar panels.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a



quaternion feedback PD controller to control the full 3-axis attitude to keep the satellite's X_B -axis in the velocity direction but optimise the roll attitude angle to track the sun vector direction with a specified body vector direction. In eclipse the satellite will be nadir pointed (zero RPY attitude). When the sun vector is aligned to the orbit velocity or anti-velocity direction the roll tracking will be momentarily stopped as the required roll angle becomes undefined.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- The eclipse sun-track flag, [sunTrackEclEn](#)
- [ConfigAdcsSatellite](#) configuration values
- Sun-tracking SBC vector ([SunPointBodyVecX \[XYZ\]](#))

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEKF) or the Gyro EKF (EstGyroEKF).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A.

2.4.4.12 Control Mode 23: ([ConRollTarget](#))

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, to keep the $+X_B$ axis in the velocity direction while controlling the roll axis to optimally track a ground target.

When:

Required to point the $+X_B$ axis towards velocity direction while rotating the roll axis to minimize a specific body direction to a target, for example wide-beam antenna pointing to a ground station that does not require direct pointing.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PD controller to control the full 3-axis attitude to keep the satellite's X_B -axis in the orbit velocity direction but optimise the roll attitude angle to track an automatically updated ground specified target vector with a specified body vector direction. The ground target will be specified beforehand by setting the geocentric latitude, longitude and altitude. If the target aligns with the velocity direction, roll tracking will be momentarily stopped as the required roll angle becomes undefined.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- [ConfigAdcsSatellite](#) configuration values
- Target-tracking SBC vector ([TgtTrackBodyVecX \[XYZ\]](#))
- [ReferenceTarget](#) reference values



- Target geocentric latitude, [CmdTargetLatitude](#)
- Target longitude, [CmdTargetLongitude](#)
- Target geocentric altitude, [CmdTargetAltitude](#)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.4.13 Control Mode 24: (ConYawTargetBest)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, to point the $+Z_B$ axis towards nadir while rotating around the yaw axis to optimally track a ground target.

When:

Required to point the $+Z_B$ axis towards nadir while rotating the yaw axis to minimize either the $+Y_B$ or $-Y_B$ body direction to a target, for example wide-beam antenna pointing to a ground station that does not require direct pointing, where antennas are mounted on the $+Y_B$ and $-Y_B$ facets.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PD controller to control the full 3-axis attitude to keep the satellite nadir pointing but optimise the yaw attitude angle to track an automatically updated ground specified target vector with either the $+Y_B$ or $-Y_B$ body facet, the best facet is chosen to minimise the yaw rotation angle. When the ground target vector is aligned to the nadir vector direction the yaw tracking will be momentarily stopped as the required yaw angle becomes undefined.

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- [ReferenceTarget](#) reference values
- Target geocentric latitude, [CmdTargetLatitude](#)
- Target longitude, [CmdTargetLongitude](#)
- Target geocentric altitude, [CmdTargetAltitude](#)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.4.14 Control Mode 25: (ConFmcTarget)

Description:



What:

A reaction wheel controller that can be used to perform FMC (Forward Motion Compensation) imaging scan operations. This mode will also perform yaw compensation, to follow a yaw reference angle so that the imager alignment around the boresight will be perpendicular with the ground scan direction (taking Earth rotation into account).

When:

Use this control mode when it is required to perform an FMC imaging scan, with the $+Z_B$ axis pointing towards the scan direction. The Unix time around the scan operation must be known: the Unix time at the center point of the scan, and the duration before and after must be specified.

How:

The controller makes use of a state machine that changes state based on Unix time. The controller will progress through the following states:

STATE	TIME START	STATE DURATION	DESCRIPTION
FmcInactive	-	-	FMC control mode is not selected
FmcWaiting	-	-	Waiting for FMC scan to start. (defaults to same control as ConXYZwheel)
FmcPrepare	$t_0 - \langle \text{scan duration} / 2 \rangle - 90\text{s}$	30 s	Control to FMC roll offset, and zero pitch angle and to yaw compensation angle
FmcInitialPitch	$t_0 - \langle \text{scan duration} / 2 \rangle - 60\text{s}$	50 s	Control initial FMC pitch angle for scan starts and yaw compensation angle
FmcRwRampUp	$t_0 - \langle \text{scan duration} / 2 \rangle - 10\text{s}$	10 s	RW ramp for 10 sec towards FMC rate
FmcScan	$t_0 - \langle \text{scan duration} / 2 \rangle$	$\langle \text{scan duration} \rangle$	FMC pitch scan at constant roll and yaw compensation angle
FmcRwRampDown	$t_0 + \langle \text{scan duration} / 2 \rangle$	10 s	Ramp Pitch RW for 10 sec back to FMC1
FmcStopping	$t_0 + \langle \text{scan duration} / 2 \rangle + 10\text{s}$	50 s	Stop FMC control
FmcEnded	$t_0 + \langle \text{scan duration} / 2 \rangle + 60\text{s}$	-	FMC scan ended. (defaults to same control as ConXYZwheel)

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- [ReferenceFmcParam](#) reference values
- FMC scanning factor, [CmdFmcFactor](#)
- FMC roll command, [CmdFmcRoll](#)
- Time at center of scan, [CmdFmcUnixSeconds](#)
- Sub-seconds to unix time, at center of scan, [CmdFmcUnixNs](#)
- FM scan duration, [CmdFmcScanDuration](#)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels



Other Requirements and Restrictions: N/A

2.4.4.15 Control Mode 26: (ConYawSunSpin)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, that will point the satellite $-Z_B$ axis to the sun while maintaining a spin rotation around the Z-axis at the configured reference spin rate ($w_{SunYawRef}$ in [ConfigAdcsController](#)). During eclipse the controller will automatically transition to nadir pointing (zero roll, pitch and yaw).

When:

Use this controller if it is required to point a payload or solar panel along the $-Z_B$ axis to the sun, while spinning around the sun vector.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value h_{Bias} aligned to the orbit normal direction (default h_{Bias} is zero). The reaction wheels are controlled with a quaternion feedback PID controller to control the full 3-axis attitude to keep the satellite's $-Z_B$ -axis in the sun direction while maintaining a spin at the reference spin rate around the body Z axis. In eclipse the satellite will be nadir pointed using a PD controller (zero RPY attitude).

(For this control mode the RW bias values should be near zero, to prevent fast sinusoidal angular momentum exchange between the X- and Y-axis.)

Configurable and Reference Values:

- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- The PID wheel controller proportional gain K_{P3} and
- The PID wheel controller derivative gain K_{D3}
- The PID wheel controller integral gain K_{I3}

The reference yaw spin rate, [wSunYawRef](#)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEKF) or the Gyro EKF (EstGyroEKF).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.4.16 Control Mode 27: (ConSunDetumble)

Description:

What:

A reaction wheel controller that will orient the solar panels to the sun, requiring minimum sensor or estimation information.

When:

Use this as a safe-mode control mode, or to detumble from low angular rates to an optimal power generation orientation.

How:



The reaction wheels are controlled with a quaternion feedback PD controller, to align the measured sun vector with the configured sun-tracking SBC vector. The controller does not use estimated attitude information but calculates the error quaternion for the feedback controller from sun measurements directly. If available, the most appropriate FSS measurement will be used, otherwise the sun vector measured by the CSS will be used. A magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value.

Configurable and Reference Values:

- [ConfigAdcsSatellite](#) configuration values
- Sun-tracking SBC vector ([SunPointBodyVecX \[XYZ\]](#))
- [ConfigAdcsController](#) configuration values
- The wheel momentum management gain, K_H (typical value = 0.5) (sun-avoidance only)
- The PD wheel controller proportional gain K_{P1} and
- The PD wheel controller derivative gain K_{D1}

Sensor/Estimator Dependency:

This mode only needs angular rate measurements and does not need position information (no orbit info/TLEs needed). The estimation mode must either be EstGyro or EstGyroEKF. Sun sensor measurements must be available, either from CSS, or FSS.

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A



2.4.4.17 Control Mode 28: (ConAstroSteer)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, that will point the satellite $-Z_B$ vector to the selected inertial target vector, while rotating the satellite around the $-Z_B$ vector so that satellite body referenced (SBC) vector specified by the `ConfigAdcsSatellite.SunPointBodyVec` [XYZ] will be optimally aligned towards the sun. Depending on the `sunTrackEclEn` flag, the controller will either continue to optimally align the configured body vector to the sun, even though it may fall below the horizon (`sunTrackEclEn` set to TRUE), or perform the smallest Euler angle rotation to point the $-Z_B$ axis to the target (no specific rotation for solar panel alignment) if `sunTrackEclEn` is set to FALSE.

When:

Payload that is mounted along the $-Z_B$ axis needs to point to an inertial target, while solar panel needs to generate optimal power.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a combination of a high bandwidth PID and low bandwidth PD quaternion feedback controller to control the full 3-axis attitude to track the inertial vector direction with the $-Z_B$ body direction, while the rotation around the $-Z_B$ axis will optimize (minimize) the angle between the configured `ConfigAdcsSatellite.SunPointBodyVec` vector and the sun.

Configurable and Reference Values:

- `ConfigAdcsController` configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel low bandwidth controller proportional gain K_{P2} and
- The PD wheel low bandwidth controller derivative gain K_{D2}
- The PID wheel high bandwidth controller proportional gain K_{P3} and
- The PID wheel high bandwidth controller derivative gain K_{D3}
- The PID wheel high bandwidth controller integral gain K_{I3}
- The eclipse sun-track flag, `sunTrackEclEn`
- `ConfigAdcsSatellite` configuration values
- Sun-pointing SBC vector (`SunPointBodyVec` [XYZ])
- Target-track SBC vector (`TgtTrackBodyVec` [XYZ])

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., only the Gyro EKF (EstGyroEKF) can be used.

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.4.18 Control Mode 29: (ConSunPayload)

Description:

What:

A magnetic and reaction wheel controller for a satellite that is 3-axis stabilised, that will point the satellite body referenced (SBC) vector specified by the `ConfigAdcsSatellite.SunPointBodyVec` [XYZ] configuration setting to the sun, while rotating the satellite around the sun vector so that satellite body referenced (SBC) vector specified by the `ConfigAdcsSatellite.TgtTrackBodyVec` [XYZ] will be optimally



aligned towards zenith. Depending on the `sunTrackEclEn` flag, the controller will either automatically transition to nadir pointing (zero roll, pitch and yaw) in eclipse (`sunTrackEclEn` set to FALSE), or continue to track the sun, even though it may fall below the horizon (`sunTrackEclEn` set to TRUE).

When:

Optimal solar power generation is required, while pointing a payload towards zenith.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PD controller to control the full 3-axis attitude to track the sun vector direction with a specified body vector direction. In eclipse the satellite will be controlled to a nadir pointing attitude, i.e., zero roll, pitch and yaw angles.

Configurable and Reference Values:

- `ConfigAdcsController` configuration values
- The wheel momentum management gain, K_H (typical value = 0.5)
- The PD wheel controller proportional gain K_{P2} and
- The PD wheel controller derivative gain K_{D2}
- The eclipse sun-track flag, `sunTrackEclEn`
- `ConfigAdcsSatellite` configuration values
- Sun-pointing SBC vector (`SunPointBodyVec [XYZ]`)
- Target-track SBC vector (`TgtTrackBodyVec [XYZ]`)

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (`EstFullEKF`) or the Gyro EKF (`EstGyroEKF`).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.4.19 Control Mode 30: (`ConlrcSunTrack`)

Description:

What:

A similar controller to `ConlrcTrack`, where an inertially referenced target, e.g., a specific star vector direction is tracked (by pointing a configurable SBC vector direction to the inertial target). This controller will additionally maintain the orientation of a solar panel to optimally generate solar power. The result is that the satellite body angular rates are controlled to near zero (relative to the inertial coordinate frame).

When:

Pointing to an inertially referenced target is required, with optimal solar generation and/or zero inertially referenced angular rates.

How:

The magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value $hBias$ aligned to the orbit normal direction (default $hBias$ is zero). The reaction wheels are controlled with a quaternion feedback PID controller to control the full 3-axis attitude to track the inertial target with a specified body vector direction. The target will be specified beforehand by setting the inertial referenced unit vector direction. The attitude reference angles are automatically adjusted to point towards the inertial target, while the rotation angle around the pointing vector will be optimised to point the user-configured solar tracking SBC vector optimally towards the sun.



Configurable and Reference Values:

- **ConfigAdcsController** configuration values
 - The wheel momentum management gain, K_H (typical value = 0.5)
 - The PID wheel controller proportional gain K_{P3} and
 - The PID wheel controller derivative gain K_{D3}
 - The PID wheel controller integral gain K_{I3}
- **ConfigAdcsSatellite** configuration values
 - IRC reference vector (**IrcTrackBodyVec** [XYZ])
 - Sun-pointing reference vector (**SunPointBodyVec** [XYZ])
- **ReferenceIrcVector** reference values
 - ECI referenced vector to track, **CmdEciVec** [XYZ]

Sensor/Estimator Dependency:

For this controller to be enabled, the angular rates and attitude quaternion must be estimated, i.e., any angular rate and attitude estimator must be active, e.g., the Full State EKF (EstFullEkf) or the Gyro EKF (EstGyroEkf).

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A.

2.4.4.20 Control Mode 31: (ConSunAvoid)

Description:

What:

A reaction wheel controller that will orient a user-configurable SBC vector, such as that of a sensitive optical payload, away from the sun, requiring minimum sensor or estimation information.

When:

Use this as a safe-mode control mode, to avoid that a sensitive optical payload points into or close to the sun.

How:

The reaction wheels are controlled with a quaternion feedback PD controller, to align the measured sun vector with the negative of the configured sun-avoidance SBC vector. The controller does not use estimated attitude information but calculates the error quaternion for the feedback controller from sun measurements directly. If available, the most appropriate FSS measurement will be used, otherwise the sun vector measured by the CSS will be used. A Magnetic controller will constantly maintain the reaction wheel angular momentum at a bias value.

This controller can be commanded, however if the sun-avoidance feature is enabled in the ADCS configuration, this control mode will be entered automatically if the sun keep out zone is breached. (See section 21.2 for further details on automatic sun avoidance.)

Configurable and Reference Values:

- **ConfigAdcsSatellite** configuration values
 - Sun-avoid SBC vector (**SunAvoidBodyVecX** [XYZ])
- **ConfigAdcsController** configuration values
 - The wheel momentum management gain, K_H (typical value = 0.5) (sun-avoidance only)
 - The PD wheel controller proportional gain K_{P2} (sun-avoidance only)
 - The PD wheel controller derivative gain K_{D2} (sun-avoidance only)

Sensor/Estimator Dependency:



This mode only needs angular rate measurements and does not need position information (no orbit info/TLEs needed). The estimation mode must either be EstGyro or EstGyroEKF. Sun sensor measurements must be available, either from CSS, or FSS.

Actuator dependency: Magnetorquers, 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.5 Wheel Management

Two control modes are defined to manage the wheel momentum in an open-loop fashion. The wheel torque and wheel momentum are still considered by the state estimators:

2.4.5.1 Control Mode 50: (ConStopRW)

Description:

What:

This is not actually a controller, but a method to have a controlled stop of the reaction wheels without upsetting the state estimators.

When:

Transitioning from a control mode that required the wheels, to a pure magnetic control mode with the wheels powered off.

How:

The wheel speeds will be ramped down to zero.

Configurable and Reference Values:

None

Sensor/Estimator Dependency:

Any estimator may be active.

Actuator dependency: 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.5.2 Control Mode 51: (ConHxyzRW)

Description:

What:

This is not actually a controller, but a method to have a controlled command of the reaction wheel XYZ angular momentum without upsetting the state estimators.

When:

Verification of reaction wheel polarity and/or pyramid mounting configuration.

How:

This mode will ramp the wheels up to achieve the specified open-loop momentum vector command. In this mode, it will also distribute the correct 3-axis momentum vector to the four-wheel pyramid configuration.

Configurable and Reference Values:

- **OpenLoopCommandHxyzRW** reference values
 - Open-loop momentum vector reference

Sensor/Estimator Dependency:

Any estimator may be active.



Actuator dependency: 3-axis wheels

Other Requirements and Restrictions: N/A

2.4.6 Required actuators and estimators for control modes

The various control modes with the actuators used and required estimated states are summarized in Table 4 below.

Table 4: Control mode actuators and estimation requirements.

CONTROL MODE GROUP	ENUMERATION VALUE	CONTROL MODE	ACTUATORS USED	REQUIRED ESTIMATED STATE
	0: ConNone	No control	None	None
Detumbling	1: ConBdot	Detumbling single axis Bdot	Y-Magnetorquer	None
	3: ConBdot3	3-axis Bdot detumbling	XYZ magnetorquers	None
	4: ConDetumble	3-axis low power fast (8 Hz) detumbling	XYZ magnetorquers	None
Magnetic Spin	2: ConYspin	Y-Thomson spin	XYZ magnetorquers	Y-body rate
	5: ConSunYspin	Sun Y-axis spin detumbling	XYZ magnetorquers	Y-body rate
	6: ConZspin	Z-Thomson spin	XYZ magnetorquers	Z-body rate
	7: ConSunZspin	Sun Z-axis spin detumbling	XYZ magnetorquers	Z-body rate
Gravity Gradient	8: ConGGboom	MTQ control of GG boom & yaw zero control	XYZ magnetorquers	XYZ rates and attitude quaternion
	9: ConGGsun	MTQ control of GG boom & Z-RW yaw sun pointing control	XYZ magnetorquers & Z-Reaction wheel	XYZ rates and attitude quaternion
Y-momentum	10: ConYwheelInit	Y-Momentum wheel (Initial pitch acquisition)	Y-Momentum wheel	XYZ rates and pitch angle
	11: ConYwheel	Y-Momentum wheel (Steady state)	XYZ magnetorquers & Y-Momentum wheel	XYZ rates and attitude quaternion
	12: ConXYZwheel	3-axis reaction wheel control	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	13: ConSunTrack	3-axis reaction wheel sun tracking control	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	14: ConTgtTrack	3-axis reaction wheel sun tracking control	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	15: ConTgtSteer	3-axis reaction wheel earth target steering control	Reaction wheels	XYZ rates and attitude quaternion
	16: ConGndTrack	3-axis reaction wheel ground station target tracking control	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	17: ConIrcTrack	3-axis reaction wheel inertial target tracking control	XYZ magnetorquers & reaction wheels	XYZ rates and attitude



CONTROL MODE GROUP	ENUMERATION VALUE	CONTROL MODE	ACTUATORS USED	REQUIRED ESTIMATED STATE
				quaternion
	18: ConMoonTrack	3-axis reaction wheel moon tracking control	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	19: ConSatTrack	3-axis reaction wheel satellite tracking control	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	20: ConYawSun	Nadir pointing and yaw towards the sun	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	21: ConYawTarget	Nadir pointing and yaw towards a ground target	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	22: ConRollSun	Nadir pointing and roll towards the sun	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	23: ConRollTarget	Nadir pointing and roll towards a ground target	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	24: ConYawTargetBest	Nadir pointing and yaw best +/-Y body facet towards a ground station	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	25: ConFmcTarget	Forward Motion Compensation Target Scan	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	26: ConYawSunSpin	RW sun yaw-spin control	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	29: ConSunPayload	RW sun tracking and rotation for payload pointing to Zenith	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
	30: ConIrcSunTrack	RW sun tracking and rotation for payload pointing to Zenith	XYZ magnetorquers & reaction wheels	XYZ rates and attitude quaternion
Safe-mode sun (with wheels)	27: ConSunDetumble	RW detumbling to sun-point control	XYZ magnetorquers & reaction wheels	XYZ rates
	31: ConSunAvoid	RW control to point sun-avoidance vector away from sun	XYZ magnetorquers & reaction wheels	XYZ rates
Astro-steering	28: ConAstroSteer	3-axis reaction wheel inertial target tracking control with optimal sun angle rotation	XYZ magnetorquers & reaction wheels	XYZ rates and attitude Quaternion (Only EstGyroEkf)
Wheel management	50: ConStopRW	Controlled stop of reaction wheels	XYZ reaction wheels	None
	51: ConHxyzRW	Open loop XYZ wheel momentum command control	XYZ reaction wheels	None



Valid estimation and control transitions

Table 5 displays the combinations of estimation and control modes which are valid, given the previous discussions of required estimated state that each control mode needs. All control modes in a particular group are considered together since they have the same estimated state requirements.

Note that this table does not consider all the requirements of the control mode (such as additional sensor inputs, operating limits etc.), but rather indicates if the estimator is suited to generate the required attitude and angular rate estimates for the control mode.

Table 5: Valid estimation and control combinations.

		ESTIMATION MODE						
		EstNone	EstGyro	EstMagRkf	EstPitchRkf	EstTriad	EstFullEkf	EstGyroEkf
CONTROL MODE GROUP	No control	OK	OK	OK	OK	OK	OK	OK
	Detumbling	OK	OK	OK	OK	OK	OK	OK
	Magnetic spin	X	OK	OK	OK	OK	NR	OK
	Gravity gradient	X	X	X	X	NR	OK	OK
	Y-momentum	X	X	X	OK	NR	OK	OK
	XYZ Wheel	X	X	X	X	NR	OK	OK
	Safe-mode Sun (with wheels)	X	OK	X	X	X	X	OK
	Astro-steering	X	X	X	X	X	X	OK
	Wheel management	OK	OK	OK	OK	OK	OK	OK

The green OK entries are the preferred combinations and reflect nominal modes for optimal performance. Combinations where the chosen estimation mode technically fulfils the controller needs, but is not recommended, are marked as NR (Not Recommended) in yellow.

The TRIAD estimation mode will lead to poor or erratic performance if attitude angles are not available due to sensor outages (eclipse, or temporary measurement errors) and it thus marked as NR. The Full-state EKF (EstFullEkf) is intended for low angular rates and might not produce accurate angular rate estimates during detumbling or spinning modes. As such it is also marked as NR for use in detumbling and spin modes.

The above table is a generalisation, and preferred combination of estimation and control modes, and their uses throughout ADCS operation is further documented in following chapters.



The ADCS firmware will not allow invalid control and estimator combinations to be commanded. This is to guard against undefined behaviour if a suitable estimator is not active to generate the required attitude and angular rate estimates. The telecommands to change estimation or control mode will apply the logic in



Table 5 and will reply with a NACK (not acknowledge) if invalid changes are commanded.

This restriction implies that the OBC will have to evaluate the above conditions before changing modes. In some conditions, it requires that the estimation mode must be changed before control mode – for example to change from magnetic detumbling using EstGyro and ConBdot3, to 3-axis wheel control, using EstGyroEKF and ConXYZwheel. In this case the OBC first must send the estimation mode change telecommand first, followed by the telecommand to change the control mode. (changing the control mode first would result in an invalid combination because EstGyro cannot be used with 3-axis wheel control). The same applies for changing to magnetic detumbling from 3-axis wheel control. In this case the control mode must be changed first (to ConBdot3) and only then can the estimation mode be changed back to EstGyro.



When sending separate telecommands to change both control and estimation modes as above, it is required to wait at least 200ms between commands. For example, after the telecommand to change estimation mode to EstGyroEKF, a delay of 200ms must be present before sending the telecommand to change control mode to ConXYZwheel. Failure to include the delay will result in the second telecommand receiving a NACK response.



In version 5.25.1 of the CubeComputer control program, an additional telecommand was included ([ConEstMode](#)), that allows sending a single telecommand that will simultaneously set estimation and control modes. The telecommand handler for this command will still perform the same logical check for valid control/estimation combinations as in Table 5, but there is no need to consider the order of estimation or control mode change, or delay between commands.

2.5 Orbit Modes

The ADCS requires knowledge of the satellite position and velocity to estimate full attitude. This information is readily available from a built-in SGP4 orbit propagator. The CubeADCS will use current Unix time and configured orbital elements to calculate position and velocity as needed. It is required that the orbital elements are set to current values, and the CubeADCS Unix time is synchronized for the estimator to work accurately.

The CubeADCS can also make use of GNSS provided position, velocity, and time measurements. Such measurements can either be transmitted to the CubeADCS as telecommands, or via direct reception of GNSS receiver data output.

The ADCS can operate in one of four orbit modes:

1. [OrbTle](#) (SGP4 TLEs only): Orbit calculated from SGP4 TLEs only
2. [OrbTleGnss](#) (GNSS primarily and SGP4 otherwise): Orbit calculated from SGP4 TLE but replaced with GNSS measurements if available
3. [OrbAsgp4](#) (SGP4 Augmented using GNSS): Orbit calculated from Augmented SGP4 TLE
4. [OrbAsgp4Gnss](#) (GNSS primarily and Augmented SGP4 otherwise): Orbit calculated from Augmented SGP4 TLE but replaced with GNSS measurements if available

The detail of how position and velocity is obtained based on the mode selection and availability of data is described in Table 6.



Table 6: Orbit mode usage

ORBIT MODE	GNSS MEASUREMENTS AVAILABLE	SOURCE OF POSITION AND VELOCITY
OrbTle	Yes	The current orbit parameters (set externally from TLEs) are used, with current time in the SGP4 propagator
	No	
OrbAsgp4*	Yes	The current orbit parameters are used, with current time in the SGP4 propagator. The orbit parameters should initially be set from TLEs but are adjusted from GNSS measurements so that SGP4 error is minimized.
	No	The current orbit parameters (refined using GNSS measurements) are used, with current time in the SGP4 propagator
OrbTleGnss	Yes	GNSS measurements are used directly
	No	The current orbit parameters (set externally from TLEs) are used, with current time in the SGP4 propagator
OrbAsgp4Gnss*	Yes	GNSS measurements are used directly. Orbit elements are updated automatically so that SGP4 error is minimized
	No	The current orbit parameters (refined using GNSS measurements) are used, with current time in the SGP4 propagator

* The ASGP4 algorithm makes use of a batch process to refine the current orbital elements so that the SGP4 propagator error compared to GNSS measurements is minimized.



3. ADCS Operational Requirements and Considerations

The ability to commission and operate the ADCS places some requirements on the satellite and ground station to download data and upload commands. This section elaborates on these requirements, and presents important information required for operating and configuring the CubeADCS.

3.1 Commands

Nominal operation of the ADCS should involve very few telecommands (set estimation & control mode, set enabled state), but throughout commissioning it will be necessary to send other commands in the command specification – see [1], [2], [3] and [4] for more details.

As stated in the beginning of Chapter 2, wherever telecommand and telemetry, or parameters in telecommand messages, or fields in telemetry messages are referred to, it will be displayed using [this style](#), for example “[TlmController](#)”. In such cases, the code name from [1], [2] and [3] will be used to identify the telecommand, telemetry, parameter or field.

3.1.1 Control and estimation mode telecommands

There exists telecommand to change the current control mode ([ConMode](#)) and Estimation Mode ([EstMode](#)).



As mentioned in the previous chapter, the ADCS firmware will not allow invalid control and estimator combinations to be commanded. It is also necessary to apply control and estimation mode changes in the correct sequence, to prevent an invalid combination. Finally, a 200ms delay is needed between commands to change estimation and/or control mode.

As of version 5.25.1 of the CubeComputer control program, an additional telecommand exists ([ConEstMode](#)), that allows sending a single telecommand that will simultaneously set estimation and control modes.

The command to change control mode (and the combined [ConEstMode](#) command) takes a parameter for control timeout ([ConTimeout](#)). If the ADCS is commanded using a non-zero value for [ConTimeout](#), the ADCS will revert to no control ([ConNone](#)) after the timeout expires.

The intention for the above timeout, is that it is used with initial activation of detumbling or magnetic spin control. The timeout is to guard against the ADCS spinning up the satellite in case of incorrect polarities or mounting configuration, and if the OBC is unable to command the ADCS back to no control mode.



When making use of the [ConTimeout](#) parameter when changing control modes, it should be noted that the control mode will change to [ConNone](#) at the end of the timeout, and not restore any previous control modes that may have been active.



When changing to a new estimation mode, if the target estimation mode is [EstFullEkf](#) or [EstGyroEkf](#), there will be a 60s convergence time before any active control takes place. The control mode will reflect as the chosen mode in telemetry; however, actuator signals (wheel torque and magnetorquer commands) will be zero until the convergence time has expired.



3.2 Telemetry

Ideally all telemetry points from the ADCS should be observable by real-time telemetry request, but the capability should also exist to log data to non-volatile storage over extended periods (typically an orbit) so that it can be downloaded whenever the satellite is in range of the ground station. The latter is often called whole orbit data (WOD).

The telemetry frames that should be logged will depend on the current commissioning or operating phase. The telemetry that should be logged will often be a selection of telemetry frames. The ADCS maintains a rolling log of historic telemetry using on-board non-volatile memory. The telemetry from this log can be extracted by issuing the appropriate telecommand to initiate the transfer and making use of the common data download sequence as described in [5].

The telemetry log may also be maintained by an on-board computer (OBC) external to the ADCS. In this case, the responsibility of maintaining the non-volatile data log shifts to the OBC. It can be accomplished from the OBC perspective either by directly requesting telemetry from the CubeADCS and storing the replied data, or by initially configuring an unsolicited telemetry output from the CubeADCS and capturing and storing the resultant output.

Regardless of which entity is responsible for maintaining the log data (OBC or CubeADCS), it is of critical importance that the telemetry log data discussed in this document, for the relevant commissioning step, is available on ground with the required log interval, with the same data integrity, resolution, and unit of measurement as in the original CubeADCS telemetry frame.

3.3 Whole orbit data logging and synchronization

It is possible to request telemetry from the ADCS at periodic intervals and then log it on the OBC, or elsewhere external to the ADCS. By logging the telemetry on the ADCS itself, the time consistency of the telemetry is guaranteed. The telemetry will be stored after each processing iteration, thus all the telemetry entries for one instance will be synchronized. If the telemetry is sampled from the ADCS to be stored externally, then care must be taken to make sure all requested telemetry is of the same epoch. The telemetry that will typically be stored spans multiple telemetry packets and each will have to be requested individually. If the time at which the frames are sampled are close to the beginning of the next ADCS processing iteration it may be possible that one sampled frame contains telemetry from the previous ADCS processing iteration, while the following sampled telemetry frame is of the next ADCS processing iteration – even though the frames were requested in succession.

To avoid such a situation, the OBC either must make use of unsolicited TLM output or synchronise its telemetry requests with the ADCS loop time. Unsolicited telemetry configuration and output is discussed in [5].

In order to perform sampling that is synchronized with the ADCS loop, the OBC may request the [TlmAcpExecution](#) telemetry frame. This telemetry frame will inform whether the processing iteration is still taking place or not. The OBC can use the [TimeSinceLoopStart](#) field in the [TlmAcpExecution](#) telemetry frame for synchronization. The OBC that samples the ADCS data should use the [TimeSinceLoopStart](#) parameter and leave enough margin before the 200ms control loop interval ends.

The ADCS loop time is synchronised to Unix time, so that control loop iterations start every 200ms relative to 0ms offset. It is thus possible for the OBC to periodically synchronise the ADCS time by sending a [UnixTime](#) telecommand. The OBC can then be sure that the ADCS loop iterations will start relative to the time that it set. The same method applies to GNSS time synchronisation. It both the OBC and CubeADCS receives the same GNSS navigation messages and PPS signal, the OBC can also be sure that the ADCS loop iterations will start at 200ms slices relative to 0ms GNSS time.



3.4 Events

The CubeADCS will log events in case of significant occurrences and state changes. Events are classed by their severity. It is important that the OBC can capture and store events as they are generated or download a log of historic events so that the information is available for interpretation on the ground. Details on events and how to download them can be found in [5].

Events other than Normal class should be considered during all parts of operation, including commissioning.

3.5 Image Downloads

In order to commission the sun sensor(s) (FSS), horizon sensor(s) (HSS) and star tracker (CubeStar), it is required that the OBC implements the ability to download images from these sensors and make them available for analysis and post-processing on the ground. The procedure that the OBC must implement to download images is described in [5].

3.6 Configuration Values and Reference Values

The CubeADCS makes use of configuration settings to govern the behaviour of the control and estimation logic. Table 7 lists the configurable settings that affect estimation and control behaviour, that can be changed by telecommand (and read using a telemetry request).

Table 7: ADCS configuration messages and their content

COMMAND/TELEMETRY NAME	CONTENT
ConfigDefaultModes (Default mode configuration)	Power-on default control mode and operational state
ConfigMount (Mounting configuration)	Mounting directions and mounting angles that define the mounting transforms between sensor/actuator coordinate frames and Spacecraft Body Coordinate (SBC) frame
ConfigMag0OrbitCal (Main magnetometer in-orbit calibration configuration)	In-flight calibration values for magnetometer identified as Mag0
ConfigMag1OrbitCal (Redundant magnetometer in-orbit calibration configuration)	In-flight calibration values for magnetometer identified as Mag1
ConfigAdcsController (ADCS controller configuration)	Controller gains and parameters
ConfigAdcsEstimator (ADCS estimator configuration)	- Estimator system noise and sensor noise covariances - Flags to control sensor use in estimators
ConfigAdcsSatellite (ADCS satellite configuration)	- Satellite moments of inertia and products of inertia - Vector directions to influence controller pointing behaviour
ConfigOrbitSatParams (Satellite orbit parameter configuration)	Orbital elements for satellite orbit
ConfigOrbitTargetParams (Target satellite orbit parameter configuration)	Orbital elements for target satellite orbit



COMMAND/TELEMETRY NAME	CONTENT
ConfigNodeSelection (Node selection configuration)	Selection flags to control which sensors are to be used, if multiple sensors of the same type are present
ConfigActuatorRwl (Wheel configuration)	Wheel hardware configuration (wheel sizes and parameters, and mounting selection – pyramid vs. three orthogonal wheels)
ConfigActuatorMtg (Magnetorquer configuration)	Magnetorquer hardware configuration (torquer strength and maximum on-time)
ConfigMagSensingElement (Mag sensing element configuration)	Selection between primary and redundant sensing element (CubeMagDeploy only)
ConfigAsgp4	Configuration parameters that influence the adaptive SGP4 algorithm

It is important that all the above configuration settings can be adjusted either directly by the OBC, or from the ground via the radio communications link and the OBC.

Changes that are made to the configuration via telecommand are only temporary – until the CubeADCS is reset or power cycled. To persist config changes to non-volatile memory, a [ConfigPersist](#) telecommand must be sent.

The control modes of the ADCS further make use of “Reference Values”, as parameters to the controllers. The reference values are listed in Table 8. Note that the reference values do not persist through power cycle or resets and should ideally be commanded prior to setting the relevant control mode.

Table 8: Reference Values for ADCS controllers

COMMAND/TELEMETRY NAME	CONTENT	APPLICABLE CONTROL MODE
OpenLoopCommandMtg	Open-loop torquer commands, per each torquer (MTQ0, MTQ1, MTQ2)	ConNone
OpenLoopCommandRwl	Open-loop wheel speed commands, per each wheel (RWL0, RWL1, RWL2, RWL3)	ConNone
OpenLoopCommandHxyzRW	Open-loop momentum vector command for the reaction wheels to follow	ConHxyzRW
ReferenceRpy	Roll, pitch and yaw angle reference command	ConXYZwheel
ReferenceTarget	Ground target location latitude, longitude and altitude	ConTgtTrack ConTgtSteer ConGndTrack ConYawTarget ConRollTarget ConYawTargetBest
ReferenceIrcVector	Inertial pointing vector [XYZ]	ConIrcTrack
ReferenceFmcParam	FMC imaging scan parameters (FMC rate, roll offset, scan duration)	ConFmcTarget



3.7 Abstract Types and Sensor Selection

The CubeADCS interfaces to multiple sensors and actuators of the same type. The CubeADCS uses the term “abstract types” to refer to a node's ability to act as one of the fundamental sensor or actuator types required by CubeADCS. The available abstract types are listed in Table 9.

Table 9: Abstract node types used by the CubeADCS

ABSTRACT TYPE	DESCRIPTION	MAX. NUMBER OF INSTANCES
RWL	Reaction wheel	4
GYR	Gyro (rate sensor)	2
EXTGYR	External gyro (rate sensor)	2
MAG	Magnetometer	2
FSS	Fine sun sensor	4
HSS	Horizon Sensor	2
STR	Star Tracker	2
EXT	External Sensor	2
GNSS	Global Navigation Satellite System	1

Up to four fine-sun sensors (FSS) can be used, two horizon sensors (HSS), two magnetometers (MAG), two on-board gyros (GYR), two external 3-axis gyros (EXTGYR) and two star trackers (STR). The CubeADCS can also command four reaction wheels (RWL) and accept input for fine steering feedback via two external sensors (EXT).

The node-selection configuration settings, adjustable via the [ConfigNodeSelection](#) configuration telecommand, will inform the ADCS which sensors to use. The selection per node, in the [ConfigNodeSelection](#) telecommand or telemetry message, is a bitmask indicating which instance of that node is available for the CubeADCS to use.

During the commissioning phase, the sensor selection will be changed to force the ADCS to use specific sensors for their respective commissioning. The node-selection flags thus do not necessarily represent the number of sensors connected to the CubeADCS, but rather the subset that should be used at the time. If all sensors and actuators have successfully been commissioned, the node-selection flags will represent all the connected nodes in the system.

In situations where more than one sensor or actuator is selected via the bitmask, the behaviour will be as in Table 10.

Table 10: Handling of multiple sensor selections in ConfigNodeSelection

ABSTRACT TYPE	DESCRIPTION	CONFIGNODESELECTION BITMASK	BEHAVIOUR WITH MULTIPLE INSTANCES SELECTED BY BITMASK
RWL	Reaction wheel	0b0001 – RWL0 0b0010 – RWL1 0b0100 – RWL2 0b1000 – RWL3	The ConfigNodeSelection.SelectRwl parameter is not used and reserved for future use. The ConfigActuatorRwl.RwlScheme , and ConfigMount.MountRwl<no.> configuration settings jointly identify which wheels are being used. A mounting value of AxisNotUsed indicates that the wheel is not used. Pyramid configurations must have mounting specified for all four wheels, but ConfigActuatorRwl.RwlFailId may be specified to indicate a failed reaction wheel, in which case the



ABSTRACT TYPE	DESCRIPTION	CONFIGNODESELECTION BITMASK	BEHAVIOUR WITH MULTIPLE INSTANCES SELECTED BY BITMASK
			CubeADCS will affect the control torque using the remaining three wheels.
GYR/EXTGYR	Gyro (rate sensor)	0b0001 – EXTGYR0 0b0010 – EXTGYR1 0b0100 – GYR0 0b1000 – GYR1	If more than one gyro is enabled via the selection bitmask, the first gyro indicating a valid measurement (as reflected by the TlmSensorRawGyro.Gyr0IsValid flag) will be used, beginning with EXTGYR0.
MAG	Magnetometer	0b01 – MAG0 0b10 – MAG1	If both MAG0 and MAG1 are selected, the MAG0 measurement will be used in the estimator if it is a valid measurement (as reflected by the TlmSensorRawMag.Mag0IsValid flag). If MAG0 does not have a valid measurement, MAG1 will be used. The eventual used magnetometer will be reflected by the TlmSensorCalMag.Mag0IsBest or TlmSensorCalMag.Mag1IsBest flags.
FSS	Fine sun sensor	0b0001 – FSS0 0b0010 – FSS1 0b0100 – FSS2 0b1000 – FSS3	Of all the selected FSS, the sensor that produces the valid measurement closest to the sensor's boresight (smallest angle) will be selected for use the estimator. The eventual used sun sensor will be reflected in the TlmSensorCalFss.Fss<no.>IsBest flags
HSS	Horizon Sensor	0b01 – HSS0 0b10 – HSS1	The horizon sensors are only used by the EstFullEkf and EstGyroEkf estimators. All horizon sensors producing valid measurements will be used by the estimator.
STR	Star Tracker	0b01 – STR0 0b10 – STR1	The star trackers are only used by the EstFullEkf and EstGyroEkf estimators. All star trackers producing valid measurements will be used by the estimator.
EXT	External Sensor	0b01 – EXT0 0b10 – EXT1	For EXT0, the measurements will be interpreted as feedback error angles for the ConTgtSteer controller. Only EXT0 is supported at present.
GNSS	Global Navigation Satellite System	0b1 – GNSS0	This parameter is not used at present.

It should further be noted that after the logic in Table 10 has been applied, the [ConfigAdcsEstimator.EkfUseFss](#), [ConfigAdcsEstimator.EkfUseCss](#), [ConfigAdcsEstimator.EkfUseHss](#) and [ConfigAdcsEstimator.EkfUseStr](#) flags further restrict whether the entire sensor group is used by the [EstFullEkf](#) and [EstGyroEkf](#) estimators. Sensors for which the corresponding bitmask position is cleared will not yield any calibrated sensor measurements in their respective [TlmSensorCal*](#) telemetry frame.

3.8 Description of sensor telemetry, with details for “Valid” flag logic

Each of the raw sensor telemetry frames contain a [IsValid](#) flag, indicating if the raw sensor telemetry is valid, and can be used for estimation or other purposes.

In general, the [IsValid](#) flag will be set to FALSE if either the internal communications or sampling of the sensor failed, or the sensor itself reports invalid measurement in its own telemetry response to the CubeComputer.



The specific logic that determines the state of the **IsValid** flag for raw telemetries is summarized in Table 11.

Table 11: Raw sensor telemetry “Valid” flag logic

TELEMETRY	FLAG NAME(S) IN TELEMETRY	LOGIC DESCRIPTION
TlmSensorRawCss	CssIsValid	Raw CSS IsValid flag will be set to FALSE if there was an error reading the ADC peripheral (to sample any of the CSS photodiodes). TRUE otherwise
TlmSensorRawMag	Mag<x>IsValid	Valid flag will be set to TRUE if communications with MAG was successful, and the MAG reports valid measurement in its telemetry. (FALSE if either the internal communication to the MAG failed, or the MAG reports invalid measurement in its own telemetry response). The MAG will report invalid measurement in its reply to the CubeComputer if there was an internal error in its sampling. The flags will also be FALSE if the magnetometer is not powered on or failed to initialise.
TlmSensorRawGyro	Gyro<x>IsValid	Raw GYR IsValid flag(s) will be FALSE if the gyro is not powered on or failed to initialise. It will also be FALSE if there was an internal error in communicating with the IMU.
TlmGnssRaw	GnssValidData	Raw GNSS data valid flag will be TRUE if a navigation message was received from a connected GNSS receiver (or sent via telecommand). The ValidData flag will remain TRUE for a duration of 1.5s after the last navigation message was output from the GNSS, thus the flag is an indication that the GNSS receiver is actively communicating. It does not indicate that the contents of the navigation message is valid, or that the GNSS receiver has lock (the GnssLock field in the TlmGnssRaw frame will indicate this).
TlmSensorRawRwl	Rwl<x>IsValid	Valid flag will be set to TRUE if communications with RWL was successful, and the RWL reports valid measurement in its telemetry. (FALSE if either the internal comms to the RWL failed, or the RWL reports invalid measurement in its own telemetry response). The RWL will report invalid measurement in its reply to the CubeComputer if there was an internal error in its sampling. The flags will also be FALSE if the wheel is not powered on or failed to initialise.
TlmExtSensorRaw	ExtSensor<x>IsValid	Valid flag will be set to TRUE if communications with the generic external sensor was successful, and the sensor reports valid measurement in its telemetry. (FALSE if either the internal communication to the external sensor failed, or if it reports invalid measurement in its own telemetry response). The flags will also be FALSE if the external sensor is not present, not powered on, or failed to initialise.
TlmRwlNssRwlRaw	Rwl<x>IsValid	Valid flag will be set to TRUE if communications with CubeNode connected to the wheel was successful, and the RWL reports valid measurement in its telemetry. (FALSE if either the internal comms to the CubeNode failed, or the RWL reports invalid measurement in its own telemetry response). The RWL will report invalid measurement in its reply to the CubeComputer if there was an internal error in its sampling. The flags will also be FALSE if the wheel is not powered on or failed to initialise, or not present in the system.



Sensor data passes through a processing step whereby the raw telemetry from each sensor is converted to a measurement that can be used by the EKF (typically a conversion to measured vector observation in the satellite body coordinate frame). The processed sensor measurements are available in the “calibrated” telemetry frames.

In this processing step, additional logic is applied to determine the IsValid flag status for calibrated telemetry, as shown in Table 12.

Table 12: Calibrated sensor telemetry “Valid” flag logic

TELEMETRY	FLAG NAME(S) IN TELEMETRY	LOGIC DESCRIPTION
TlmSensorCalFss	Fss<x>IsValid	Calibrated FSS IsValid flag will be TRUE if the raw measurement is valid AND angle between measured sun angle and sensor boresight is < 80 deg
TlmSensorCalCss	CssIsValid	Calibrated CSS IsValid flag will be TRUE if the raw measurement is valid, AND a sun-vector could be constructed from it: the configuration must allow for it, because there must be a CSS measurement for each of +/-X, Y, Z, and the magnitude of the measured sun vector must be more than 1/5 of the full range of the ADC.
TlmSensorCalMag	Mag<x>IsValid	Set to TRUE if raw MAG telemetry is valid, AND after applying in-flight calibration correction, mag field length is in the range 15 to 65 uT.
TlmSensorCalGyro	Gyro<x>IsValid ExtGyro<x>IsValid	GYR IsValid flags will reflect the status of the raw GYR valid flags. If the measured angular rate exceeds the possible range for the respective IMU (+/-450 deg/s for GYR0, +/-250 deg/s for GYR1) the IsValid flag in calibrated telemetry will be set to FALSE.
TlmSensorCalHss	Hss<x>IsValid	Calibrated HSS IsValid flag will be TRUE if the raw measurement is valid, AND raw measurement angles fall in acceptable range for the sensor (abs(tilt angle) < 45 deg, and -180 < rotation angle < 180 deg).
TlmSensorCalStr	Str<x>IsValid	Calibrated STR IsValid flag will be TRUE if the raw measurement is valid and contains either a valid quaternion, or at least two matching star vector pairs in the raw telemetry.
TlmSensorCalGnss	GnssIsValid	Calibrated GNSS IsValid flag will be TRUE if the raw GNSS measurement is valid and the GNSS reported time is not older than current RTC (real-time clock)
TlmSensorCalRwl	RwlIsValid	Calibrated wheel IsValid flag will be TRUE if the measured wheel speed is within maximum momentum range for the wheel (as per configuration), and the difference between current and last measurement falls within torque limits of the wheel (as per configuration).

The FSS and MAG sensors may potentially produce multiple valid measurements (if there are more than one of each type of sensor) where the ADCS estimation code only requires a single measurement. (Other sensors such as the STR and HSS may also produce multiple valid measurements from multiple sensor instances, but the EKF will use all these measurements and don't have to select a single instance).

The “IsBest” flags in the CSS and MAG calibrated telemetry frames will reflect which instance is currently being used by the estimation code. The logic that the ADCS applies to make this selection is detailed in Table 13.



Table 13: “Best for Estimators” logic in calibrated telemetry

TELEMETRY	BEST FOR ESTIMATOR FLAG NAME(S)	LOGIC DESCRIPTION
TlmSensorCalFss	Fss<x>IsBest	For all FSS that are enabled via the NodeSelection flag, the “best for estimators” decision is made based on which sensor has a sun vector measurement with smallest angle to its boresight. (The FSS is more accurate with sun close to the bore-sight). Only one FSS will have “best for estimators” flag set to TRUE.
TlmSensorCalMag	Mag<x>IsBest	If MAG0 is selected via the NodeSelection configuration (bit 0 of the mask == 1), and MAG0 produces valid measurements, “best for estimators” for Mag0 will be set to TRUE (and “best for estimators” for Mag1 will be FALSE). Otherwise, if Mag1 is enabled via the NodeSelection config (bit 1 of the mask == 1) and Mag1 produces valid measurements, “best for estimators” for Mag1 will be set to TRUE. If neither MAG is enabled in the NodeSelection config, or neither of them produces valid measurements, “best for estimators” for both will be FALSE

3.9 Satellite configuration

The satellite configuration, [ConfigAdcsSatellite](#), includes the moments- and products of inertia of the satellite, as well as SBC referenced vector directions that influence target tracking and sun pointing behaviour. These settings should already be configured on the ground and should also be captured on the hardware configuration and client mission overview document, [8], as this information becomes available.

Whenever there is a change to the satellite configuration in-flight, for instance because of solar panel deployment, the configuration should be updated to ensure that the estimator is using the most accurate values for the moment of inertia matrix, when propagating the attitude.

3.10 Mounting configuration

All actuators and sensors each have their own local coordinate systems. The mounting configuration for the ADCS must be specified correctly for the ADCS to function as intended. Mounting settings for the magnetometer, gyro and magnetorquers are critical during detumbling since incorrect values might cause control activations to increase angular rates of the satellite rather than lowering them.

Errors in mounting configurations for other sensors and wheels will result in large estimation errors and an inability to perform 3-axis control or stabilization.

Mounting settings are either specified as an enumeration that identifies an axis direction, or the angles in a Euler 3-2-1 sequence. For sensors and actuators where only a single axis definition matters, such as wheels and CSS photodiodes, only an axis direction is specified.

The axis direction is specified by selecting a value from the enumerations in Table 14.

Table 14: AxisSelect enumeration values

VALUE	NAME	DETAIL
0	AxisNotUsed	Sensor or actuator is not used
1	AxisPosX	Sensor or actuator axis aligns with +X _B axis
2	AxisNegX	Sensor or actuator axis aligns with -X _B axis
3	AxisPosY	Sensor or actuator axis aligns with +Y _B axis
4	AxisNegY	Sensor or actuator axis aligns with -Y _B axis



VALUE	NAME	DETAIL
5	AxisPosZ	Sensor or actuator axis aligns with +Z _B axis
6	AxisNegZ	Sensor or actuator axis aligns with -Z _B axis
7	AxisThd1	Wheel in 1 st position of tetrahedral configuration
8	AxisThd2	Wheel in 2 nd position of tetrahedral configuration
9	AxisThd3	Wheel in 3 rd position of tetrahedral configuration
10	AxisThd4	Wheel in 4 th position of tetrahedral configuration
11	AxisPmdA	Wheel in 1 st position of pyramid configuration
12	AxisPmdB	Wheel in 2 nd position of pyramid configuration
13	AxisPmdC	Wheel in 3 rd position of pyramid configuration
14	AxisPmdD	Wheel in 4 th position of pyramid configuration

Sensors that require a full alignment transform will have an angle triplet, labelled as alpha, beta and gamma in the configuration settings. The angles are used in an Euler 3-2-1 sequence, with alpha (α) applied first, around the Z-axis. The transform direction cosine matrix (DCM) can be computed from:

$$\mathbf{M} = \begin{bmatrix} \cos \alpha \cos \beta & \cos \beta \sin \alpha & -\sin \beta \\ \cos \alpha \sin \beta \sin \gamma - \cos \gamma \sin \alpha & \cos \alpha \cos \gamma + \sin \alpha \sin \beta \sin \gamma & \cos \beta \sin \gamma \\ \sin \alpha \sin \gamma + \cos \alpha \cos \gamma \sin \beta & \cos \gamma \sin \alpha \sin \beta - \cos \alpha \sin \gamma & \cos \beta \cos \gamma \end{bmatrix}$$

This DCM will then be used to transform a vector in the sensor or actuator local coordinate frame to the SBC frame, if left-multiplied with the untransformed vector (in local coordinate frame).

The local coordinate frame and axes definitions for all CubeProducts are detailed in [9]. The mounting orientations of sensors and actuators should also be captured on the hardware configuration and client mission overview document, [8], as this information becomes available.

In general, telemetry frames marked as 'Raw' will have measurements in the local sensor or actuator coordinate frame, while 'Cal' telemetry will be transformed to the (ADCS) SBC coordinate frame.

3.11 Sensor and actuator firmware configuration changes

Although CubeProduct firmware configurations are set with due care, it may be necessary that configuration settings that are part of the CubeProduct's firmware configuration must be adjusted during commissioning. Examples of such settings include CubeSense Sun (FSS) exposure and detection parameters, and CubeSense Earth (HSS) detection thresholds.

There is no direct method to change these configuration values since they are stored and maintained by the CubeProduct itself. The options that are available to update these values are:

- 1 The Telecommand and Telemetry Pass-through feature of the CubeComputer interface is used (see [5] for details), to send commands to the CubeProduct through the CubeComputer main communications interface. This method exposes the CubeProduct's telecommand and telemetry interface, and all settings that can be adjusted by sending commands directly to the node may be used. The [ConfigPersist](#) telecommand must be sent to the node after making configuration adjustments, so that the updated setting is persisted in the non-volatile memory of the node. It is recommended that the OBC implements a general pass-through telecommand and telemetry mechanism with parameters that can be changed from the ground, to avoid having to implement



all the possible telecommands and telemetry message formats for each node. (This is the preferred approach)

- 2 The firmware update method described in [6] is used to upload a new configuration binary file for the node in question. This will require a compiled configuration file with the updated settings, and CubeSpace will have to be consulted to obtain such a file.

3.12 Orbital Elements and GNSS

The orbit SGP4 parameters must be regularly updated using the [ConfigOrbitSatParams](#) telecommand, to account for a changing orbit. This action is performed from the OBC however the command will originate from the ground segment since it will be required to obtain the latest TLEs from www.celestrak.org or www.space-track.org. It is recommended that the TLEs be updated once daily.

If available, a GNSS that supplies position and velocity measurements to the CubeADCS will improve overall position accuracy.



It must be noted that even if the user relies on the adaptive SGP4 mode, with occasional GNSS measurements, it is still necessary to update the orbit parameters from TLEs occasionally, since the ASGP4 corrections do not update all the orbit parameters, and the orbit will eventually drift. It is also necessary for the orbit parameters to be set initially, because the ASGP4 corrections can only be applied if the GNSS measurements do not differ from the estimated SGP4 position by a threshold.

3.12.1 Recommended ASGP4 Settings

In systems where a GNSS is present, it is possible to make use of occasional GNSS measurements to improve the accuracy of the SGP4 propagator output. The Adaptive SGP4 (ASGP4) feature of the CubeADCS makes this possible by adjusting the orbital parameters used in the SGP4 propagator, using the GNSS measurements as input.

The ASGP4 algorithm offers improvement to the unmodified TLEs obtained from NORAD, although the extent of the improvement depends on use case.

The ASGP4 algorithm performs an update to the current orbit parameters after a batch of GNSS measurements. The parameters of the ASGP4 feature that may be adjusted include the following:

Table 15: ASGP4 parameters

PARAMETER	DESCRIPTION	RECOMMENDED VALUE
BatchSize	The number of GNSS position and velocity datapoints to use in the batch update	• 100 (for continuous GNSS measurements) • One orbit duration (for occasional GNSS measurements, where it is possible to gather GNSS data for a full orbit)
MaxTimeBetween	Maximum time between successive GNSS measurements for a batch update to be successful	1000 s
MaxPosErr	Maximum allowed position error between current SGP4 output and GNSS position, for batch update to be successful	100 km
TimeGain	Time offset compensation gain	1.0
UpdateRaanIncl	Update SGP4 RAAN and Inclination angles from GNSS	TRUE



UpdateEccen	Update SGP4 Eccentricity from GNSS	FALSE
UpdateApMa	Update SGP4 Arg. Perigee and Mean Anomaly from GNSS	FALSE
UpdateTime	Update SGP4 Epoch from GNSS	TRUE

The best performance can be achieved for a small batch size. This makes the ASGP4 output adapt to the real-time orbit faster, but the performance is only realized if a new batch of GNSS measurements can be processed right after the previous (i.e. continuous GNSS measurements are available).

If the ASGP4 algorithm will be used with intentional intermittent GNSS measurements (i.e. the GNSS receiver is switched off to conserve power), it will be better to perform an ASGP4 batch update using a full orbit worth of GNSS data. For subsequent orbits, the GNSS can then be switched off, and the ASGP4 output using the modified orbit parameters will offer an improvement to the unmodified TLEs.

3.13 Time synchronisation

The CubeADCS requires that its real-time clock is synchronised to current Unix time, to provide accurate attitude estimation when using EKF estimation modes. The options that are available for performing time synchronisation are:

- [UnixTime](#) telecommand sent by OBC: this will require that the OBC time is periodically synchronised to external (GPS or other) time
- On-board GNSS receiver provides navigation messages (which include current time) to CubeADCS directly, through the [GnssSensorCmd](#) command.
- On-board GNSS receiver provides navigation messages (which include current time) and PPS (Pulse-per Second) signal to the CubeADCS directly

3.13.1 Unix Time and Orbit Epoch Changes

It is necessary to restrict orbit parameters that the SGP4 propagator uses, to avoid situations where the ADCS will compute an obviously incorrect orbit state. One such logic test that is in place, is that the orbit epoch provided in the [ConfigOrbitSatParams](#) telecommand must fall within two weeks of the currently set unix time of the CubeComputer. If this is not the case, the [ConfigOrbitSatParams](#) command will receive a NACK response. The correct sequence is to first set the Unix time to current time, followed by the [ConfigOrbitSatParams](#) command.



When receiving a NACK to the [ConfigOrbitSatParams](#) telecommand, please always consider this restriction first as it is often the reason for the failed telecommand. This restriction also prevents successful connections when performing HIL simulations. For HIL simulations please ensure the simulation start time and configured orbit epoch is within 2 weeks from each other.

3.13.2 Unix Time and GNSS Navigation Messages

The ADCS will reject GNSS navigation messages, and respond with a NACK to [GnssSensorCmd](#) telecommands, if the current Unix time of the CubeComputer differs from the GNSS time in the navigation message or [GnssSensorCmd](#) command parameters, by more than 1 year. This is to prevent upsets to the attitude estimation if the GNSS time is incorrectly measured or reported.

3.14 Inertial and Earth-fixed Reference Frames

For purposes of converting between inertial and earth-fixed reference frames, the exact reference frames that the ADCS uses are detailed in this section.



3.14.1 Earth-centered inertial (ECI) reference frame

The ECI position and velocity vectors that the ADCS reports ([TlmModels](#) telemetry frame) are relative to the J2000 ECI reference frame. The J2000 ECI frame is defined by the Earth's Mean Equator and Mean Equinox (MEME) at 12:00 Terrestrial Time on 1 January 2000. The x-axis is aligned with the mean vernal equinox. The z-axis is aligned with the Earth's rotation axis (or equivalently, the celestial North Pole) as it was at the specified epoch.

Barring very small errors, the J2000 ECI reference frame and the International Celestial Reference Frame (ICRF) are considered equivalent.

- SGP4 provided position and velocity is converted from TEME (True Equator Mean Epoch) to the above ECI frame.
- Where satellite position and velocity are obtained from GNSS measurements, the measurements in ECEF are converted to J2000 ECI frame.
- Star tracker quaternions (where applicable) relate/transform from J2000 ECI frame to sensor coordinate frame.

3.14.2 Orbit Referenced Coordinate (ORC) reference frame

The Orbit reference frame (ORC) is defined by the J2000 ECI referenced position and velocity vectors. The ORC Z-axis is the unit vector in the negative of the position vector (+Z_ORC points to nadir). The ORC Y-axis is the cross product of position and velocity vectors and points in the orbit anti-normal direction, and the X-axis completes the right-hand coordinate frame (points towards velocity vector for a circular orbit).

3.14.3 Earth-centered Earth-Fixed (ECEF) reference frame

ECEF referenced position and velocity reported by GNSS telemetry ([TlmGnssRaw](#) and [TlmGnssCal](#)) are reported in the ITRF (International Terrestrial Reference Frame).

Where coordinates are converted to/from latitude, longitude, and altitude, the WGS84 reference ellipsoid is used ([TlmModels](#) and [ReferenceTarget](#)).



4. Commissioning and Operations Overview

In the flow diagrams that follow in subsequent sections, the convention shown in Figure 1 is used.

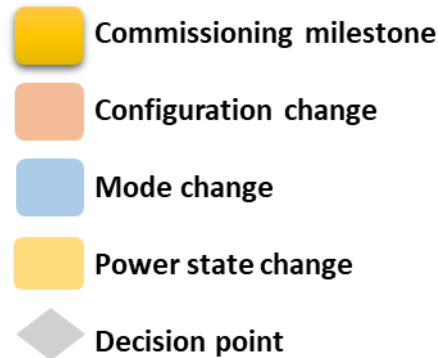


Figure 1: Flow diagram block convention

The commissioning sequence is illustrated in Figure 2.

It is recognised that there are multiple ADCS use-cases and the modularity and configurable nature of the CubeADCS results in different commissioning routes with different operational end states. The process described in this manual attempts to address the variability by identifying the following commissioning outcomes:

- Y-spin or Z-spin stabilized
- Sun-pointing spin
- Y-momentum wheel stabilized
- 3-axis controlled
- Gravity-gradient stabilized

This manual further presents a general strategy that can be followed under normal operations, to transition from initial uncertain attitude state into the same outcomes as listed above. The sequence that can be followed for normal (post-commissioning) operations is shown in Figure 3.

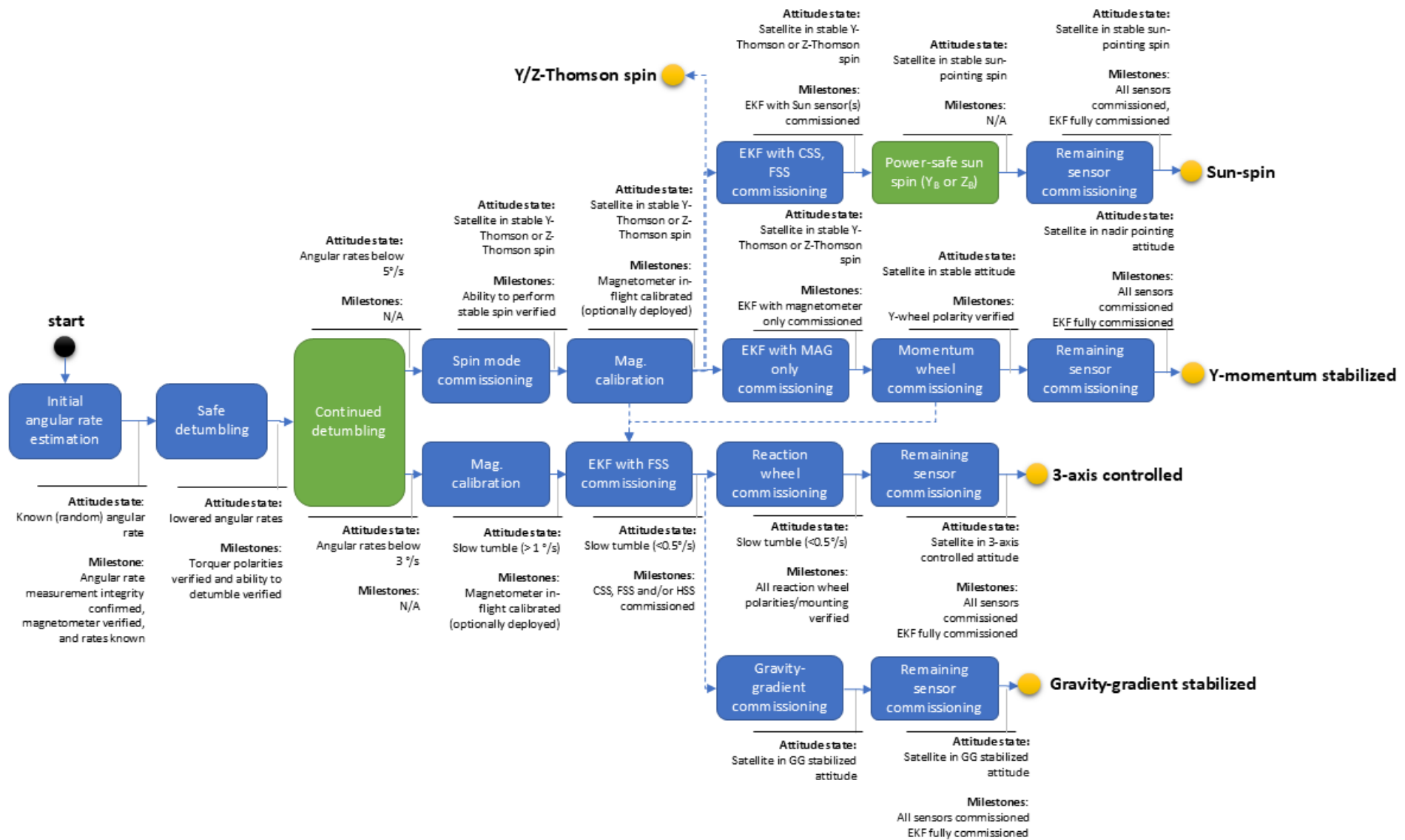
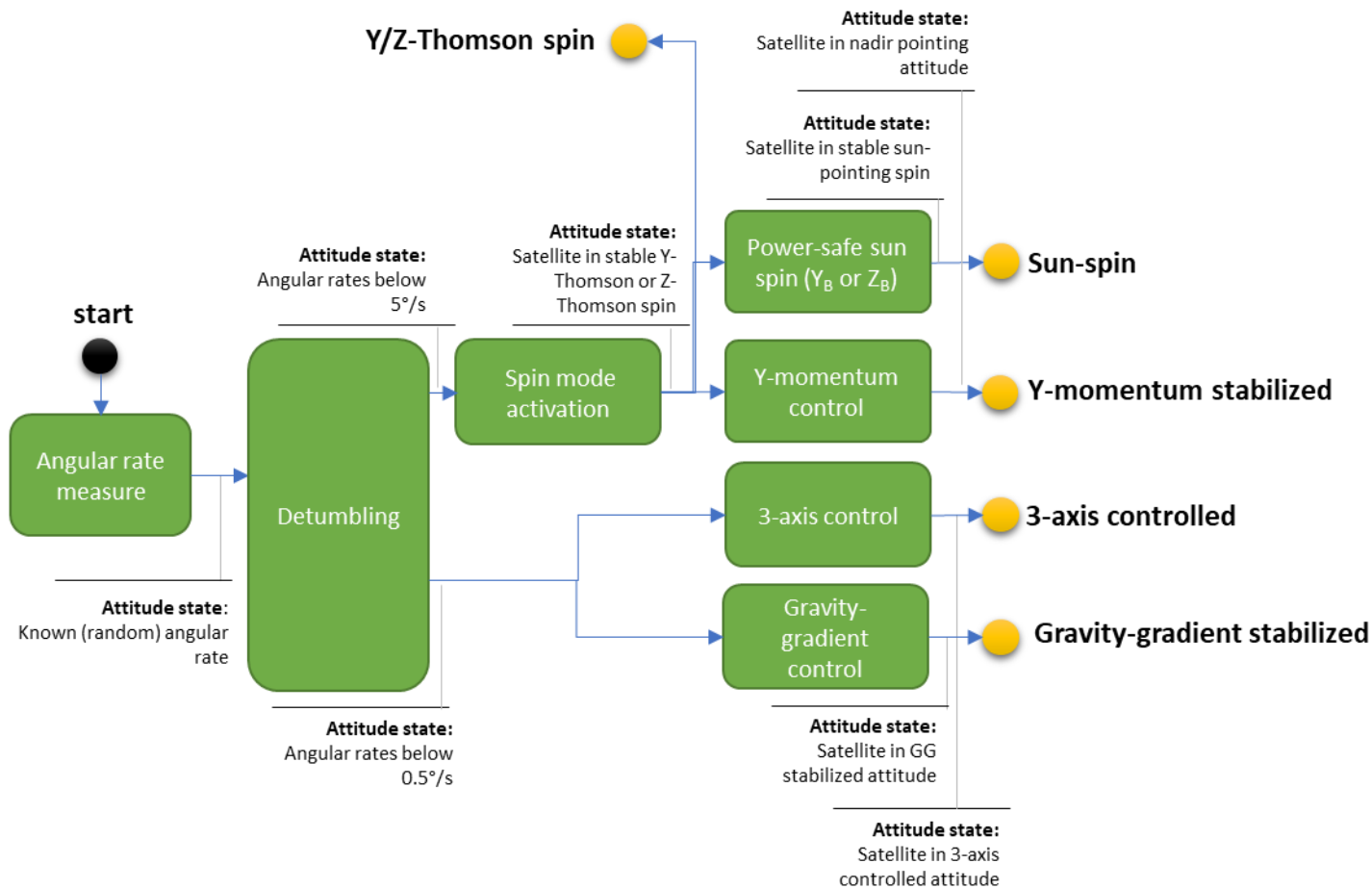


Figure 2: Commissioning high-level sequence

**Figure 3: Normal operations sequence**



The steps indicated in Figure 2 and Figure 3 are further elaborated in the sections that follow. Table 16 lists the steps and which document section can be referenced for the necessary detail.

Table 16: Commissioning and operations steps.

NO.	STEP	PARAGRAPH
1.	Determine initial angular rates	5
2.	Initial safe detumbling	6.1
3.	Detumbling	6.2
4.	Spin mode commissioning	7.1
5.	Spin mode activation	7.2
6.	Magnetometer calibration and deployment	8
7.	EKF commissioning	9
8.	CSS commissioning	10
9.	FSS commissioning	11
10.	Sun-spin control	12
11.	Momentum wheel commissioning	13.1
12.	Y-momentum control	13.2
13.	3-axis Reaction wheel commissioning	14
14.	3-axis Reaction wheel control modes	15, 18, 19
15.	HSS commissioning	16
16.	Star Tracker commissioning	17
17.	GNSS Receiver commissioning	20

Example plots and data contained in this manual make use of real-world commissioning data from satellites that carry a CubeADCS as far as possible. In cases where such data is not available, a simulation of a typical satellite using a CubeADCS is used.

The ADCS data that will be required throughout commissioning is listed in Table 17 below. The numeric entries in the table are the required logging interval (in seconds) for the respective telemetry frames.



Table 17: Typical Telemetry logging requirements.

TLM NAME	REQUIRED MINIMUM SAMPLING PERIOD (S)													
	DETERMINE INITIAL ANGULAR RATES	DETUMBLING	SPIN STABILISATION	MAGNETOMETER DEPLOYMENT	MAGNETOMETER CALIBRATION	EKF COMMISSIONING	CSS COMMISSIONING	FSS COMMISSIONING	SUN-SPIN CONTROL	WHEEL RAMP/POLARITY TESTS	ALL 3-AXIS / Y- WHEEL CONTROL	HSS COMMISSIONING	STR COMMISSIONING	GNSS RECEIVER COMMISSIONING
TlmModels (Models Telemetry)					10						10			10
TlmEstimatorMain (Main estimator telemetry)						10	10	10	10	1	10	10	10	10
TlmEstimatorBackup (Backup estimator telemetry)	10													
TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10	10	10	1	10	10	10	10	10	1	10	10	10	10
TlmController (Controller telemetry)		10	10	10	10	10	10	10	10	1	10	10	10	10
TlmSensorRawCss (Raw CSS sensor telemetry)	1*	10	10	1	10	10	10	10	10	1	10	10	10	
TlmSensorRawMag (Raw MAG sensor telemetry)	1*	10	10	1	10									
TlmSensorRawGyro (Raw GYR sensor telemetry)	10	10	10	1					10					
TlmHssCubeSenseEarthRaw (Raw CubeSense Earth telemetry)												10		
TlmGnssRaw (Raw GNSS sensor telemetry)						10								10
TlmSensorRawRwl (Raw RWL sensor telemetry)										1	10			
TlmExtSensorRaw (Raw external sensor telemetry)						***								
TlmStrPst3sRaw (Raw PST3S star tracker telemetry)						***							***	
TlmStrCubeStarRaw (Raw CubeStar telemetry)													***	
TlmFssCubeSenseSunRaw (Raw CubeSense sun telemetry)								10	10					
TlmSensorCalFss (Calibrated FSS sensor telemetry)						10		10	10					
TlmSensorCalCss (Calibrated CSS sensor telemetry)						10	10	10	10					
TlmSensorCalMag (Calibrated MAG sensor telemetry)				1	10	10				1	10			



TLM NAME	REQUIRED MINIMUM SAMPLING PERIOD (S)													
	DETERMINE INITIAL ANGULAR RATES	DETUMBLING	SPIN STABILISATION	MAGNETOMETER DEPLOYMENT	MAGNETOMETER CALIBRATION	EKF COMMISSIONING	CSS COMMISSIONING	FSS COMMISSIONING	SUN-SPIN CONTROL	WHEEL RAMP/POLARITY TESTS	ALL 3-AXIS / Y- WHEEL CONTROL	HSS COMMISSIONING	STR COMMISSIONING	GNSS RECEIVER COMMISSIONING
TlmSensorCalGyro (Calibrated GYR sensor telemetry)					10	10				1	10			
TlmSensorCalHss (Calibrated HSS sensor telemetry)						10						10		
TlmSensorCalStr (Calibrated STR sensor telemetry)						10							10	
TlmSensorCalGnss (Calibrated GNSS sensor telemetry)						10								
TlmSensorCalRwl (Calibrated RWL sensor telemetry)						10				1	10			
TlmTorquerCurrents (Torquer current measurements)		10	10	10	10	10					10			
TlmCubeComputerHealth (CubeComputer Health)	10	10	10	10	10	10	10	10	10	10	10	10	10	10
TlmFssCubeSenseSunHealth (Health telemetry from CubeSense Sun)						10								
TlmHssCubeSenseEarthHealth (Health telemetry from CubeSense Earth)						10								
TlmCubeMagHealth (Health telemetry from CubeMag magnetometer)		10	10	1	10	10	10	10	10	10	10	10	10	10
TlmRwlHealth (Health telemetry for reaction wheels)						10				10	10			
TlmPst3sHealth (Health telemetry for CubeNode PST3S)						***							***	
TlmStrHealth (Health telemetry for CubeStar)													***	

- *Only if tumbling rates > 20 °/sec are expected, else a sampling period of 10s will be sufficient.
- **It is recommended to keep a continuous log of health telemetry for the ADCS CubeComputer, and for all the nodes that are enabled. A logging interval of 10s during commissioning, or 1min post-commissioning should be sufficient.
- *** Logging of EXT sensor data, and PST3S raw telemetry is dependent on the inclusion of such sensors in the CubeADCS



5. Initial Angular Rate Estimation

The outcome of initial angular rate estimation is used as a decision point – to determine if it is required to make use of a high initial angular rate detumbling control mode, or regular B-dot detumbling.

The initial angular rate is determined from the CubeComputer onboard angular rate sensor. Angular rate estimation is enabled by selecting an estimation mode of 1 ([EstGyro](#)) when sending the [EstMode](#) telecommand. It is also possible to make use of estimation mode 2 ([EstMagRkf](#)), which will use successive magnetometer measurements in a Kalman filter to estimate the angular rate. This method is however valid for smaller angular rates, and also less accurate compared to gyro measurements. It is however useful to compare the magnetometer rate estimator to the gyro measurements. To do this, [EstMagRkf](#) can be specified as backup estimation mode when using the [EstMode](#) telecommand.

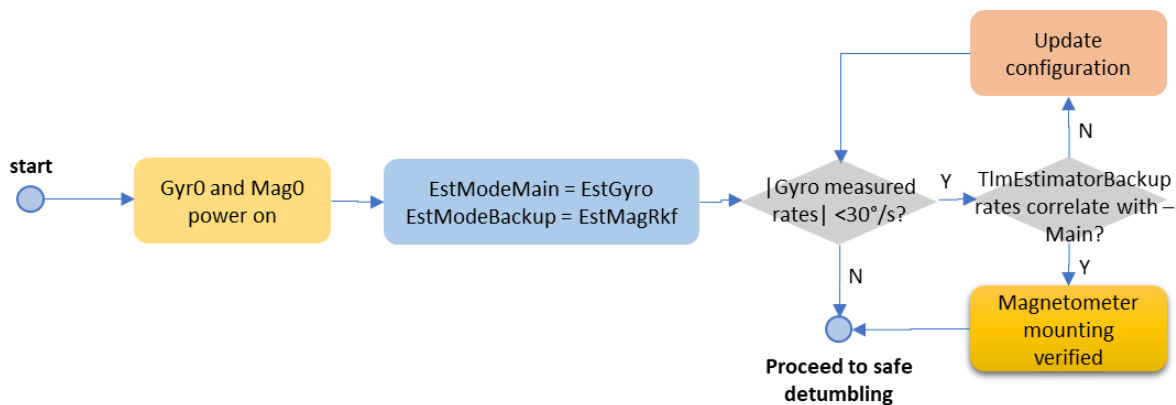


Figure 4: Initial angular rate estimation

It is also possible to confirm gyro measured angular rates by interpreting a log of magnetometer measurements, or a log of CSS measurements.

The on-board gyro should be powered on by default at start-up, however if this is not the case, or if a non-default configuration is used, the gyro should be powered-up by sending a [PowerState](#) command with the [Gyro0Power](#) parameter set to [PowerOn](#). If the magnetometer rate estimator is to be used, MAG0 or MAG1 should also be powered on.

Table 18: Initial rate estimation.

COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	Gyro0Power set to PowerOn . All others PowerOff Optionally set Mag0Power or Mag1Power to PowerOn
TELEMETRY LOGGING	EstMode (Estimation mode)	EstModeMain set to EstGyro Optionally, EstModeBackup set to EstMagRkf
	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmEstimatorBackup (Backup estimator telemetry)	10s
	TlmSensorRawGyro	10s



	(Raw GYR sensor telemetry)	
	Optional: <code>TlmSensorRawMag</code> (Raw MAG sensor telemetry)	10s
DURATION	5 mins (nominal case)	
TEST CONDITION	- Gyro estimated angular rate and magnetometer estimated rate is equal (to within 1 deg/s) and follows the same trend. - Raw magnetometer measurements change with period corresponding to estimated rate.	

An example of magnetometer estimated rates, and gyro measured Y-rate is shown in the graph below.

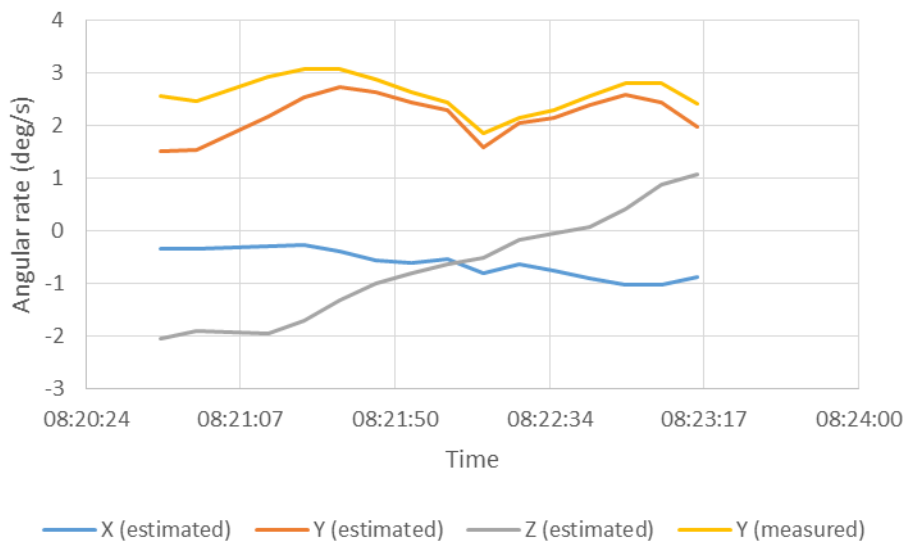


Figure 5: Estimated angular rates and Y-axis measured angular rate example.

The corresponding magnetometer measurements (in the body coordinate frame) for this scenario is displayed in the figure below.

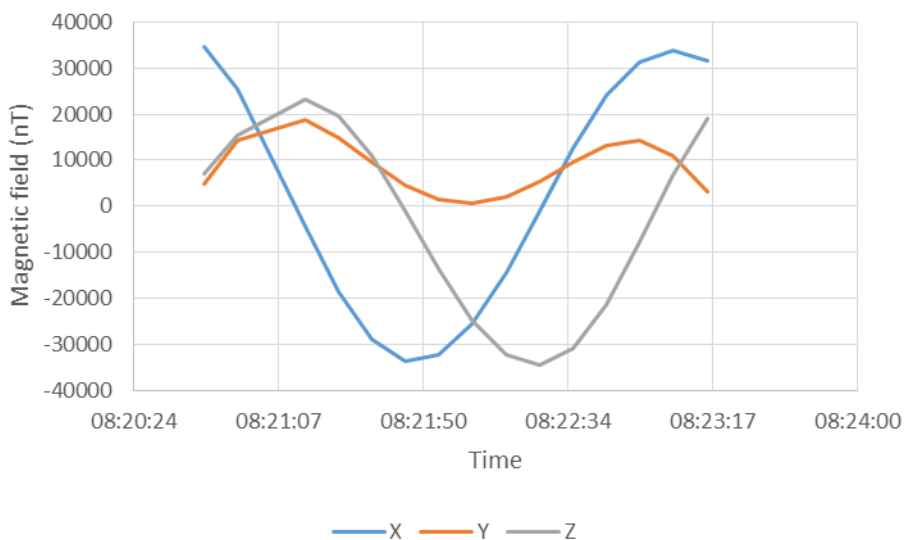


Figure 6: Magnetometer measurements for initial rate estimation.



The magnetometer shows large variations in X and Z components which indicates a predominantly Y-spin. The period of the X and Z sinusoidal plots is about 150 seconds, which corresponds to a Y-spin rate of 2.4 deg/s. Further, the fact that the Z sine plot lags behind the X by 90 degrees indicates that it is a positive spin about the Y-axis.

5.1 Rate estimation troubleshooting

If the gyro rate measurement and magnetometer estimated rate vector shows poor correlation, the cause could be one of the following:

5.1.1 Faulty main magnetometer

If the main magnetometer develops a fault, the sampled magnetic field signal will give an indication. Possibly one or more channels will have a constant output instead of varying with time. The [Mag0IsValid](#) or [Mag1IsValid](#) field in the [TlmSensorRawMag](#) will also indicate if there is a problem with either magnetometer.

The CubeMagDeploy has a redundant magnetic field sensing element. If the primary magnetometer is deemed to be faulty, the redundant sensing element can be selected by sending a [ConfigMagSensingElement](#) telecommand with the [Mag0SensingElement](#) or [Mag1SensingElement](#) set to [MagRedundant](#).

If a second magnetometer is present in the CubeADCS, the second magnetometer can be selected to be used by the estimator by using the [ConfigNodeSelection](#) telecommand and selecting an appropriate mask for [SelectMag](#).

5.1.2 The magnetometer calibration is incorrect or inaccurate

If the magnetometer mounting is incorrectly specified, the X, Y and Z estimated rates will be interchanged. In this case, one of the estimated rate vector components should match the rate sensor measurement (but could have opposite sign). It could also be that the sensitivity matrix or offset vector was incorrectly entered when programming the ADCS configuration. However, at this point during commissioning, the sensitivity matrix is expected to be the identity matrix, and the offset vector is expected to be a zero vector. (Specified in [ConfigMag0OrbitCal](#) and [ConfigMag1OrbitCal](#))

5.1.3 The satellite is spinning too fast

The RKF estimator ([EstMagRkf](#)) is only accurate if the satellite is spinning below 35 °/s. If any component of the rate vector is larger than this, the estimated rate from the RKF will be inaccurate. The CubeComputer on-board gyro (rate sensor) will measure accurately to ± 450 °/s for GYRO and ± 1000 °/s for GYR1. If the satellite is spinning at high angular rates, it will not be possible to correlate magnetometer rate measurements with the gyro measurements. It is still possible to continue with detumbling, however the mode that is used will depend on the angular rate measurement.

Once the satellite has been detumbled to low enough angular rates, the RKF estimation comparison may be repeated to correlate magnetometer measurements with the gyro.



6. Detumbling

6.1 Safe Detumbling (detumbling with unverified torquer polarity)

During magnetic control commissioning it is necessary to determine torquer polarities. A safe approach is recommended to prevent accidental spin-up of the satellite in case of incorrect torquer polarity. The recommended approach is demonstrated by Figure 7.

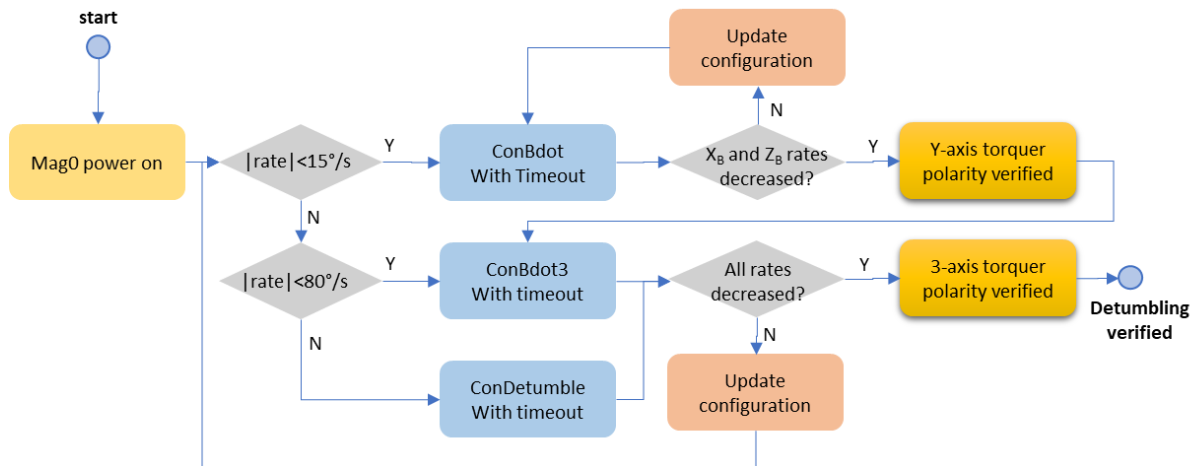


Figure 7: Initial detumbling during commissioning

If the magnetometer was not enabled in the previous step, it should be enabled using the [PowerState](#) telecommand.

It is recommended to use the [EstGyro](#) estimation mode for detumbling. If the measured angular rates are low enough ($<15^\circ/\text{s}$), the preferred approach is to first use the [ConBdot](#) control mode.



Throughout this manual, if mention is made of an angular rate upper threshold (i.e. $|\text{rate}| < 15^\circ/\text{s}$), it is meant to be interpreted as: if **all** of the angular rate vector components have absolute values **below** the threshold. If a low threshold is specified (i.e. $|\text{rate}| > 15^\circ/\text{s}$) it should be interpreted as: if **any** of the rate vector components have absolute value **above** the threshold.

The [ConBdot](#) control mode uses only the Y_B axis aligned magnetorquer to lower the X_B and Z_B angular rates, however the controller only works for angular rates up to $20^\circ/\text{s}$. It should thus be observed that with this control mode active, the X_B and Z_B angular rates decrease over time. If this is indeed the case, it is confirmed that the torquer polarity for the Y_B aligned torquer rod is correct and it is acceptable to continue to use the [ConBdot3](#) control mode. The latter can detumble from higher initial rates ($\sim 100^\circ/\text{s}$), which is why if the angular rates are too high for the [ConBdot](#) mode, this controller should be used instead. The [ConBdot3](#) mode will make use of all three magnetorquers, thus if the angular rates do not decrease as expected, or increases, it will be more difficult to diagnose which magnetorquer is incorrect polarity (This is why the Y_B torquer polarity is ideally verified first).

If the initial angular rate is even too high for the [ConBdot3](#) control mode, the 8Hz detumbling controller, [ConDetumble](#) should be used.

Regardless of the control mode being used, a control timeout should be specified in the [ConTimeout](#) parameter when changing the control mode using the [ConMode](#) telecommand. The suggested timeout value is 600s, although this may be made longer for higher moments of inertia (larger satellites). Once correct torquer polarities have been established, operations can proceed to the normal detumbling sequence in section 6.2.



The raw magnetometer measurements and CSS photodiode measurements may optionally be included in TLM log files since it is possible to correlate spin rates with these measurements.

Table 19: Initial Safe-detumbling

COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	Gyr0Power set to PowerOn. Mag0Power or Mag1Power to PowerOn
	EstMode (Estimation mode)	EstModeMain set to EstGyro
	ConMode (Control mode)	ControlMode set to ConBdot, ConBdot3 or ConDetumble ConTimeout set to 600s
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmSensorRawGyro (Raw GYR sensor telemetry)	10s
	Optional: TlmSensorRawMag (Raw MAG sensor telemetry)	10s
	Optional: TlmSensorRawCss (Raw CSS sensor telemetry)	10s
	TlmController (Controller telemetry)	10s
DURATION	Minimum 10 minutes	
TEST CONDITION	- ConBdot: Decrease in X_B and Z_B rates. - ConBdot3: Decrease in all components of angular rate vector - ConDetumble: Decrease in all components of angular rate vector	

6.2 Post-commissioning detumbling

Once torquer polarities have been established, the sequence in this section can be used to detumble reliably from any initial angular rates. This is also the recommended procedure to detumble during normal or post-commissioning operations if the satellite ends up in a random initial state. The sequence is illustrated in Figure 8.

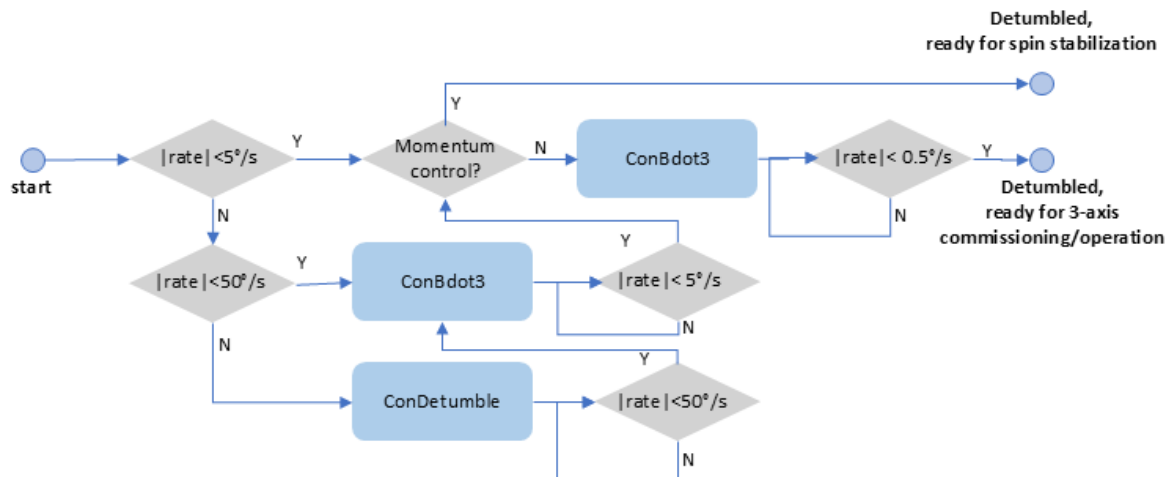


Figure 8: Post-commissioning detumbling sequence



Note that this sequence does not guarantee a safe mode “known” attitude, since the ConBdot3 controller will merely reduce angular rates. This sequence serves as preparation for safe-mode spin control that should follow from the end states in Figure 8.



Note also, that to transition from Figure 8 directly to 3-axis commissioning or operations, it must be shown that the satellite will be power-positive in any orientation or spin rate. This will likely only be true if the satellite has body mounted solar panels only, or if the solar panels have not yet been deployed.

Depending on the initial angular rate (established previously either during commissioning or by enabling the EstGyro estimation mode), a decision is made to either use ConBdot3, or ConDetumble. The latter control mode is required for spin rates more than 100°/s, although a threshold of 50°/s is specified as a safety margin, since the ConBdot3 controller becomes less efficient as the upper limit is approached.

If the angular rates are lower than 5°/s, a further decision is made – based on the overall control outcome. If the desired end control state is a spin control mode, or Y-momentum wheel control mode, it is sufficient at this point to proceed to a magnetic spin control mode. If the end control state is a 3-axis controlled or gravity gradient stabilized control mode, detumbling must continue until angular rates below 0.5°/s are observed.

Table 20: Continued detumbling

PREREQUISITE	Torquer polarities relative to magnetometer mounting verified	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	Gyr0Power set to PowerOn. Mag0Power or Mag1Power to PowerOn
	EstMode (Estimation mode)	EstModeMain set to EstGyro
	ConMode (Control mode)	ControlMode set to ConBdot3 or ConDetumble ConTimeout set to 0s (disabled)
TELEMETRY FRAME		PERIOD



TELEMETRY LOGGING	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmSensorRawGyro (Raw GYR sensor telemetry)	10s
	Optional: TlmSensorRawMag (Raw MAG sensor telemetry)	10s
	Optional: TlmSensorRawCss (Raw CSS sensor telemetry)	10s
	TlmController (Controller telemetry)	10s
DURATION	Dependent on initial angular rates	
TEST CONDITION	- Control outcome is spin stabilized or Y-momentum wheel : Angular rate vector components $< 5^\circ/\text{s}$. - Control outcome is 3-axis wheel control or gravity gradient: Angular rate vector components $< 0.5^\circ/\text{s}$.	

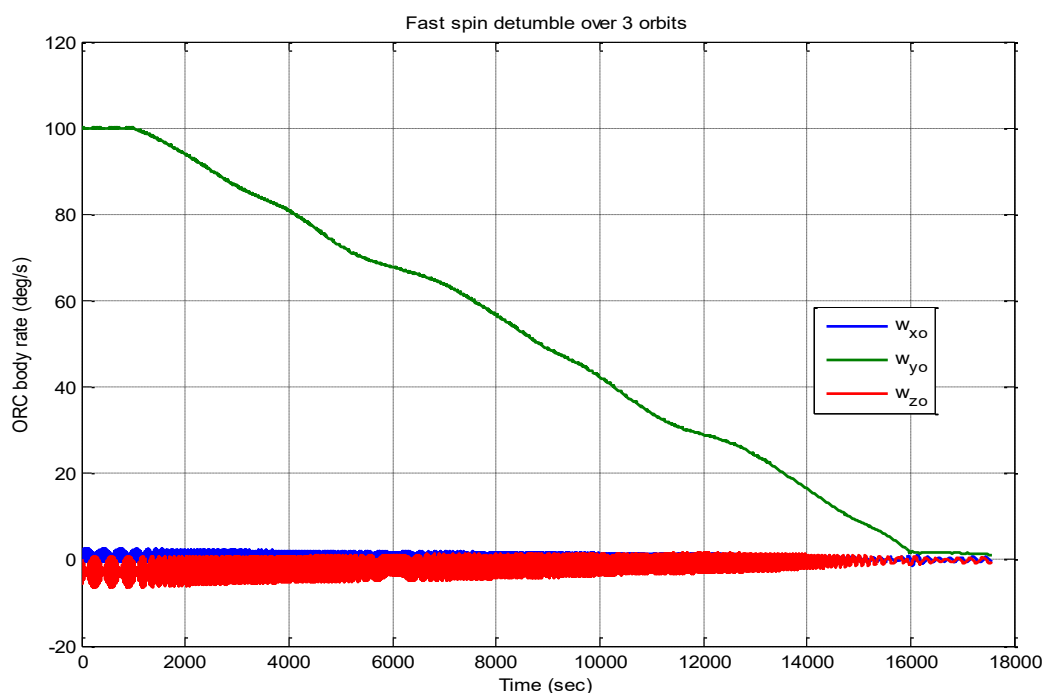


Figure 9: Example of high initial rate detumbling.

6.3 Detumbling troubleshooting

Ineffective or undesired detumbling performance could be caused by incorrect magnetometer calibration/mounting or incorrect torquer configuration. Both these causes contribute to the same problem, since it results in a torquer firing to achieve an outcome based on a mismatched magnetic field vector.



If the estimated rates are within the controllable range of normal detumbling, the default torquer configuration will be correct as verified on the ground. The default torquer configuration and polarity must be tested thoroughly for correctness in the satellite body axes, after integration of the ADCS unit in the satellite. Discrepancies in mounting configuration should ideally be addressed prior to launch.

The commissioning steps in sections 5 and 6.1 should also be followed to ensure correct polarity. Use the [ConfigMount](#) telecommand to adjust the mounting configuration if it is deemed to be incorrect.



7. Spin stabilization (Y-Thomson or Z-Thomson)

If the intention is to end up in a stable or safe-mode spinning motion, or if Y-momentum wheel stabilisation is to be used, it is required to enter a Y-Thomson or Z-Thomson spin state.

A Y-Thomson state is characterized by a spinning motion around just the Y_B satellite axis, and with the spin axis aligned with the orbit normal vector for a negative spin rate ($wy_ref < 0$) and orbit anti-normal vector for a positive spin rate ($wy_ref > 0$). Z-Thomson is the same, except the Z_B satellite axis will be the spin axis. In Y-Thomson and Z-Thomson control, the spin rate is controlled to a reference value, [wy_ref](#). These control modes make use of magnetic control only and do not require sensors other than a magnetometer and gyro.

7.1 Spin mode commissioning

During initial spin mode activation, it is likely that TLE data for the satellite is not yet known. It might thus not be possible to make use of one of the EKF estimation modes since they require position information to obtain model IGRF or sun model vector equivalents. It is still possible to make use of spin control by selecting the [EstGyro](#) estimator. The gyro rate estimator can still be used to determine angular rates, and the magnetometer should be sampled to confirm the Y- or Z-spin state.

Table 21: Spin stabilisation.

PREREQUISITES	Satellite in slow spin ($ rates < 5^\circ/s$)	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn
	EstMode (Estimation mode)	EstModeMain set to EstGyro
	ConMode (Control mode)	ControlMode set to ConYspin or ConZspin
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmSensorCalMag (Calibrated MAG sensor telemetry)	10s
DURATION	Until steady-state y-Thomson or z-Thomson spin reached	
TEST CONDITION	Y_B (Y-Thomson) or Z_B (Z-Thomson) spin rate matches reference rate (to within 10%). - Magnetic field vector shows spin vector aligned to orbit normal	

The expected angular rate telemetry for a steady-state Y-Thomson spin is shown in the graph in Figure 10. The Y-spin reference was set at $-2.2^\circ/s$.

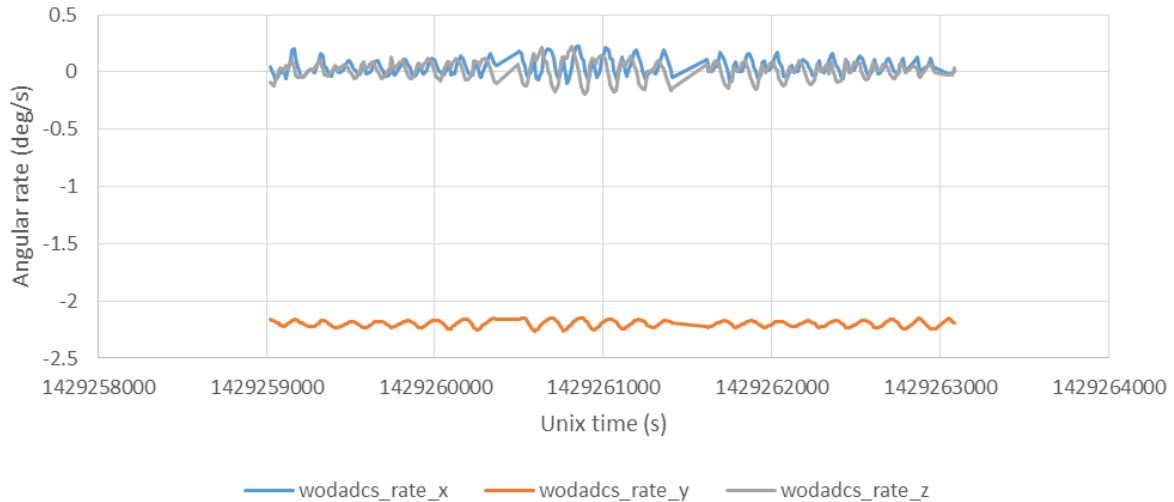


Figure 10: Example of Y-Thomson spin state: angular rates.

In absence of full attitude (roll, pitch and yaw angles) estimation, the alignment of the spin axis regarding the orbit normal or orbit anti-normal can be determined from the sampled magnetic field vector. For a Y-spin, the magnetic field will vary sinusoidally in the X and Z components with a smaller slow varying Y-component. Figure 11 below displays that the satellite motion is converging to such a state, but the Y-component of the magnetic field vector is still too large for a polar orbit (small component normal to the orbit plane, i.e. Y-component measurement) so more time is needed to settle into an accurate Y-Thomson state.

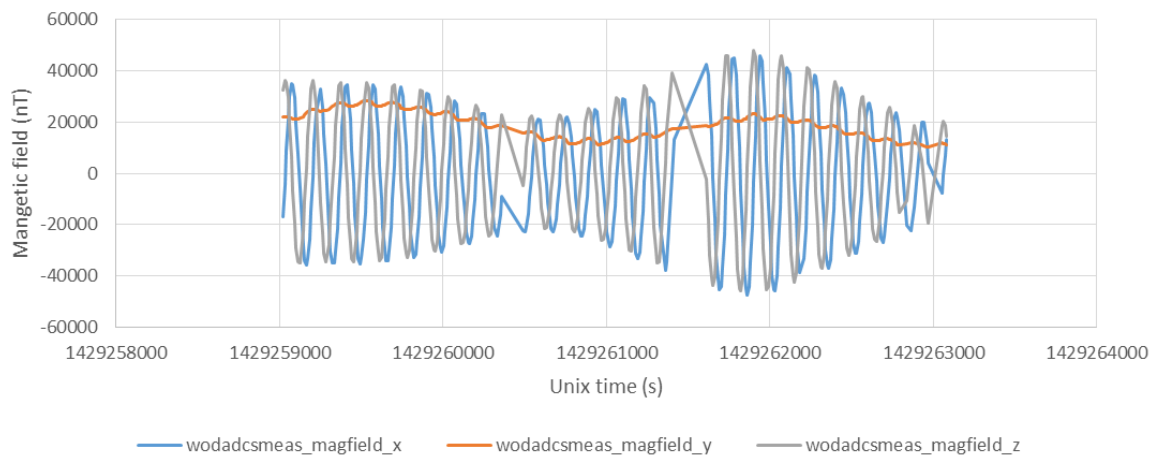


Figure 11: Example of Y-Thomson spin state: magnetic field vector.

7.2 Post-commissioning spin mode activation

If TLEs are available, the Gyro EKF (EKF that estimates attitude angles and gyro bias) should be used once the satellite angular rates are sufficiently low. This will enable the user to confirm if the Y-Thomson or Z-Thomson spin is stable.

Prior to using the [EstGyroEKF](#) mode, it is necessary to set the orbital elements of the satellite, by using the [ConfigOrbitSatParams](#) telecommand.



It will be necessary to set the orbit parameters to recent TLEs provided by NORAD on www.space-track.org or www.celestrak.com. For accurate satellite position and velocity prediction, it is also important that the on-board time is synchronized to the current Unix time using the [UnixTime](#) telecommand.

The [EkfUseFss](#), [EkfUseCss](#), [EkfUseHss](#) and [EkfUseStr](#) flags in the [ConfigAdcsEstimator](#) telecommand message must all be set to false, to ensure the EKF is only using the magnetometer measurements. Also see section 8 for further details on EKF commissioning.

The expected attitude angles for a Y-Thomson and Z-Thomson spin are shown in Figure 12 and Figure 13

Table 22: Post-commissioning spin stabilisation.

PREREQUISITES	Satellite in slow spin ($ \text{rates} < 5^\circ/\text{s}$)	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn
	ConfigOrbitSatParams (Satellite orbit parameter configuration)	SGP4 orbital elements
	UnixTime (Unix time)	Current Unix time
	ConfigAdcsEstimator (ADCS estimator configuration)	EkfUseFss , EkfUseCss , EkfUseHss and EkfUseStr flags set to false
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf
	ConMode (Control mode)	ControlMode set to ConYspin or ConZspin
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
DURATION	Until steady-state Y-Thomson or Z-Thomson spin reached	
TEST CONDITION	Y_B (Y-Thomson) or Z_B (Z-Thomson) spin rate matches reference rate (to within 10%). - Roll and yaw angles $< 10^\circ$ (Y-Thomson), or $\text{Roll} \pm 90^\circ < 10^\circ$ (Z-Thomson)	

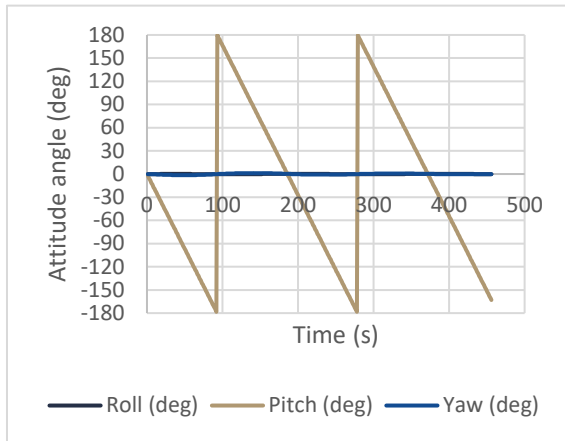


Figure 12: Roll, pitch and yaw attitude angles in stable Y-Thomson spin

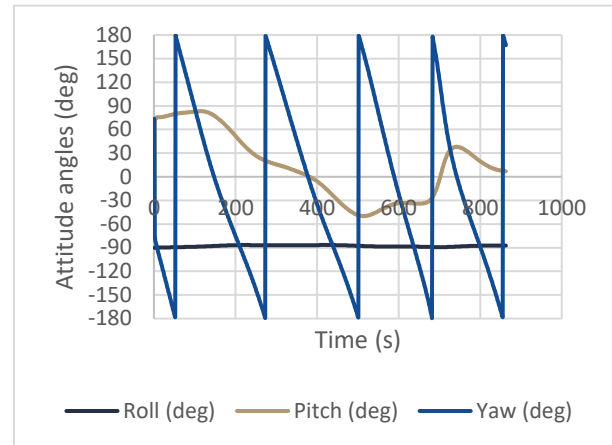


Figure 13: Roll, pitch and yaw attitude angles in stable Z-Thomson spin

7.3 Moment of Inertia considerations for stable spin

For the ADCS to achieve a stable spinning motion it is required to spin about the principal axis with either the smallest or the largest moment of inertia. A stable Y-Thomson spin will not be possible if Y_B is an intermediate moment of inertia axis, and likewise for Z-Thomson if Z_B moment of inertia is not the smallest or the largest.

The CubeADCS further requires that the moment of inertia for the chosen spin axis is different from the other axes' moments of inertia by at least 10%. This margin is in place as moments of inertia are often determined from CAD models, and these have inherent uncertainty.

The above must be ensured through a proper mechanical design and if it cannot be achieved easily by proper placement of the satellite components, extra mass can be added. For example, if $I_{ZZ} > I_{YY}$, extra mass can be added symmetric to the satellite CoM, but close to the body Z-axis and as far as possible from the satellite body Y-axis to ensure I_{YY} exceeds I_{ZZ} by more than 5%.

7.4 Stable spin aided by momentum bias

If the above MOI requirement cannot be practically met, it may still be possible to execute a Y-Thomson or Z-Thomson spin, but it will require use of one or more reaction wheel. An angular momentum bias on the intended spin axis will increase the “gyroscopic stiffness” of the satellite body, which will aid the Y-spin or Z-spin controller in maintaining a stable spinning motion.

Such a momentum bias is achieved by commanding an open-loop wheel speed (for 3-axis aligned wheels, or a single Y-axis wheel) or an open-loop momentum vector (for a reaction wheel pyramid). A negative angular momentum equal to 10% to 20% of the maximum wheel capacity will be sufficient. Example: For a typical 3U CubeSat with a CubeWheel CW0057 Y-axis wheel, where $I_{ZZ} > I_{YY}$, a reference Y-wheel speed of -800 to -1600 rpm will be sufficient.

The procedure for commanding the Y-axis wheel offset speed is described below.



Table 23: Detumbling with Y-wheel offset speed.

PREREQUISITES	Satellite in slow spin ($ \text{rates} < 5^\circ/\text{s}$)	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn 3-axis wheels, or single Y-axis wheel: Rwl<?>Power to PowerOn (<?> identifies the wheel aligned with the Y_B axis or Z_B axis) Pyramid wheel configuration: Rwl0Power , Rwl1Power , Rwl2Power and Rwl3Power to PowerOn
	ConfigOrbitSatParams (Satellite orbit parameter configuration)	SGP4 orbital elements
	ConfigAdcsEstimator (ADCS estimator configuration)	EkfUseFss , EkfUseCss , EkfUseHss and EkfUseStr flags set to false
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf
	3-axis wheels, or single Y-axis wheel: OpenLoopCommandRwl (Open-loop wheel speed command) Pyramid wheel configuration: OpenLoopCommandHxyzRW (Open-loop wheel momentum command)	Wheel speed or momentum reference
TELEMETRY LOGGING	ConMode (Control mode)	ControlMode set to ConYspin or ConZspin
	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
DURATION	TlmSensorCalRwl (Calibrated RWL sensor telemetry)	10s
	Until steady-state Y-Thomson or Z-Thomson spin reached	
TEST CONDITION	Y_B (Y-Thomson) or Z_B (Z-Thomson) spin rate matches reference rate (to within 10%). - Roll and yaw angles $<10^\circ$ (Y-Thomson), or $\text{Roll} \pm 90^\circ <10^\circ$ (Z-Thomson)	

7.5 External disturbance torque preventing a stable Y-Thomson

In rare situations some satellites will still not allow control into a stable Y-Thomson spin with small roll and yaw angles. This situation is mainly due to significant external disturbance torques, i.e. aerodynamic torques or unmodelled large internal magnetic moments. In this case the Y-Momentum stabilised control mode will also not behave optimal. Care must thus be taken when designing the satellite with such a control strategy in mind. Satellites that are meant to be stabilized using Y-wheel (and transition from Y-Thomson) should be designed to have symmetric layout of deployable solar panels – this will minimize aerodynamic and SRP disturbance. Care must also be taken to ensure no permanent magnets or magnetizable material is used, and current loops must be designed or laid out to cancel each other out.



If it is discovered in-flight that the intended approach does not work and the ADCS includes 3-axis wheels or a pyramid wheel configuration, it should be considered to transition to 3-axis control instead (see section 14).



8. Magnetometer Calibration and Deployment

Each magnetometer is calibrated on the ground to account for misalignment and temperature-dependent sensitivity and offset. These correction coefficients are stored in the magnetometer's own configuration and should not be changed.

Ideally the magnetometer measures the undistorted ideal magnetic field of the Earth once in space. Once the magnetometer is mounted on the satellite, the magnetic field is however distorted by certain parts of the satellite - like ferrous or magnetic materials and wires carrying large currents.



Although an in-flight calibration step must be performed to remove offsets and distortion of the magnetic field, **it is important to characterise the impact of interfering spacecraft elements on the magnetometer measured magnetic field before launch.** This can be done by logging magnetometer measurements from the satellite after integration, while maintaining a uniform magnetic field around the satellite. The various sub-systems of the satellite should be turned on and exercised to represent operational scenarios. It is also important to assess the impact of RF transmissions on the measured magnetic field. If any of these are found to cause a significant change in measured magnetic field, the offending sub-system must be shielded, or the magnetometer must be moved to a different location on the satellite where there is less interference. This should be performed for all magnetometers that the CubeADCS uses

An in-flight magnetometer calibration correction is further applied by the ADCS control program to ensure minimal errors. The in-flight calibration correction uses a user-configurable sensitivity matrix and offset vector to correct for distortion and offset in the measurement. These settings are adjusted via the [ConfigMag0OrbitCal](#) and [ConfigMag10OrbitCal](#) configuration messages.

The procedure to determine this “delta” sensitivity matrix and offset involves sampling the magnetic field vector over one orbit and post-processing of the logged data. The satellite does not have to be controlled in this period, but it should ideally be doing a slow tumble ($< |3 \text{ deg/s}|$). During commissioning, the preferred step to perform magnetometer calibration is after the initial detumbling, if the angular rates have been lowered to below $3.0^\circ/\text{s}$, or if the satellite is in a steady-state spin i.e. a Y-Thomson state.

To ensure adequate calibration measurements, the satellite should spin through all attitude angles relative to the magnetic field vector. It is thus not ideal for the satellite to be in a stable (non-changing) attitude while measurements are being gathered.

The magnetometer calibration procedure can be repeated as often as needed and should be repeated when the magnetic signature of the satellite changes – for instance after deploying solar panels.

During the calibration process, the amount of noise and distortion that the magnetometer experiences should be assessed and based on this, a decision must be made on which magnetometer to use, and optionally whether deploying the magnetometer (CubeMagDeploy only) is required.

The calibrated measured magnetic field vector should be sampled at a period of 10s for at least one full orbit (more data can also be used).

The sampled magnetometer data is then processed by making use of a batch linear least-squares method to find the bias and orthogonality (sensitivity) matrix. The calibration processing is incorporated in the CubeSupport application and is described in [7]. References on the procedure can be found in:

- John L. Crassidis, Kok-Lam Lai, and Richard R. Harman. “Real-Time Attitude-Independent Three-Axis Magnetometer Calibration”, Journal of Guidance, Control, and Dynamics, Vol. 28, No. 1 (2005), pp. 115-120.



- Roberto Alonso, Malcolm D. Shuster, "TWOSTEP, a Fast Robust Algorithm for Attitude-Independent Magnetometer Bias Determination", The Journal of the Astronautical Sciences, Vol. 50, No. 4. (December 2002), pp. 433-451.

It is also important to know the accurate position of the satellite at the times the samples were taken. An SGP4 orbit propagator is used by the CubeSupport magnetometer calibration tool and the closest TLE to the sampled data should be used. The time at which samples were taken should also be accurately known. The calibration tool will also output the RMS error between measured and modelled magnetic field vector, giving an indication of the measurement accuracy of the magnetometer. Once the updated magnetometer offset and sensitivity matrix parameters have been computed, it can be programmed onto the ADCS.

NOTE: The magnetometer calibration procedure should be carried out separately for Mag0 and Mag1 magnetometers (if the latter is available). The calibration procedure should also be repeated when switching from the primary to the redundant sensing element of the CubeMagDeploy.

See Figure 14 and Figure 15 for the B-field magnitude comparison before and after calibration of the magnetometer measurements with the matched IGRF model output.

Table 24: Magnetometer calibration.

PREREQUISITES	Satellite in slow spin ($ \text{rates} < 3^\circ/\text{s}$)	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn. Mag0Power or Mag1Power to PowerOn
	EstMode (Estimation mode)	EstModeMain set to EstGyro
	ConMode (Control mode)	ControlMode set to ConBdot3 or ConYspin or ConZspin
	Delay	1 Orbit (at least)
	Download telemetry log	
	On-ground Processing	
	ConfigMag0OrbitCal or ConfigMag1OrbitCal	Updated sensitivity matrix and offset vector
	ConfigPersist	
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmSensorCalMag (Calibrated MAG sensor telemetry)	10s
DURATION	1 Orbit (logging)	
TEST CONDITION	Magnetometer measurement error has been determined and updated gain and sensitivity has been computed.	

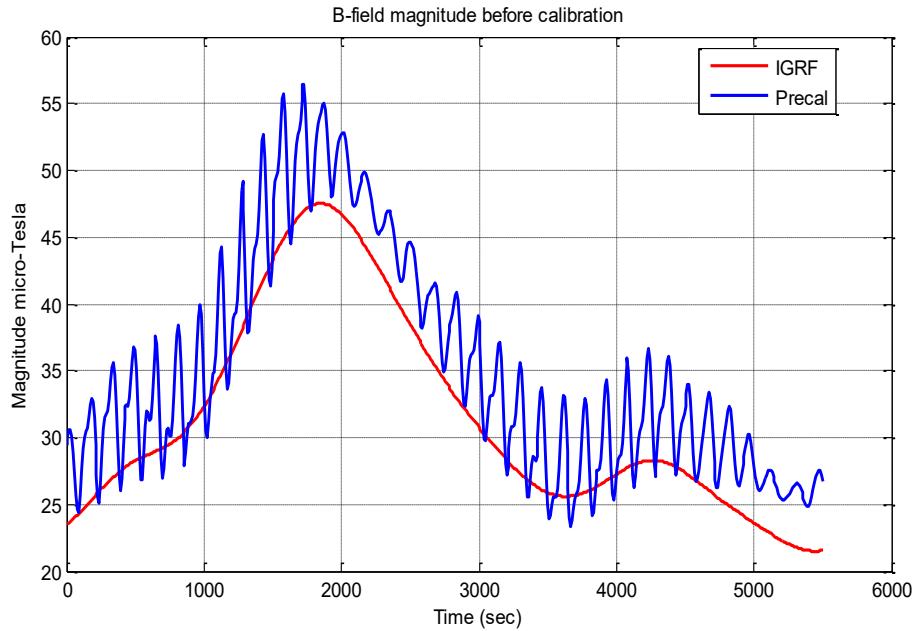


Figure 14: Before calibration B-field magnitude comparison ($\sigma = 2.848 \mu\text{T}$)

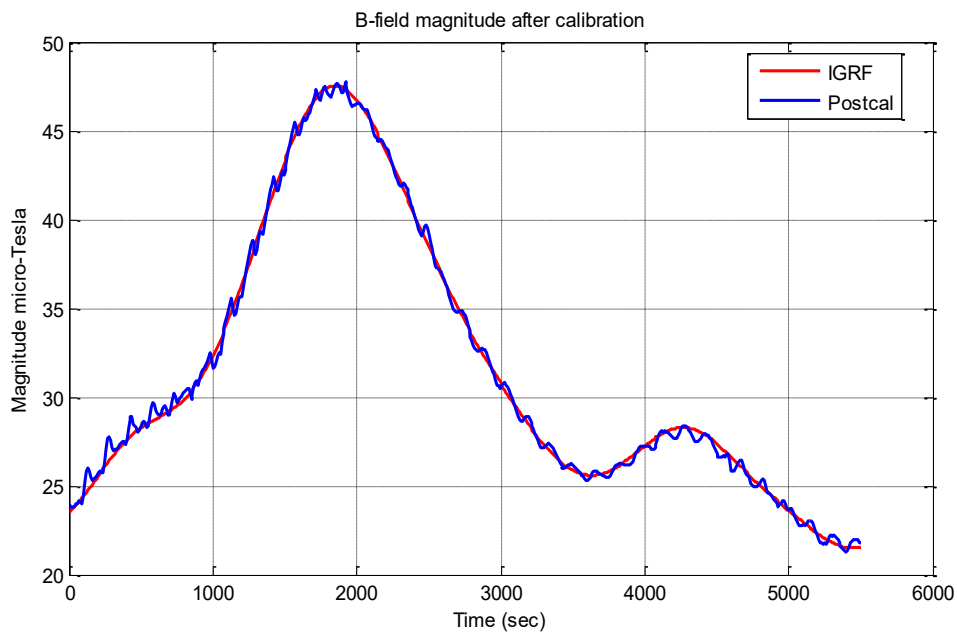


Figure 15: After calibration B-field magnitude comparison ($\sigma = 0.365 \mu\text{T}$)

The calibration processing is performed on the calibrated magnetometer measurements – the measured magnetic field vector is already transformed to the SBC coordinate frame. It is important to take note of the existing [ConfigMag0OrbitCal](#) or [ConfigMag1OrbitCal](#) settings prior to obtaining the log of magnetic field measurements. If the settings that are used are not an identity matrix for sensitivity and zero offset vector, the calibrated measurements on which the post-processing will occur, will already include a delta calibration. It will then be necessary to compute a new offset and sensitivity matrix from:

$$\mathbf{d}_{new} = \mathbf{d}_{\Delta} + \mathbf{S}_{\Delta} \mathbf{d}_0$$
$$\mathbf{S}_{new} = \mathbf{S}_{\Delta} \mathbf{S}_0$$



Where $\mathbf{d}_0$ and $\mathbf{S}_0$ is the in-flight offset vector and sensitivity matrix that was used when the log file was sampled. $\mathbf{d}_\Delta$ and $\mathbf{S}_\Delta$ is the offset vector and sensitivity matrix that results from the calibration process, and $\mathbf{d}_{new}$ and $\mathbf{S}_{new}$ is the new offset vector and sensitivity matrix to use (for the in-flight calibration settings).

The magnetometer calibration is considered “good” if the error between measured and modelled magnetic field vector is less than 0.3 μT . Errors of up to 0.5 μT may still be acceptable.

8.1 Troubleshooting magnetometer calibration

Electromagnetic interference (EMI) has the potential to cause upsets in the measured magnetic field. If the magnetometer calibration process returns unfavourable results, usage of EMI sources (e.g. transmitting over UHF) should be investigated as a possible source of disturbance. Mitigation techniques can include transmitting with lower power, or temporarily suspending non-critical transmissions while collecting magnetometer samples.

If disturbances to the main magnetometer cannot be mitigated successfully, the user should consider one of the following options:

- Deploying the CubeMagDeploy,
- Switching from primary to redundant magnetic field sensing element (CubMagDeploy only),
- Switch to using Mag1 (if available).

8.2 Deployment

Deployment of the magnetometer is required to limit the magnetic disturbances caused by the satellite bus, e.g. currents flowing in solar panels, wire loops and reaction wheel magnets. Deployment should ideally occur while ground contact with the satellite has been established (during a ground station overpass). Magnetometer health telemetry at 1 second intervals is required to evaluate if the deployment was successful. It is also better that the satellite is still in a slow tumble (e.g. Y-Thomson spin).

The CubeMag Compact and Deploy magnetometers have sensing switches to determine whether the boom has deployed. Upon deployment the magnetometer should rotate through a 90° angle. The CubeMag will use the feedback from the sensing switch to automatically transform its raw output, so that the measurements are still in the local coordinate frame shown in [9]. It is thus not necessary to make any immediate configuration changes to the CubeADCS, however as a precaution the control mode should be set to none (no control must be taking place) during the deployment.

Table 25: Magnetometer boom deployment.

PREREQUISITES	Satellite in slow spin ($ \text{rates} < 3^\circ/\text{s}$)	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn
	EstMode (Estimation mode)	EstModeMain set to EstGyro
	ConMode (Control mode)	ControlMode set to ConNone
	DeployCommandMag (Mag deploy command)	DeployMag0 or DeployMag1 set to true
	Delay	5s
	ConMode (Control mode)	ControlMode set to ConBdot3 , ConYspin or ConZspin (restore previous mode)
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD



	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	1s
	TlmSensorRawMag (Raw MAG sensor telemetry)	1s
	TlmCubeMagHealth (Health telemetry for CubeMag magnetometer)	1s
DURATION	1 min	
TEST CONDITION	• Mag0BurnCurrent or Mag1BurnCurrent shows increase • Mag0BurnPinState or Mag1BurnPinState changes from false, to true, to false • Mag0DeployPinState or Mag1DeployPinState changes from false to true (and remains true) • Mag0BurnUnderCurrent and Mag0BurnOverCurrent (or likewise for Mag1) remains clear • RAW magnetometer measurements do not show any step increases or discontinuities.	

The magnetometer deployment activation signal will be active for a maximum of 3s. An increase to the burn current in the [TlmCubeMagHealth](#) telemetry frame should be observed, and the deployment sensing should return true after successful deployment.

If deployment failed, the [MagDeployError](#) event will be registered. The [ErrorCode](#) of this event can be used to further determine the cause of failed deployment. The deployment may be re-attempted in case of failure.

As the CubeMag will automatically transform its measurements based on the deployed status, the raw magnetometer measurements should not show any discontinuities. If the raw measurements do display jumps or discontinuous measurements, it might be that the magnetometer did not rotate through a full 90°. The raw measurements should then be further analysed to determine the actual deployment angle, and a configuration change to the CubeMag may be necessary.



9. EKF commissioning

The Full-state EKF ([EstFullEkf](#)) and gyro EKF ([EstGyroEkf](#)) both provide full attitude information – attitude angles (and quaternion) as well as ORC (orbit) referenced and IRC (inertial) referenced angular rates.

One of the EKF modes should be used for stabilized and 3-axis attitude control modes. The EKF estimation modes will incorporate as many sensor inputs as are available based on currently enabled nodes and the [EkfUseFss](#), [EkfUseCss](#), [EkfUseHss](#) and [EkfUseStr](#) flags of the configuration [ConfigAdcsEstimator](#) message.

The EKF estimation modes require satellite position knowledge. The satellite position serves as input to determine model IGRF magnetic field vector and sun vector, and star vectors in ORC frame. These are required since the EKF measurement update requires a matching modelled and observed (measured) vector pair.

The satellite position is determined from an SGP4 propagator. The propagator is always active, but requires up-to-date orbital elements, extracted from TLEs to provide useful position information. The SGP4 orbital elements are set by the OBC, by sending an [ConfigOrbitSatParams](#) telecommand.

It will be necessary to periodically set the orbit parameters to recent TLEs provided by NORAD on www.space-track.org or www.celestrak.com. For accurate satellite position and velocity prediction, it is also important that the on-board time is synchronized to the current Unix time using the [UnixTime](#) telecommand.

The estimated satellite position should be verified after uploading TLEs, by logging the [TlmModels](#) telemetry. The latitude, longitude and altitude of this log can be compared to third party applications such as Orbitron¹.

Throughout the rest of the commissioning steps, the various sensors will be verified for correct operation, and afterwards included in the EKF computation. The [EkfUseFss](#), [EkfUseCss](#), [EkfUseHss](#) and [EkfUseStr](#) flags of the configuration [ConfigAdcsEstimator](#) message, as well as the node selection flags in the [ConfigNodeSelection](#) message should be enabled as needed, to include the relevant sensor in the EKF measurement update. For the very first activation, all the [EkfUse*](#) flags must be set to false so that only magnetometer measurements are used.

The EKF estimation accuracy should be checked after each new sensor inclusion, by logging the estimated attitude (RPY angles and angular rates) with the satellite in a nadir pointing or stable spin attitude. The raw sensor measurements may also be logged for comparison.

Table 26: EKF commissioning.

PREREQUISITES	Satellite in steady-state y-Thomson or z-Thomson spin, or very low angular rates ($ \text{rates} < 0.5^\circ/\text{s}$)	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn
	ConfigOrbitSatParams (Satellite orbit parameter configuration)	SGP4 orbital elements
	UnixTime (Unix time)	Current Unix time

¹ <http://www.stoff.pl/>



	ConfigAdcsEstimator (ADCS estimator configuration)	EkfUseFss , EkfUseCss , EkfUseHss and EkfUseStr flags set to include necessary sensors
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf or EstFullEkf
	ConMode (Control mode)	ControlMode set to ConYspin , ConZspin , ConBdot3 , ConYwheel , ConXYZwheel , ConGGboom
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmModels	10s
	Calibrated sensor TLM (as needed)	10s
DURATION	Typically 1 orbit	
TEST CONDITION	• Latitude, longitude, and altitude match with third-party TLE viewer • Calibrated sensor TLM shows working sensor (see separate sections on sensor commissioning) • Roll, pitch and yaw, and angular rates match with expected attitude control behaviour	

9.1 Gyro EKF vs Full-state EKF

Since both estimation modes return the same information, it must be decided which mode to use by default. The Gyro EKF does not estimate the angular rates but uses the gyro measurements directly (with estimated bias removed). Since the Gyro EKF does not rely on model information to propagate the angular rates, it will be more accurate in the presence of modelling errors (large moment of inertia errors) and external disturbances such as aerodynamic and SRP torque, or unmodelled magnetic disturbance torques caused by internal or solar panel magnetic moments.

The rate sensors used on the CubeADCS have relatively low noise and temperature-sensitive bias drift compared to other COTS MEMS rate sensors, thus the recommended default is to use the Gyro EKF estimation mode. The full-state EKF can be used during commissioning as the backup estimator for comparison purposes, to determine if it provides better accuracy.

If the primary rate sensors of the CubeADCS (GYR0) are faulty it might be that the redundant IMU (GYR1) noise and bias drift is unsatisfactory high. In this case the Full-state EKF might also be considered.



10. CSS Commissioning

The ADCS calculates a Coarse Sun Sensor (CSS) vector measurement from (ideally) 6 photodiodes, placed to point along each positive and negative principal axis direction. The CSS measurement is hampered by Earth Albedo, photodiode mounting angle offsets and obscuration and reflections from the satellite. As such, the CSS measurement is not accurate enough to provide high accuracy pointing in 3-axis modes (the CSS vector angle error is typically 10°), however it is useful when sun-pointing spin modes are going to be used. Fine sun sensors have limited field of view, but the CSS vector can be computed for all satellite orientations, provided that there is a CSS photodiode in each axis direction with full hemispherical FOV, i.e. no shadowing ($\pm X_B, \pm Y_B, \pm Z_B$).

The CSS measurement can be included in the EKF measurement update, however the sun spin control modes, [ConSunYspin](#) and [ConSunZspin](#), require that either a CSS vector or FSS measurement is available regardless of which estimation mode is active.

The CSS is commissioned by logging raw and calibrated CSS measurements over multiple orbits (2 or 3 orbits ideally) and by correlating the measurements with measured angular rates and magnetometer-only EKF estimated attitude angles.

Initially the [ConfigAdcsEstimator.EkfUseCss](#) flag must be set to false so that the measurement is not yet used in the EKF. (At this stage in commissioning, the GyroEKF will typically be used with only magnetometer measurements). The flag may be enabled after confirmation of correct CSS vector calculation.

Table 27: CSS commissioning

PREREQUISITES	Satellite in steady-state Y-Thomson or Z-Thomson spin, or very low angular rates ($ \text{rates} < 0.5^\circ/\text{s}$)	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn
	ConfigOrbitSatParams (Satellite orbit parameter configuration)	SGP4 orbital elements
	UnixTime (Unix time)	Current Unix time
	ConfigAdcsEstimator (ADCS estimator configuration)	EkfUseFss , EkfUseCss , EkfUseHss and EkfUseStr flags set to false
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf or EstFullEkf
	ConMode (Control mode)	ControlMode set to ConYspin , ConZspin , ConBdot3
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmSensorRawCss (Raw CSS sensor telemetry)	10s
	TlmSensorCalCss (Calibrated CSS sensor telemetry)	10s
DURATION	2 to 3 orbits	



PREREQUISITES	Satellite in steady-state Y-Thomson or Z-Thomson spin, or very low angular rates ($ \text{rates} < 0.5^\circ/\text{s}$)
TEST CONDITION	• RAW CSS measurements correlate with mounting direction of photodiode • Calibrated sensor TLM valid flag is true for entire sunlit duration • Calibrated CSS vector correlates with sun direction and RPY attitude angles

10.1 Adjustment to CSS parameters

If the mounting of a CSS photodiode is found to be incorrect, it can be corrected by sending a [ConfigMount](#) telecommand with updated settings. If it is found that a particular CSS photodiode is not measuring as expected, despite correct mounting (for instance if the sensor is faulty, or contaminated by reflection or obscuration), the particular CSS mounting must be set to [AxisNotUsed](#). The CSS vector computation will ignore photodiodes with such configuration settings and as long as enough photodiodes remain it may still be possible to compute a CSS vector (although not necessary for all orientations).

10.2 CSS use by Estimators

If the CSS measured vector is confirmed to be correct the [ConfigAdcsEstimator.EkfUseCss](#) flag may be enabled when using the gyro EKF or full-state EKF, although due to the low accuracy of the CSS it might not have a significant improvement to the estimated attitude.

The CSS measurement can also be used in the [EstTriad](#) estimation mode. This mode is however only intended for testing purposes and should ideally not be used for nominal operations.



11. FSS Commissioning

The CubeSense Sun sensors (FSS) are tested by first obtaining a log of measurements over at least one orbit. This should ideally take place while the satellite is already in a stable nadir pointing attitude (if Y-momentum wheel control is active), although if the intention is to proceed to a sun-spin control mode from a steady-state Y-Thomson or Z-Thomson spin, or if the intention is to transition directly from B-dot control at low rates (below $0.5^\circ/\text{s}$) to 3-axis control, then it will be acceptable to commission the FSS while in these initial states.

If the satellite is still tumbling, it may be necessary to log data for longer, as the FSS might not have the sun in its field of view for the entire sunlit duration.

It is Important that the Sun sensors remain unused by the EKF estimator (the `ConfigAdcsEstimator.EkfUseFss` flag must be false). Only once they have been verified, can the flag be changed to true, for use in the EKF.

If more than one FSS is present, they can be commissioned at the same time by enabling all of them. The `ConfigNodeSelection.SelectFss` mask must be set to enable all the FSS which are being commissioned – this is needed so that a calibrated vector is computed for each FSS.

Table 28: FSS commissioning

PREREQUISITES	Satellite in nadir pointing attitude, or steady-state y-Thomson or z-Thomson spin, or very low angular rates ($ \text{rates} < 0.5^\circ/\text{s}$)	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	<code>PowerState</code>	If not already enabled, <code>Gyr0Power</code> set to <code>PowerOn</code> . <code>Mag0Power</code> or <code>Mag1Power</code> to <code>PowerOn</code> <code>Fss<X>Power</code> to <code>PowerOn</code> (for all FSS to be commissioned)
	<code>ConfigOrbitSatParams</code> (Satellite orbit parameter configuration)	SGP4 orbital elements
	<code>UnixTime</code> (Unix time)	Current Unix time
	<code>ConfigAdcsEstimator</code> (ADCS estimator configuration)	<code>EkfUseFss</code> , <code>EkfUseHss</code> and <code>EkfUseStr</code> flags set to false
	<code>EstMode</code> (Estimation mode)	<code>EstModeMain</code> set to <code>EstGyroEkf</code> or <code>EstFullEkf</code>
	<code>ConMode</code> (Control mode)	<code>ControlMode</code> set to <code>ConYspin</code> , <code>ConZspin</code> , <code>ConBdot3</code> , <code>ConYwheel</code>
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	<code>TlmEstimatorMainHighRes</code> (Main estimator high-resolution telemetry)	10s
	<code>TlmSensorRawCss</code> (Raw CSS sensor telemetry)	10s
	<code>TlmSensorCalCss</code> (Calibrated CSS sensor telemetry)	10s
	<code>TlmFssCubeSenseSunRaw</code> (Raw CubeSense sun telemetry)	10s



PREREQUISITES	Satellite in nadir pointing attitude, or steady-state y-Thomson or z-Thomson spin, or very low angular rates ($ \text{rates} < 0.5^\circ/\text{s}$)	
	TlmSensorCalFss (Calibrated FSS sensor telemetry)	10s
DURATION	2 to 3 orbits	
TEST CONDITION	- Raw and calibrated FSS measurements correspond with CSS measurements. TlmFssCubeSenseSunRaw.Fss<X>CaptureResult Shows SunCaptureSuccessNormal or SunCaptureSuccessFlipped. TlmFssCubeSenseSunRaw.Fss<X>DetectionResult shows SunSuccess when sun measurements are expected in the sensor FOV, and SunNoDetect or SunNotFound otherwise (and in eclipse).	

Once the sensors have been verified, they can be enabled for inclusion in the EKF by sending a [ConfigAdcsEstimator](#) telecommand with the [EkfUseFss](#) parameter set to *true*.

11.1 Mounting transform

If the raw detection results are valid, but the calibrated Sun vector does not match with expected values, it could be because of incorrect mounting transform. Consult section 3.10 for details on mounting angles and how to configure them, along with the CubeSense coordinate frame definition in [9].

11.2 Diagnosing problems

The best way to diagnose problems with the CubeSense Sun is to capture and download an image where the sun is expected in the field of view.

The procedure that the OBC must implement to download images is described in [5]. CubeSense sun images can either be captured and buffered locally on the CubeComputer for later download to the OBC (making use of the CubeADCS Image Log) or can be streamed through the CubeComputer to the OBC. If the latter option is used (stream directly) the CubeSense sun must be enabled in passthrough mode, by sending a [PowerState](#) command with the relevant sensor setting set to [PowerOnPass](#).

While the image download is taking place, the sensor will not be used for estimation purposes.

CubeSense sun images have dimensions of 1024 x 1024 pixels, and the complete raw image will occupy 1MB of memory. To conserve the satellite to ground data link, the OBC may implement compression or down-sampling of the images.

The download images can then be analysed to determine if adjustment to sun sensor parameters is necessary.

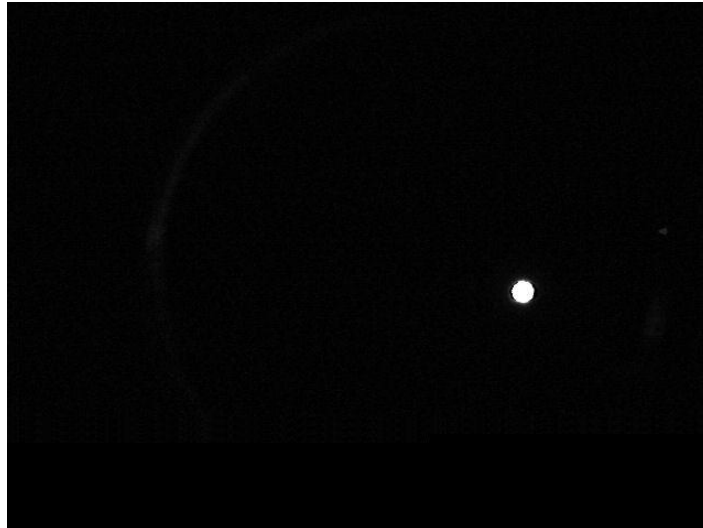


Figure 16: Downloaded CubeSense Sun image with good exposure

11.3 Adjustments to Sun sensor parameters

11.3.1 Camera and detection settings

If the CubeSense Sun sensor reports detection error results, when it was supposed to report a valid result, it could be due to incorrect camera gain and/or exposure settings, or detection threshold setting. These settings should ideally be adjusted after capturing and downloading images to compare.

The sensor's exposure, gain and detection threshold settings can only be adjusted by changing the configuration for the CubeSense itself – see section 3.11 for details.

The CubeSense Sun's **Config** telecommand will allow changing the camera exposure and gain, while the **CamThreshold** telecommand can be used to change the detection threshold.

Sensor boresights and lens distortion coefficients are determined prior to delivering the units and should not have to be changed.

11.3.2 Satellite elements in field of view

Figure 17 shows an example of FSS telemetry where the sun sensor operates nominally for a portion but then reflections of the sun from objects that protrude into the FOV causes erroneous detections. This is seen towards the left of the plot where the raw Alpha angle jumps to +80 deg. The sensor would still report a valid result because it cannot discern between the true sun and the reflection, and this will ultimately result in estimation errors in the EKF, if this sensor was included for use in the EKF.

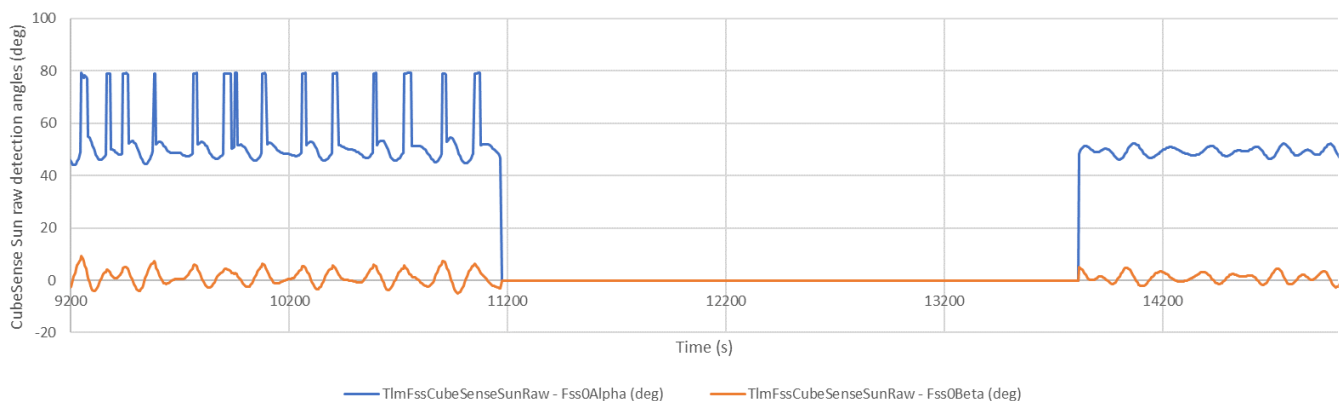


Figure 17: CubeSense Sun detection results with errors present



Towards the right side of Figure 17 the sun sensor mask parameters were modified to ignore detection results where the sun was detected inside a masked region. Such mask regions can only be determined from images captured by the sensor and downloaded for processing.

If the downloaded image reveals objects in the field of view of the CubeSense which cause the sun to reflect, it is possible to instruct the CubeSense to ignore certain parts of the image. Ideally such problems should be identified during integration onto the satellite – by using non-reflective coatings and considering placement of the sun sensor, the problem of reflections can be prevented. Setting masked regions in the sensor image plane reduces the operating range of the sensor.

It is also beneficial to determine CubeSense mask regions prior to launch of the satellite, as downloading CubeSense photos from in-flight satellites might cause delays to the commissioning process. The mask regions can be determined by capturing a photo using the sun sensor, while mounted inside the satellite. All deployable solar panels and antennas must be extended so that the scene the sensor images is that of the flight configuration. It may be necessary to adjust the exposure setting for the sun sensor to allow a photo with adequate lighting conditions to be captured. The latter statement is also true when capturing CubeSense Sun photos from in-flight satellites.

Figure 18 shows an example of a CubeSense Sun sensor photo that was captured in flight. The contrast was artificially enhanced using commercial image editing software. From this image, it is possible to determine the mask regions by finding the coordinates of rectangular areas.

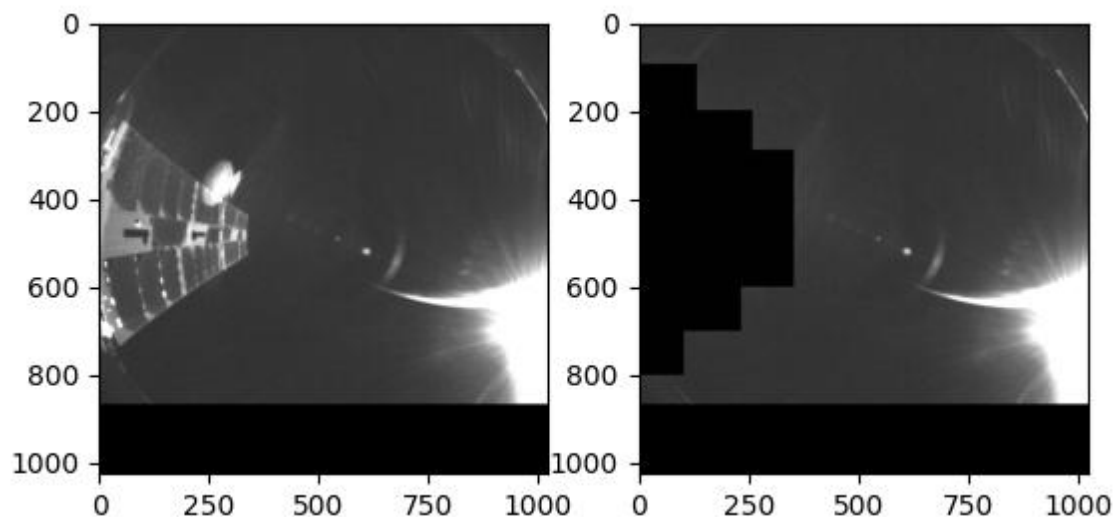


Figure 18: Example of CubeSense Sun masking, using photo captured from in-flight satellite

To set a masked region it will also be necessary to change the configuration of the CubeSense itself (see section 3.11 for details). The [CamMask](#) telecommand can be used to set mask regions, and the [CamMasks](#) telemetry request will inform which masked regions have been set.



12. Sun-Spin control

The Y-sun spin and Z-sun spin controllers will continue to maintain the spin rate from Y-Thomson and Z-Thomson spin and at the same time precess the angular momentum vector to point towards the sun. It is thus required to transition into sun-spin control from initial Y-Thomson or Z-Thomson spin.

Such a control strategy is useful if most solar panels are mounted (or deployed) with normal vectors aligned to the spin vector. The fact that the [ConSunYspin](#) and [ConSunZspin](#) controllers only use magnetic control with minimal sensor inputs makes it suitable as a safe-mode control strategy.

It is required to make use of the CSS or ideally CSS and a FSS with boresight mounted along the spin direction, for the controller to be able to rotate the spin axis towards the sun. The CSS and/or FSS must already be commissioned at this time.

Table 29: Transition to sun-spin mode

PREREQUISITES	Satellite in steady-state Y-Thomson or Z-Thomson spin. CSS and/or FSS commissioned	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn Fss<X>Power to PowerOn (for all FSS to be commissioned)
	ConfigOrbitSatParams (Satellite orbit parameter configuration)	SGP4 orbital elements
	UnixTime (Unix time)	Current Unix time
	ConfigAdcsEstimator (ADCS estimator configuration)	EkfUseHss and EkfUseStr flags set to false. EkfUseCss and EkfUseFss set to true.
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf
	ConMode (Control mode)	ControlMode set to ConSunYspin or ConSunZspin
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmSensorCalCss (Calibrated CSS sensor telemetry)	10s
	TlmSensorCalFss (Calibrated FSS sensor telemetry)	10s
DURATION	As required	
TEST CONDITION	N/A	



12.1 Limitations to sun-spin

12.1.1 Satellite configuration and moments of inertia

The same moment of inertia implications exists as for pure Y-Thomson and Z-Thomson spin – the chosen spin axis cannot be an intermediate moment of inertia axis. In addition, the moment of inertia about the spin axis should be different from the nearest other axis' moment of inertia by at least 10%.

It is also required to have products of inertia $<10\%$ of the smallest moment of inertia. Satellites with unsymmetric deployable structures will have difficulty in maintaining a stable spin and track the sun.

12.1.2 Orbit

Sun-spin control is best achieved in sun-synchronous orbits (polar orbits) with sun angles (the angle between the sun vector and the orbital plane) above 45° . For small sun angles, such as a 12:00/24:00 sun-synchronous orbit, the angular momentum vector must rotate through a large angle to end up tracking the sun.

Disturbances in low Earth orbit further hinder the ability to end up in the desired spin state. If the satellite has large deployable surfaces at low altitudes ($< 450\text{km}$) it will be difficult to maintain a sun-spin condition.

12.2 Other safe-mode spin options

If the intended orbit is not conducive to a sun-tracking spin as mentioned above – for instance in a 12:00/24:00 sun-synchronous orbit, it should be considered to simply make use of a Z-Thomson or Y-Thomson spin as the safe mode. If the largest area of solar panels is mounted to face along the $+Y_B$ or $-Y_B$ direction, a Z-Thomson spin will result in the solar panels rotating in-and-out of the sun continuously (a sun-barbeque spin) which will also ensure a consistent or known solar generation capability. Likewise, if the solar panels deploy so that their normal vector aligns with the $-Z_B$ axis, a Y-Thomson spin in such an orbit will achieve the same sun-barbeque result.

Finally, if the satellite is placed in a dawn-dusk sun-synchronous orbit, with solar panels mounted along the $+Y_B$ or $-Y_B$ direction, a Y-Thomson spin will ensure optimal power generation (no need for additional sun tracking).



13. Y-momentum control

Y-momentum stabilization makes use of a Y_B momentum bias to achieve a stable attitude. This requires use of single Y_B -axis aligned reaction wheel, or in the case of a pyramid wheel configuration the wheels in the pyramid will have joint wheel speeds to result in a net Y-momentum bias.

The momentum bias increases the “gyroscopic stiffness” of the satellite, and external disturbance torques will not rotate the satellite but rather cause a precession of the momentum vector. In this control mode the X_B and Z_B rates will be damped to zero while the wheel bias momentum is maintained. The pitch angle can optionally be controlled to a reference value if needed. Roll and yaw angles cannot be controlled to reference values and will remain close to zero.

13.1 Y-momentum commissioning

Before enabling the Y-momentum mode, it must be verified that the momentum wheel can be commanded and that its angular momentum has the correct sign. The wheel ramp-up test is carried out while the satellite is already in a Y-Thomson spin. The gyro-EKF estimation should be active before commanding the wheel, so that the effect in angular rates and pitch angle can be observed (see Section 8 for more details).

To begin, a Y momentum bias must be commanded by sending an open-loop momentum bias ([OpenLoopCommandHxyzRW](#)) telecommand. The appropriate RWL(s) should be enabled first. If the CubeADCS has a single Y_B aligned wheel, then this is the wheel that should be enabled. If the CubeADCS has 3-axis reaction wheels, with each wheel aligned with a principal axis, the particular wheel (RWL0, RWL1, RWL2) should be identified first from the mounting configuration. For instance, if [ConfigMount.MountRwl0](#) is either [PosY](#) or [NegY](#), then this (RWL0) is the wheel that should be enabled. If the CubeADCS makes use of a reaction wheel pyramid, all four wheels must be enabled.

The Y-Thomson controller that should be active at this stage (to maintain the Y-Thomson spin) should be deactivated temporarily by setting the control mode to [ConHxyzRW](#).

The wheel or pyramid will now be commanded to a constant momentum value to absorb the spinning satellite angular momentum. The Y-momentum bias should be set to the following reference value:

$$h_y = -I_{yy}\omega_{y-ref}$$

where I_{yy} is the satellite moment of inertia about the Y-body axis (passing through the center of mass) in kg.m^2 units, ω_{y-ref} is the reference spin rate in deg/s for the detumbling controller (default value is -1 or -2 deg/s).

The wheel, or pyramid, is then allowed to spin at constant speed for a period of 2 minutes. During this time the measured wheel speed, estimated angular rates, measured Y-rate and estimated pitch angle is logged at a 1s period. After the 2-minute delay period, the wheel(s) is commanded to stop (open-loop speed wheel momentum command), and the Y-Thomson controller is activated again to maintain the Y-Thomson spin state.

Table 30: Y-wheel ramp-up test.

PREREQUISITES	Y-Thomson spin state held for at least 1 orbit prior to this, gyro EKF estimation using magnetometer measurements enabled	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn



PREREQUISITES	Y-Thomson spin state held for at least 1 orbit prior to this, gyro EKF estimation using magnetometer measurements enabled	
		Rwl<X>Power to PowerOn (for wheel that corresponds to Y_B aligned wheel, or all 4 wheels for pyramid)
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf
	ConMode (Control mode)	ControlMode set to ConHxyzRW
	OpenLoopCommandHxyzRW (Open-loop wheel momentum command)	$H_y = -I_{yy}\omega_{y-ref}$
	Delay	2 mins
	OpenLoopCommandHxyzRW (Open-loop wheel momentum command)	$H_y = 0$
	ConMode (Control mode)	ControlMode set to ConYspin
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	1s
	TlmSensorRawRwl (Raw RWL sensor telemetry)	1s
	TlmSensorCalRwl (Calibrated RWL sensor telemetry)	1s
DURATION	1 pass	
TEST CONDITION	- Observe the satellite estimated angular Y-rate go to zero when the wheel(s) ramps up and go back to reference spin rate when wheel is stopped. - Observe pitch angle change to near-constant (or slowly varying) when wheel is spinning.	

The outcome of the wheel ramp-up test should be to check that the wheel (or wheel pyramid) achieves the commanded wheel speed (or commanded momentum). As a result, the satellite should stop spinning about the Y-axis, since all the angular momentum will be absorbed by the spinning wheel(s). When the wheel stops spinning, the momentum is transferred back to the satellite, and it will continue in a Y-Thomson motion.

If the satellite's Y-axis rate increased during the test rather than decreasing, the polarity of the Y-wheel is incorrect and must be adjusted by updating the ADCS configuration. If the angular rate of the other axes increased, the incorrect wheel was identified, or the mounting configuration is incorrect.

13.2 Y-momentum Control

Since Y-momentum mode requires knowledge of the pitch angle, one of the EKF estimation modes, [EstFullEkf](#) or [EstGyroEkf](#), should be used. It will be necessary to set the orbit parameters to recent TLEs provided by NORAD on www.space-track.org or www.celestrak.com. For accurate satellite position and velocity prediction it is also important that the on-board time is synchronized to current Unix time. (See section 8 on commissioning of the EKF.)



As in the previous step, the Y_B axis aligned reaction wheel (or all 4 wheels in the case of a pyramid configuration) must be enabled. Y-momentum mode is entered by changing the control mode to ConYwheelInit.

The control mode will change automatically to the steady-state Y-momentum mode (ConYwheel) when the body Y-momentum has been transferred to the wheel and the pitch angle becomes less than 50° from the nadir direction.

Table 31: Y-wheel ramp-up test.

PREREQUISITES	Y-Thomson spin state held for at least 1 orbit prior to this, gyro-EKF estimation using magnetometer measurements enabled	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn. Mag0Power or Mag1Power to PowerOn Rwl<X>Power to PowerOn (for wheel that corresponds to Y_B aligned wheel, or all 4 wheels for pyramid)
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf
	ConMode (Control mode)	ControlMode set to ConYwheelInit
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmSensorRawRwl (Raw RWL sensor telemetry)	10s
	TlmSensorCalRwl (Calibrated RWL sensor telemetry)	10s
DURATION	As needed	
TEST CONDITION	- Pitch angle is controlled to zero. - Angular rates are controlled to zero. - Wheel momentum stays around reference value	

One such activation from Y-Thomson spin at -2 deg/s to Y-Momentum wheel control for nSight-1 (a 2U CubeSat in the QB50 ISS orbit) is shown in the following plots (Figure 19 to Figure 21). The results contain the Y-momentum control mode performance for about 10 orbits. The maximum RPY attitude errors were about ± 10 deg.

From the last plot in Figure 21, it can be seen that the wheel speed did not remain constant (or near-constant) around the reference value as it was intended. This is because of an external disturbance torque that causes the satellite to rotate away from nominal zero roll, pitch and yaw attitude. The wheel controller



counteracts the disturbance and keeps the satellite stable, but it results in momentum build-up on the wheel.

In this case, the performance can be improved by increasing the gain for the magnetic momentum management controller (see next sub-section). Although these results were obtained with the Full State EKF using only magnetometer measurements, it could still be improved by recalibration of the magnetometer and adding more accurate attitude sensors, e.g. Sun & horizon sensors and Star Trackers.

It is likely that it will be necessary to adjust gain values and estimation noise covariance parameters, which is why logging is essential at this stage. Gain values, estimation modes, and estimation settings can be adjusted based on observation, but simulation testing is also strongly advised.

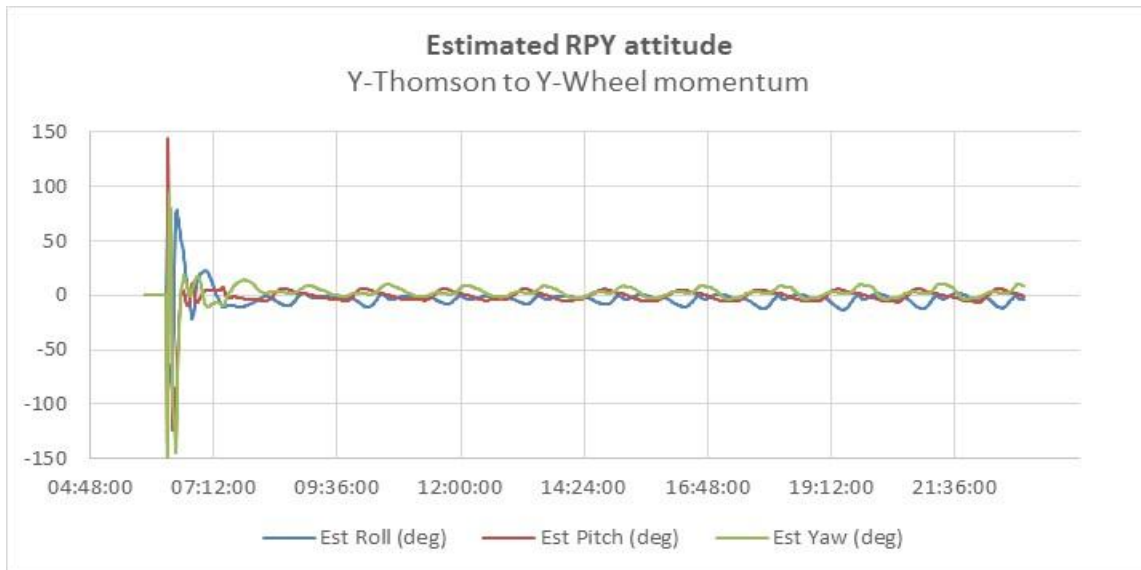


Figure 19: Estimated RPY angles after Y-momentum activation.

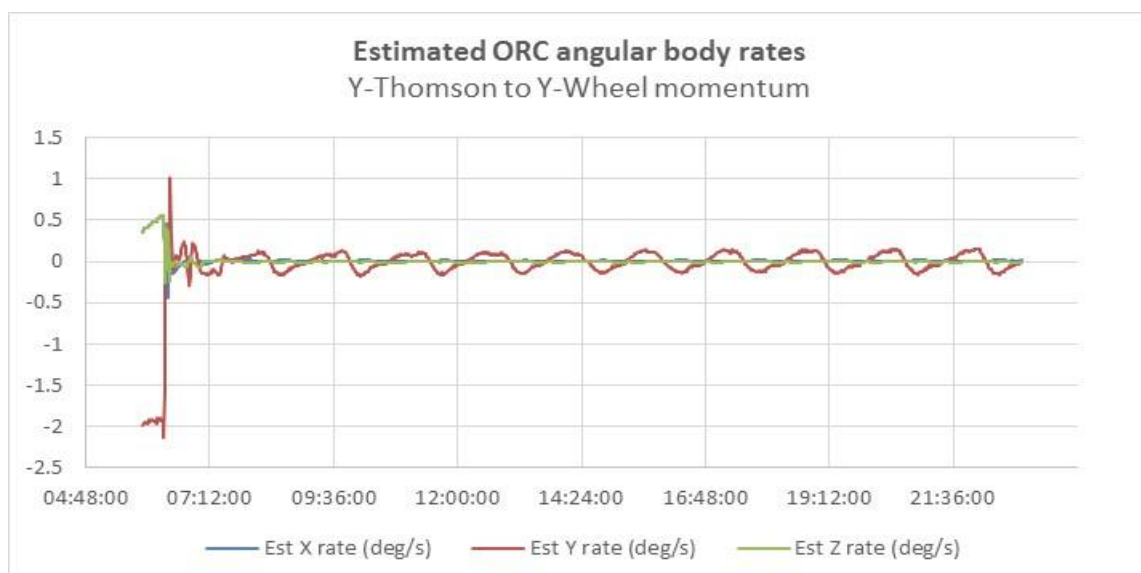


Figure 20: Estimated Body angular rates after Y-momentum activation.

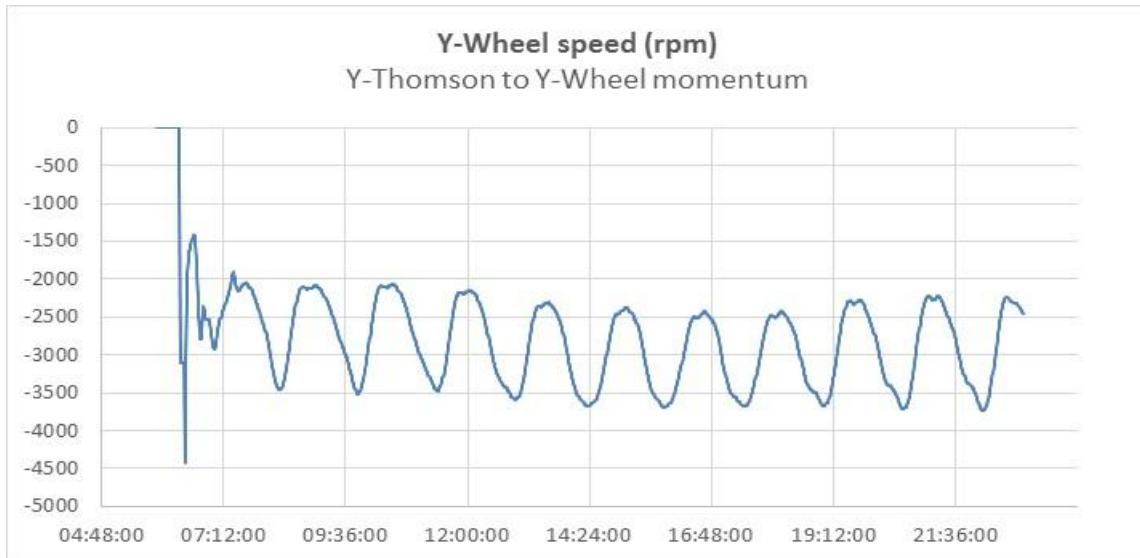


Figure 21: Measured Y-Wheel speed after Y-momentum activation.

13.3 Adjustments for Y-momentum mode

The following adjustments can be made in the ADCS configuration to improve on the Y-momentum performance.

13.3.1 Estimation modes & other sensors

Estimation accuracy can be improved by:

- Incorporating other sensor measurements in the EKF
- Re-calibrating the magnetometer

Magnetometer calibration was described in Chapter 8

It is also possible to adjust the system noise and measurement noise covariance used by the EKF to further improve convergence.

13.3.2 Y-Wheel controller gains

All controller gains, including the wheel proportional and derivative gains, are determined by CubeSpace based on the information provided in [8]. These gains are verified through simulation. It may however be that due to different dynamic situations, such as the presence of high aerodynamic disturbance torque or disturbances that could not be modelled in simulation, that different gains have to be used. In such a case, the gains are adjusted by sending the [ConfigAdcsController](#) telecommand.

13.3.3 Y-Wheel reference momentum

If there are large external disturbance torques, the user can consider increasing the reference wheel momentum to provide more gyroscopic stiffness and lessen the effect of disturbance torques.

13.3.4 Y-Momentum management gain

If Y-momentum build-up is observed, increasing the magnetic momentum dumping gain can help reduce the build-up. It may also be possible to fly at a different non-zero pitch angle to reduce the Y-axis aerodynamic disturbance torque. A realistic simulation test with full aerodynamic modelling may give an indication if such an angle exists.



14. 3-Axis Reaction Wheel Commissioning

The reaction wheel commissioning step should be undertaken with the satellite experiencing low angular rates, ideally with known attitude profile such as with a Y-Thomson or Z-Thomson spin.

The commissioning procedure involves actuating the wheels individually, or per-axis, and observing the expected changes to gyro-measured angular rates.

The procedure for a 3-axis orthogonal wheel is similar to that of a reaction wheel pyramid with 4 wheels, however the pyramid requires an initial check to confirm mounting configuration for the wheels.

14.1 Initial pyramid reaction wheel commissioning

Prior to attempting to rotate about an axis, the pyramid is first commissioned by ramping all wheels to their bias values, and confirming that the angular rates of the satellite is not disturbed by this. If the pyramid is configured correctly, there will be zero nett angular momentum transfer as all four wheels are ramped to the same speed.

To begin, the satellite must be rotating with low angular rates with Gyro or Gyro EKF estimation (Mode 1 or 6). All four reaction wheels (RWL0, RWL1, RWL2 and RWL3) should be enabled using the [PowerState](#) command.

The control mode is set to [ConHxyzRW](#), which will cause the wheels to spin up to their configured bias level. This is carried out for 1 or 2 minutes after which the control mode is changed to [ConStopRW](#) – to let all wheel speeds go to zero, before resuming the previous control mode (spinning control mode, or detumbling).

It is also possible to transition to testing the rest of the axis rotations directly, in which case the control mode can remain as [ConHxyzRW](#).

Table 32: Pyramid bias ramp-up test.

PREREQUISITES	Low angular rates (< 1 deg/s) and preferably a stable spinning motion such as Y-Thomson or Z-Thomson	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	if not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn Rwl<X>Power to PowerOn (for wheel that corresponds to Y _B aligned wheel, or all 4 wheels for pyramid)
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf or EkfGyro
	ConMode (Control mode)	ControlMode set to ConHxyzRW
	Delay	2 mins
	ConMode (Control mode)	ControlMode set to ConStopRW
	Delay	1 min
	ConMode (Control mode)	ControlMode restored to previous control mode
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	1s



PREREQUISITES	Low angular rates (< 1 deg/s) and preferably a stable spinning motion such as Y-Thomson or Z-Thomson	
	TlmSensorRawRwl (Raw RWL sensor telemetry)	1s
	TlmSensorCalRwl (Calibrated RWL sensor telemetry)	1s
DURATION	1 pass	
TEST CONDITION	- Observe wheel speeds increase with two wheels reaching positive bias values and two wheels reaching negative bias values. - Observe little to no change in the satellite estimated angular rate	

The expected behaviour is illustrated using telemetry plots in Figure 22 and Figure 23. At 35s into the plot, the **ConHxyzRW** control mode was enabled causing the wheels to spin up to their bias values. The traces for RWL0 and RWL2 overlap in the plot, as well as RWL1 and RWL3.

The angular rate plot in Figure 23 shows now change to the gyro measured angular rates as a result, indicating correct pyramid configuration.

At 170s an open-loop momentum command of $H_z = 0.01$ Nms was commanded (see next section). The reaction wheel speeds change, and as a result the satellite angular rate Z-component decreases. In this case the satellite was performing a spinning motion and due to the spin dynamics the Y-axis rate slightly increase positive as well.

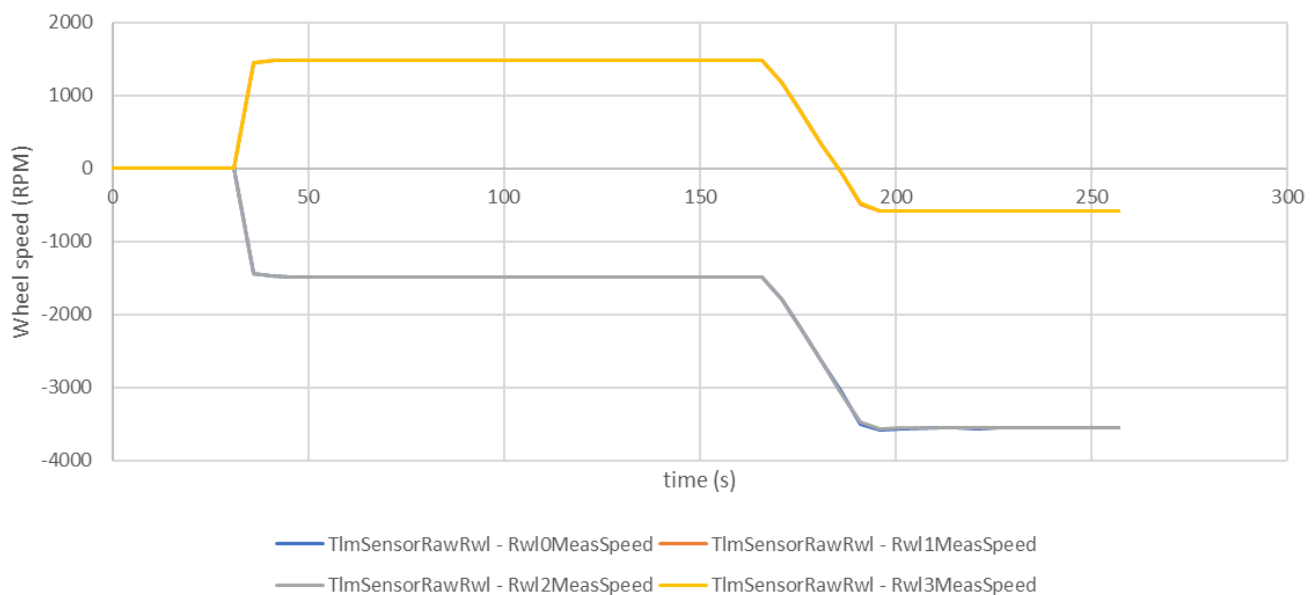


Figure 22: Pyramid reaction wheel commissioning - wheel speeds

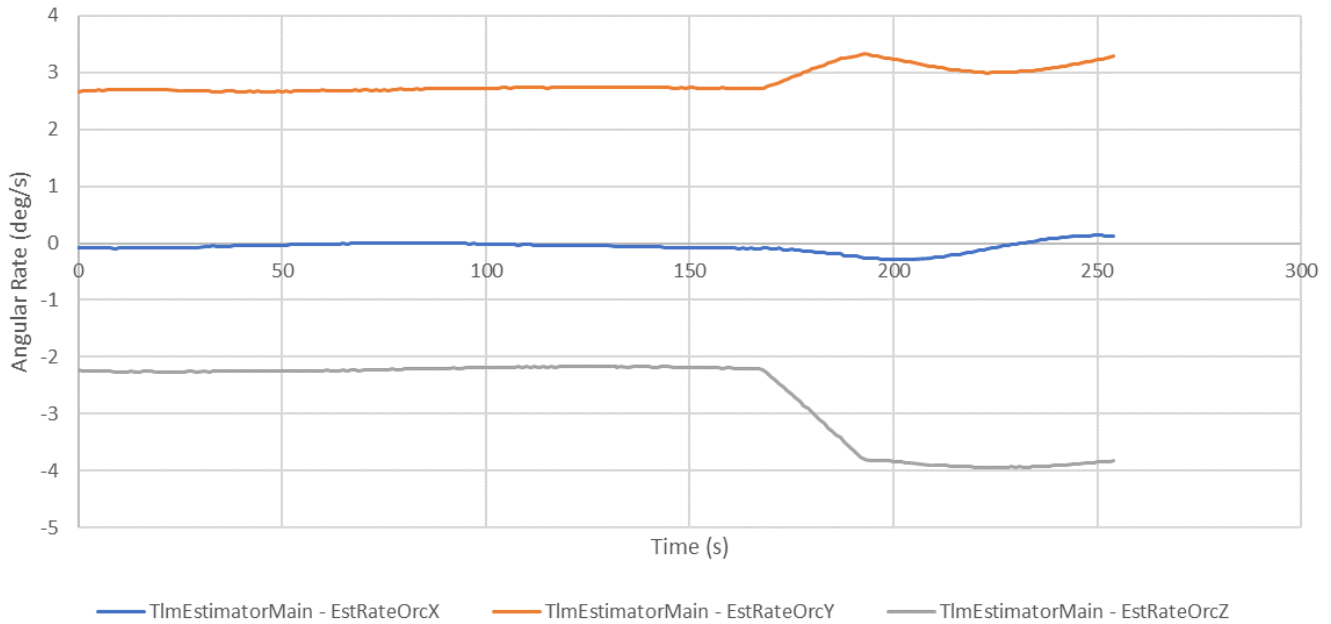


Figure 23: Pyramid reaction wheel commissioning - angular rates

14.2 Three-axis orthogonal wheel commissioning

The purpose of this part of the wheel commissioning step is to determine that the wheel configuration result in the expected angular rotation of the satellite.

To begin, the satellite must be rotating with low angular rates with Gyro or Gyro EKF estimation (Mode 1 or 6). All the reaction wheels (RWL0, RWL1 and RWL2) should be enabled using the [PowerState](#) command.

The ADCS must not perform active control using the wheels, and this is achieved by setting the control mode to [ConHxyzRW](#). For pyramid commissioning, it is possible to transition to this step directly from previous step.

The wheel or pyramid will now be commanded to a constant momentum value:

$$h_{x/y/z} = -I_{xx/yy/zz}\omega_{test}$$

where $I_{xx/yy/zz}$ is the satellite moment of inertia about the particular body axis (passing through the center of mass) in kg.m^2 units, ω_{test} is the expected change in spin rate in that should be observed.

The wheel, or pyramid, is then allowed to spin at constant speed for a period of 2 minutes. During this time the measured wheel speed, estimated angular rates is logged at a 1s period. After the 2-minute delay period, the wheel(s) is commanded to stop (control mode set to [ConStopRW](#)), and the previously active controller is activated again to maintain the current state (spin control or detumbling).

Table 33: X-wheel polarity test.

PREREQUISITES	Satellite must be in a stable y-momentum mode with low angular rates in the body axes as estimated and measured by the mems rate sensors	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn



PREREQUISITES	Satellite must be in a stable y-momentum mode with low angular rates in the body axes as estimated and measured by the mems rate sensors	
		Rwl<X>Power to PowerOn (for wheel that corresponds to Y _B aligned wheel, or all 4 wheels for pyramid)
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf or EstGyro
	ConMode (Control mode)	ControlMode set to ConHxyzRW
	OpenLoopCommandHxyzRW (Open-loop wheel momentum command)	$H_{x/y/z} = -I_{xx/yy/zz}\omega_{test}$
	Delay	2 mins
	OpenLoopCommandHxyzRW (Open-loop wheel momentum command)	$H_{x/y/z} = 0$
	ConMode (Control mode)	ControlMode restored to previous control mode
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	1s
	TlmSensorRawRwl (Raw RWL sensor telemetry)	1s
	TlmSensorCalRwl (Calibrated RWL sensor telemetry)	1s
DURATION	1 Pass	
TEST CONDITION	- Observe measured wheel speed match the expected speed (equivalent from open-loop command). - Observe the satellite corresponding measured angular rate changes inverse to the H command that was given (positive H command results in decrease in angular rate for that axis).	

An example of a momentum ramp for a 3-axis orthogonal wheel configuration is shown below in Figure 24 to Figure 26. An X-axis momentum is commanded to result in the X-axis aligned wheel is spinning up as in Figure 25. The effect in angular rates can be seen in Figure 24. In Figure 26 the same is shown for a Z-axis momentum command.

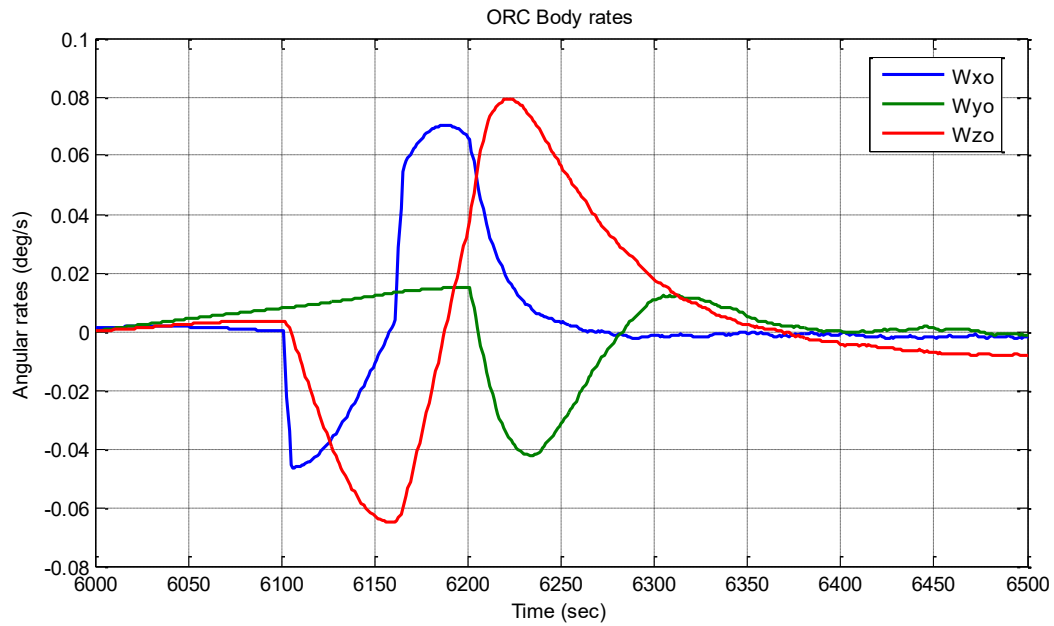


Figure 24: Orbit referenced body angular rate disturbance with X-axis wheel +200 rpm pulse disturbance.

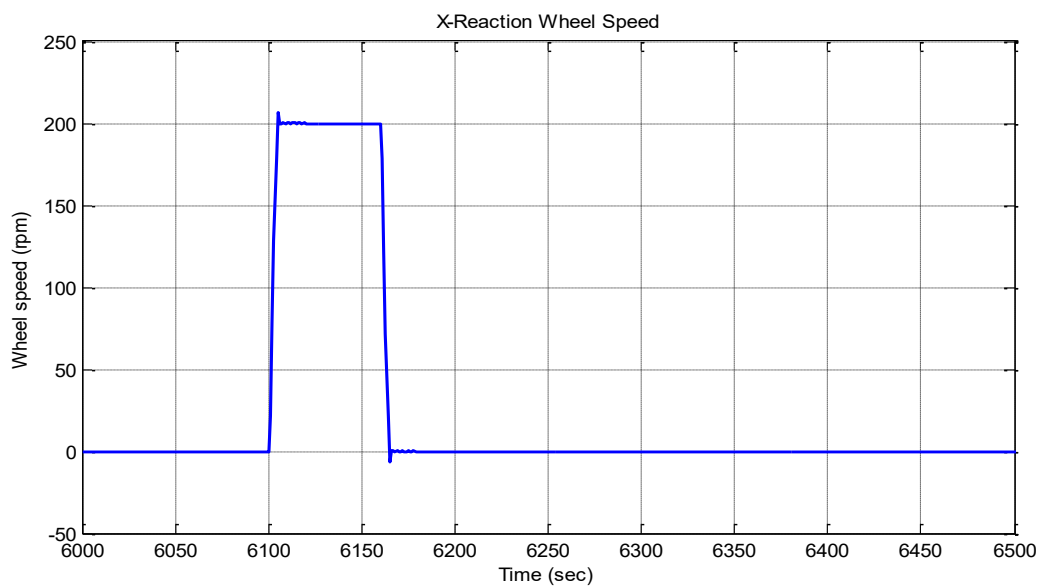


Figure 25: X-axis reaction wheel speed reference pulse.

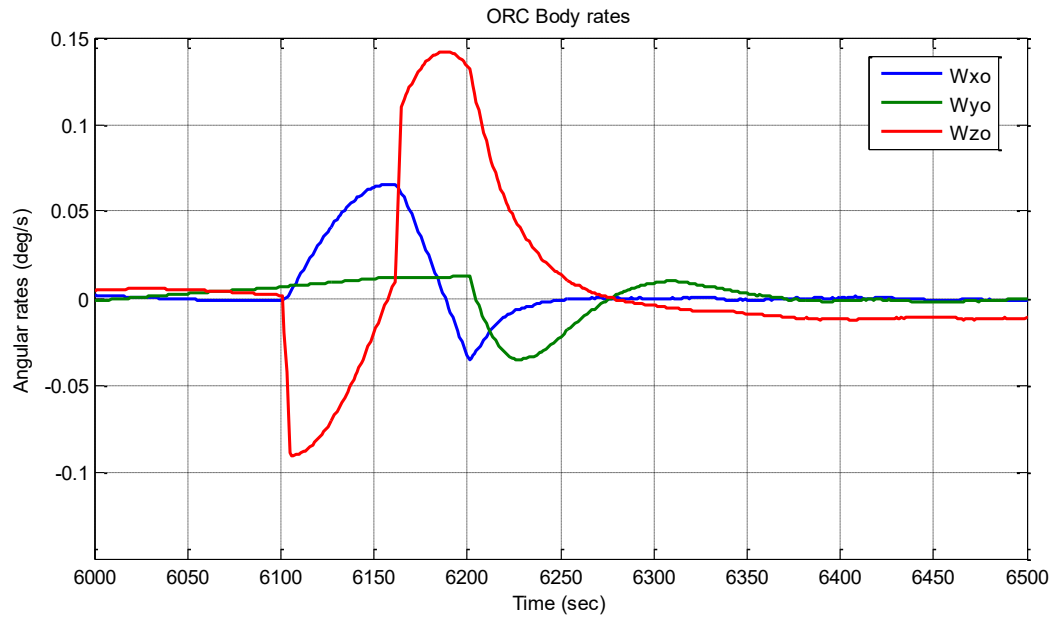


Figure 26: Orbit referenced body angular rate disturbance with Z-axis wheel +200 rpm pulse disturbance.



15. 3-axis Reaction Wheel Control

After reaction wheel configuration and operation has been verified in the previous commissioning step in chapter 14, the 3-axis RW control modes can be used.

The transition to 3-axis wheel control mode can take place from low angular rate initial conditions, such as with the [ConDetumble](#) control mode. It is however possible in this situation for attitude sensors to have invalid measurements due to them pointing in non-optimal directions. It is therefore better to transition to 3-axis control from one of the spinning control modes, or Y-momentum control mode.

These modes must be used with estimation mode set to either [EstFullEkf](#) or [EstGyroEkf](#) (mode 5 or 6). The [EstGyroEkf](#) mode is preferred since it results in better attitude estimates in the presence of unmodelled disturbances.

The Gyro EKF will continuously estimate any bias drift and the satellite's attitude quaternion using the magnetometer and FSS. If star tracker measurements and/or HSS are available and already commissioned (see next chapters), the star vector measurements and/or horizon angles can be utilized throughout the orbit for more accurate estimates in both the Full-state EKF and the Gyro EKF.

Both full-state EKF and gyro EKF estimation modes require knowledge of the satellite position and orbit. It is required to set orbit parameters in the CubeADCS configuration from up to date TLEs. GNSS may also be used, but because GNSS data may become unavailable it is important that orbit parameters are also kept up to date. Finally, the CubeADCS Unix time must be synchronized to current time.

As before, the procedure for pyramid RW configurations require an intermediate sub-step to [ConHxyzRW](#) control mode first. This is necessary for the wheels in the pyramid to first reach their bias values, without destabilizing the satellite attitude. The control mode can be changed to [ConXYZwheel](#) after the wheels have ramped to their bias speed.

See Table 34 for the typical initial commissioning steps of the zero net bias 3-axis reaction wheel controller.

Table 34: XYZ reaction wheel control commissioning.

PREREQUISITES	Satellite must have low angular rates, but preferably be in a stable spinning mode or already in Y-axis momentum wheel control mode. Orbit parameters must be set from TLEs, and Unix time synchronised.	
Command sequence	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn Rwl<X>Power to PowerOn (for wheel that corresponds to Y_B aligned wheel, or all 4 wheels for pyramid) FSS<X>Power to PowerOn (for FSS that have been commissioned)
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf or EstFullEkf
	Pyramid only: ConMode (Control mode)	ControlMode set to ConHxyzRW
	Pyramid only: Delay	As necessary, for wheels to ramp to bias level (determined by PyrHnom and NDel configuration settings)
	ConMode (Control mode)	ControlMode set to ConXYZwheel
Telemetry logging	TELEMETRY FRAME	PERIOD



PREREQUISITES	Satellite must have low angular rates, but preferably be in a stable spinning mode or already in Y-axis momentum wheel control mode. Orbit parameters must be set from TLEs, and Unix time synchronised.	
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	1s
	TlmSensorRawRwl (Raw RWL sensor telemetry)	1s
	TlmSensorCalRwl (Calibrated RWL sensor telemetry)	1s
Duration	1 pass	
Test condition	- Observe angular rates are controlled to zero - Observe attitude angles are controlled close to their reference values - Observe that reaction wheel momentum build-up does not occur	

The next six figures (Figure 27 to Figure 31) show the simulation result of a 3U CubeSat in a 550 km Sun-sync LTDN 10h30 orbit. During the first orbit the satellite started at $t = 100$ sec in a Y-Momentum wheel control mode from an initial RPY attitude = $[0^\circ, 2^\circ, -2^\circ]$. A Gyro EKF was used to estimate the attitude from magnetometer- and CubeSense Sun measurements starting at $t = 0$ sec. During the second orbit at $t = 6000$ sec, the zero bias XYZ reaction wheel controller was enabled.

Figure 27 displays the true satellite's attitude and it can be seen that during the eclipse period (approx. 1900-3900 sec and 7500-9500 sec) the attitude estimation error increases due to magnetometer only measurements. In Y-Momentum wheel control mode the attitude errors stay below 2° and in XYZ reaction wheel control mode the attitude errors stay below 0.5° . These errors can improve significantly with star tracker measurements added to the EKF in eclipse. Figure 28 illustrates the true ORC referenced body rates. Figure 29 displays the XYZ-Wheel speeds; it is clear that the Y-Wheel momentum bias is dumped quickly once the zero bias XYZ reaction wheel controller is enabled with magnetic control momentum dumping.

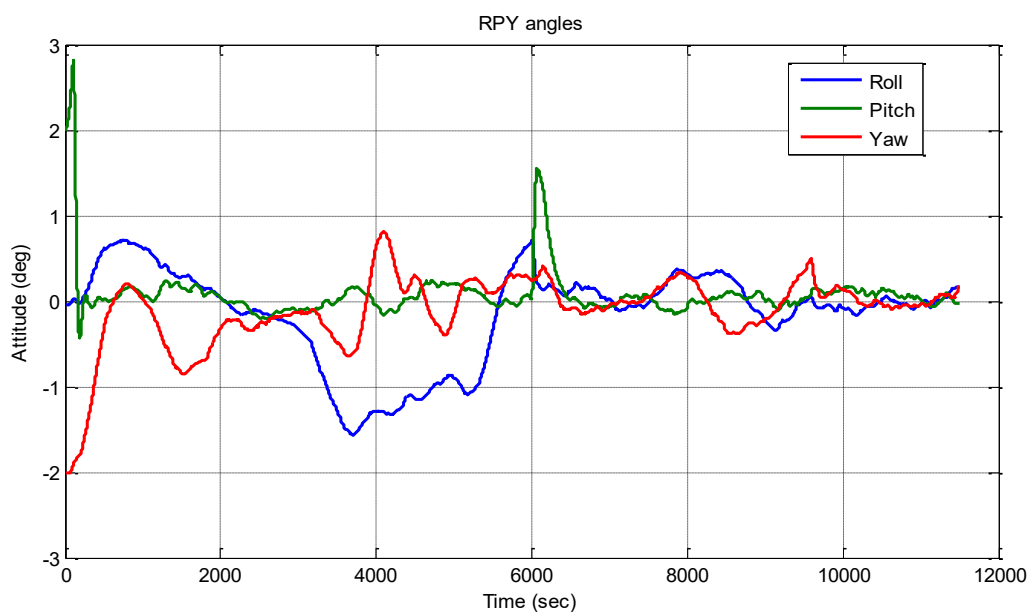


Figure 27: True RPY attitude from Y-Momentum to XYZ reaction wheel control.

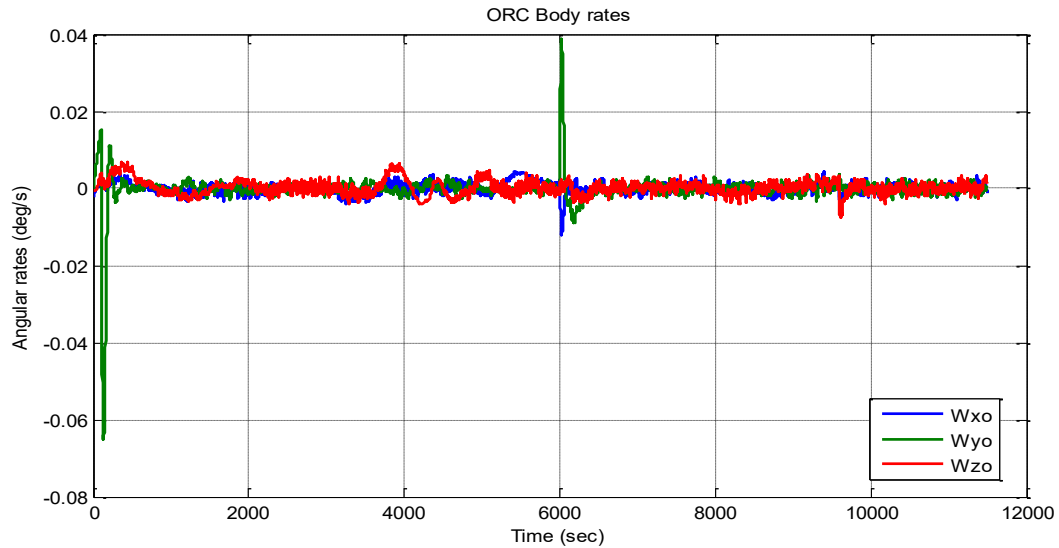


Figure 28: True ORC body rates from Y-Momentum to XYZ reaction wheel control.

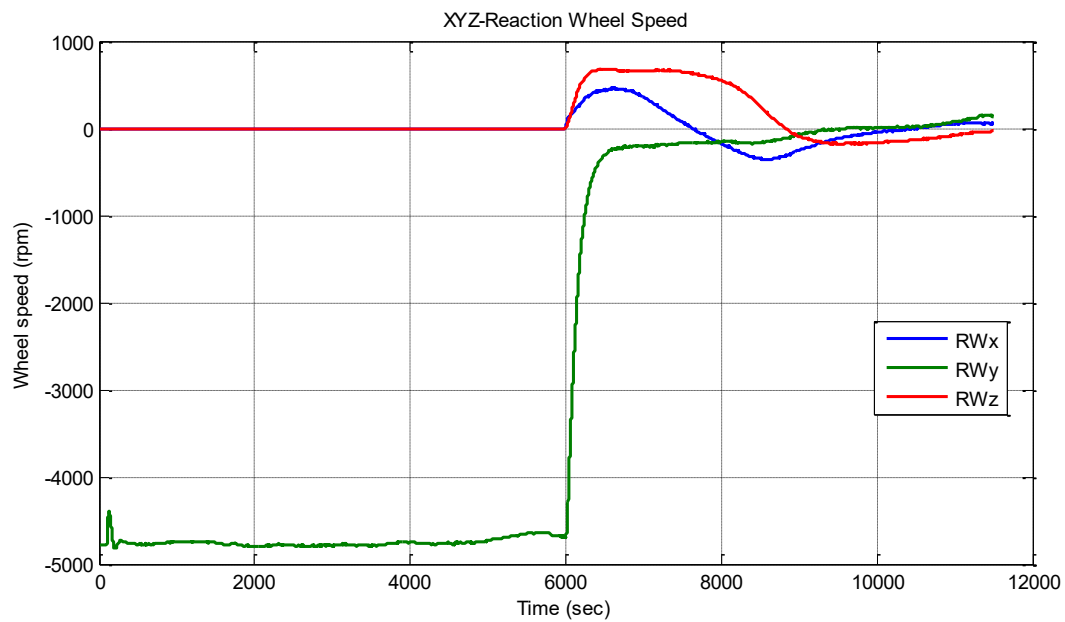


Figure 29: Wheel speeds from Y-Momentum to XYZ reaction wheel control.

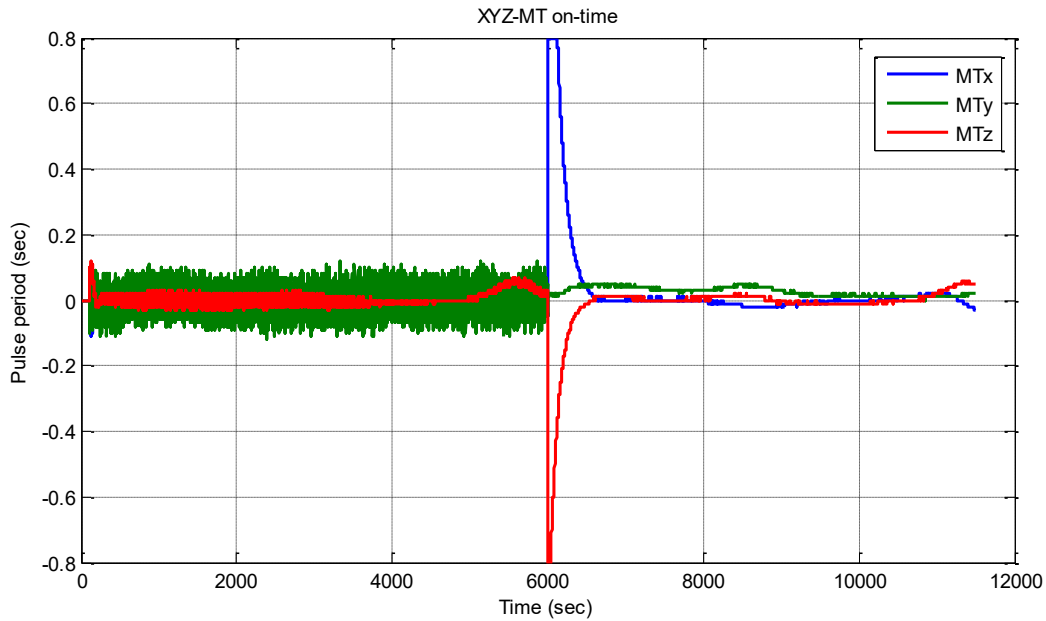


Figure 30: Magnetorquer on-time from Y-Momentum to XYZ reaction wheel control.

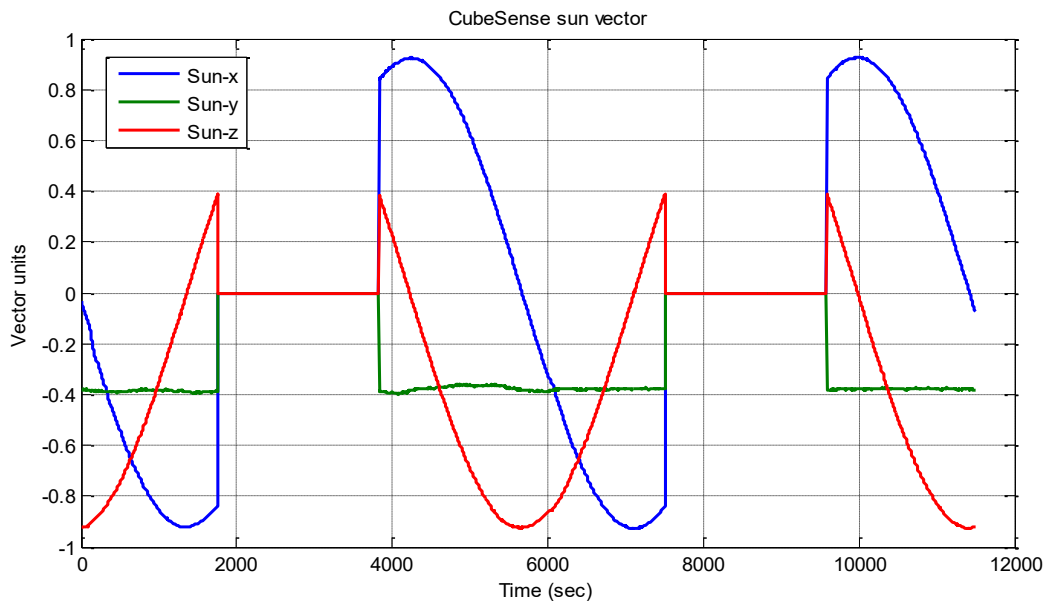


Figure 31: CubeSense Sun vector measurement from Y-Momentum to XYZ reaction wheel control.

To reduce the attitude drift in eclipse when magnetometer only measurements are used, due to attitude unobservability in the B-field vector direction (i.e. when star tracker measurements are not available), an angular momentum bias can be commanded onto the reaction wheels normal to the orbit plane. This bias may be set with the [H_bias](#) configuration parameter. The bias is a negative value in Nms units of typically about 30% of the saturation angular momentum of the reaction wheels, e.g. for CubeWheel CW0017 with $H_{sat} = 1.7$ milli-Nms, set $H_{bias} = -0.0005$ Nms.

This setting should only be used for three-axis orthogonal wheel configurations.

It is further possible to reduce attitude drift in eclipse by including sensors that can produce measurements in eclipse, such as the HSS (CubeSense Earth) or star trackers.



16. HSS (CubeSense Earth) Commissioning

The HSS is commissioned by obtaining a log of measurements from the sensor over an orbit, and correlating measurements with expected horizon profiles. If the satellite attitude is kept nadir pointing using 3-axis RW control, the measured detection angles for the CubeSense Earth should remain close to constant, and the calibrated measurement (nadir vector) should be close to $[0, 0, 1]^T$.

Throughout this process it might be necessary to download a photo that was captured using the CubeSense Earth sensor. This is necessary if the detection parameters have to be adjusted for better performance.

While the HSS are still being commissioned, they should be excluded from use in the EKF by setting the [EkfUseHss](#) flag to false.

The sequence of commands necessary to generate a log of HSS measurements is shown in Table 35.

Table 35: HSS commissioning

PREREQUISITES	Satellite in nadir pointing attitude	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn Hss<X>Power to PowerOn (for all HSS to be commissioned) Reaction wheels enabled Other sensors enabled as needed
	ConfigOrbitSatParams (Satellite orbit parameter configuration)	SGP4 orbital elements
	UnixTime (Unix time)	Current Unix time
	ConfigAdcsEstimator (ADCS estimator configuration)	EkfUseHss flag set to false. EkfUseStr and EkfUseFss flags set as needed
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf or EstFullEkf
	ConMode (Control mode)	ControlMode set to ConXYXwheel
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TlmHssCubeSenseEarthRaw (Raw CubeSense Earth telemetry)	10s
	TlmSensorCalHss (Calibrated HSS sensor telemetry)	10s
DURATION	1 to 2 orbits	
TEST CONDITION	TlmHssCubeSenseEarthRaw.Hss<X>CaptureResult Shows EarthCapSuccess. TlmHssCubeSenseEarthRaw.Hss<X>DetectionResult shows EarthSuccess when sun measurements are expected in the sensor FOV. (for nadir pointing satellite HSS measurements should be continuously available) TlmHssCubeSenseEarthRaw.Hss<x>RawElev shows constant values, close to zero. TlmHssCubeSenseEarthRaw.Hss<x>RawRot shows constant values.	



16.1 HSS troubleshooting

The CubeSense Earth has various parameters that control the behaviour of the detection algorithm. If the sensor produces detection errors but otherwise operates as normal, it is likely that the configuration must be adjusted. The ideal parameters for detection can only be learned from images that are captured using the sensor and downloaded to ground.

HSS images have low resolution, and thus should not pose any challenges for the data download.

An example of a CubeSense Earth photo that was captured in flight is shown in Figure 32. Lighter pixels in the image correspond to warmer temperatures.

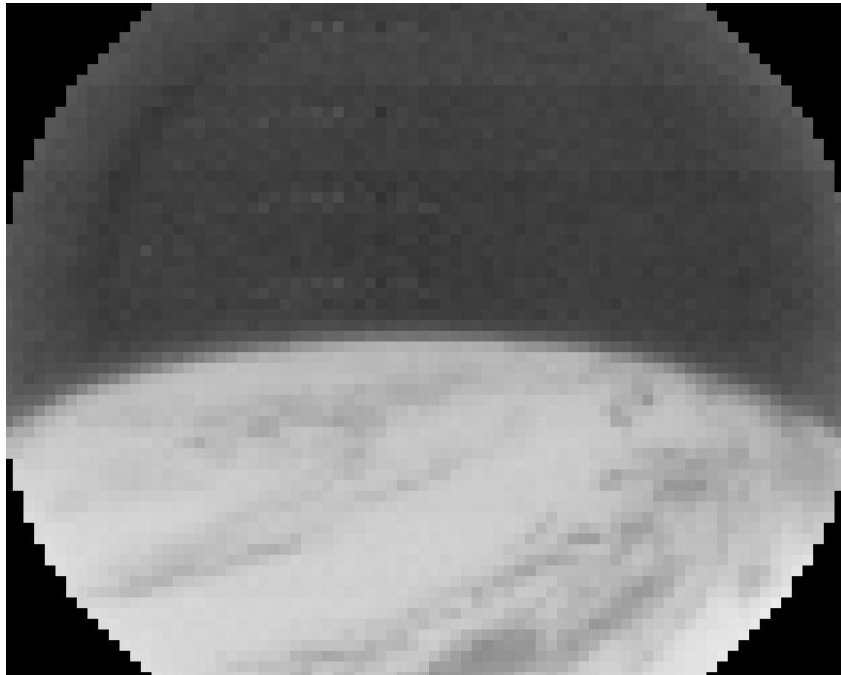


Figure 32: Example CubeSense Earth image

It may be required to adjust the following HSS configuration parameters based on the information present in the image:

- **MinimumGradient**: Minimum gradient to be considered a horizon,
- **MaxSpaceTemp**: Temperature used to calculate histogram scan boundaries for determining temperature thresholds,
- **MaxEarthTempOffset**: Offset from 'maxSpaceTemp' to determine 'maxEarthTemp',
- **MinSpaceTempOffset**: Offset from 'maxSpaceTemp' to determine 'minSpaceTemp'.

CubeSpace can be consulted where there is uncertainty in what values to use for the above, in order to achieve successful detections.

The above parameters are part of the HSS API (and not the CubeComputer control program) and must be set by changing the configuration of the HSS directly. In an integrated CubeADCS this can be achieved using pass-through telecommands. Reference [5] contains detail on how to make use of pass-through telecommands and telemetry.



17. Star Tracker Commissioning

The star tracker(s) is commissioned by obtaining a log of measurements from the sensor over an orbit and analysing measurements on the ground.

The type of star tracker that is employed influences the commissioning verification, although the sampling procedure is the same for both CubeStar and the TY-Space PST3S star tracker.

The star tracker requires that the satellite angular rates are low (< 1 deg/s for the TY-Space star tracker, and below 0.3 deg/s for the CubeStar). The attitude must be nadir pointing to prevent the Earth from appearing in the star tracker FOV.

As with other sensors, while the star trackers are still being commissioned, they should be excluded from use in the EKF by setting the [EkfUseStr](#) flag to false.

The procedure to download a log of star tracker measurements is shown in Table 36.

Table 36: Star tracker commissioning

PREREQUISITES	Satellite in nadir pointing attitude	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn Str<X>Power to PowerOn (for all STR to be commissioned) Reaction wheels enabled Other sensors enabled as needed
	ConfigOrbitSatParams (Satellite orbit parameter configuration)	SGP4 orbital elements
	UnixTime (Unix time)	Current Unix time
	ConfigAdcsEstimator (ADCS estimator configuration)	EkfUseStr flag set to false. EkfUseHss and EkfUseFss flags set as needed
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf or EstFullEkf
	ConMode (Control mode)	ControlMode set to ConXYXwheel
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	10s
	TYSpace: TlmStrPst3sRaw (Raw PST3S star tracker telemetry)	10s
	CubeStar: TlmStrCubeStarRaw (Raw CubeStar telemetry)	10s
	TlmSensorCalStr (Calibrated STR sensor telemetry)	10s
DURATION	1 to 2 orbits	
TEST CONDITION	For CubeStar:	For TY-Space PST3S:



PREREQUISITES	Satellite in nadir pointing attitude	
	• <code>TlmStrCubeStarRaw.Str<X>CaptureStatus</code> Shows <code>StarCaptureDone</code>. • <code>TlmStrCubeStarRaw.Str<X>DetectStatus</code> shows <code>StarDetectDone</code> • <code>TlmStrCubeStarRaw.Str<x>IdentStatus</code> shows <code>StarIdDone</code> • <code>TlmStrCubeStarRaw.Str<X>NumDetected</code> and <code>TlmStrCubeStarRaw.Str<X>NumIdentified</code> shows number of detected and identified stars > 4, when the sun, moon and Earth fall outside the exclusion area of the CubeStar FOV.	• <code>TlmStrPst3sRaw.Quat1 to Quat4</code> shows smooth trajectories over time • <code>TlmStrPst3sRaw.DataValid</code> shows valid measurements • <code>TlmStrPst3sRaw.IdentifiedStars</code> shows number of identified stars > 4

17.1 Expected star tracker data results

17.1.1 TY-Space PST3S

For a continuously nadir pointing satellite it is expected that the quaternion that the TY-Space star tracker outputs form smooth traces when plotted over time, as in Figure 33.

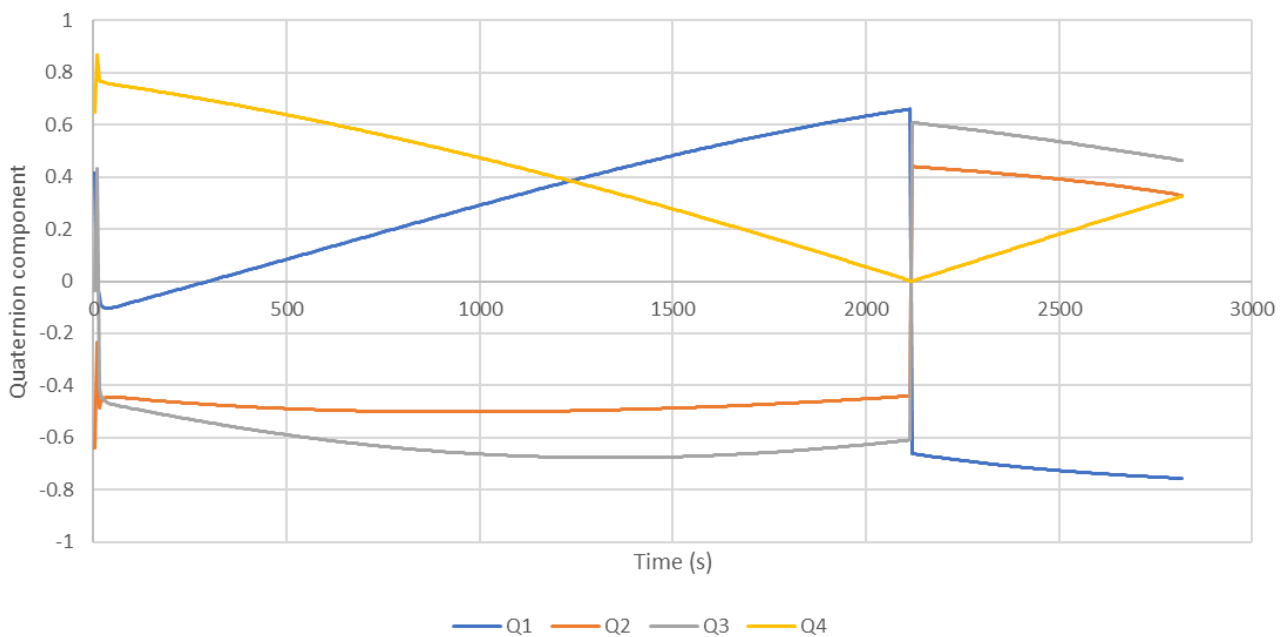


Figure 33: Star tracker quaternion output for nadir pointing satellite

The change that is observed after 2000s in the above graph is when the Q4 component of the quaternion approaches negative values. The quaternion is inverted so that Q4 is always a positive number. (This still results in the same transformation as $q = -q$ for quaternion rotations).

The PST3S telemetry should also show that enough stars were identified in order to output the measured quaternion.



17.1.2 CubeStar

The `NumDetected` field in the `TlmStrCubeStarRaw` telemetry frame will typically vary between 3 and 15 stars. The `NumIdentified` field will indicate the number of stars that were successfully identified and should have a value of 4 or more for successful identification.

The Tracking flag in the `TlmStrCubeStarRaw` telemetry frame should change to true after the first lost-in-space acquisition and then remain true until the point where the conditions to detect or identify stars are no longer conducive (sun, moon or Earth in FOV, or too high angular rates).

17.2 CubeStar troubleshooting

As with other sensors, the CubeStar also has configurable detection settings to change the behaviour of the detection algorithm.

Settings that can typically be considered when adjusting include:

- `AdTolFixed`, `AdTolR`, `AdTolRelaxed`: Angular separation tolerances when matching observed star pairs to catalogue star pairs,
- `TrackWindowSize`: Window size when searching for tracked stars,
- `AnalogGain`: Analog gain of image sensor,
- `IntTimeNumLines`: Integration time of image sensor.

Throughout the process of adjusting it might be necessary to download a photo that was captured using the CubeStar. This is necessary if the detection parameters have to be adjusted for better performance.

17.2.1 Recalibration

If identification performance does not improve, the star camera calibration may be affected by either the launch procedure or improper handling of the unit. Some further improvement might then be had if the star camera is recalibrated. In this case, it is recommended that the user captures, downloads, and sends to CubeSpace at least one star tracker-image of a populated part of the celestial sphere.

This image will be used to determine the current calibration accuracy with which further commissioning steps can be devised.

CubeStar images have a full resolution of 1280 x 1038 pixels.



Figure 34: Downloaded CubeStar image with good exposure.



18. Sun Tracking 3-Axis Control

When the XYZ reaction wheel 3-axis controller gives satisfactory performance, the Sun tracking controller can be enabled during the sunlit part of an orbit. In eclipse this control mode will automatically revert to nadir pointing.

It is possible to configure which side of the satellite must point towards the sun (for instance for solar panels mounted on different sides). This configuration setting is available in the [ConfigAdcsSatellite](#) message, by changing the [SunPointBodyVecX](#), [Y](#) and [Z](#) values.

See Table 37 for the typical initial commissioning steps of the Sun tracking 3-axis reaction wheel controller.

Table 37: Sun tracking 3-axis control commissioning.

PREREQUISITES	Satellite must be in a stable nadir pointing 3-axis reaction wheel mode with angular rates in the body axes as estimated and measured by the rate sensors and EKF. Orbit parameters must be set from TLEs, and Unix time synchronised.	
Command sequence	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn Rwl<X>Power to PowerOn (for wheel that corresponds to Y_B aligned wheel, or all 4 wheels for pyramid) FSS<X>Power to PowerOn (for FSS that have been commissioned) Other sensors powered on as needed
	EstMode (Estimation mode)	EstModeMain set to EstGyroEkf or EstFullEkf
	Pyramid only: ConMode (Control mode)	ControlMode set to ConSunTrack
Telemetry logging	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	1s
	TlmSensorRawRwl (Raw RWL sensor telemetry)	1s
	TlmSensorCalRwl (Calibrated RWL sensor telemetry)	1s
	TlmSensorCalFss (Calibrated FSS sensor telemetry)	10s
Duration	1 pass	
Test condition	- Observe FSS vector aligned with selected body vector direction - Observe the RPY attitude changes for Sun tracking - Observe reaction wheel performance	

Figure 35 illustrates the RPY attitude angles during Sun tracking in the sunlit part of an orbit, during eclipse the RPY attitude will be zero (nadir pointing). The CubeSense Sun sensor measurement with boresight and solar panel normal vector pointing to the Sun during Sun tracking is shown in Figure 36. Figure 37 displays



the XYZ reaction wheel angular momentum during Sun tracking, the large excursions at the start and end of the sunlit period of the orbit are the result of manoeuvres from nadir to Sun pointing and back.

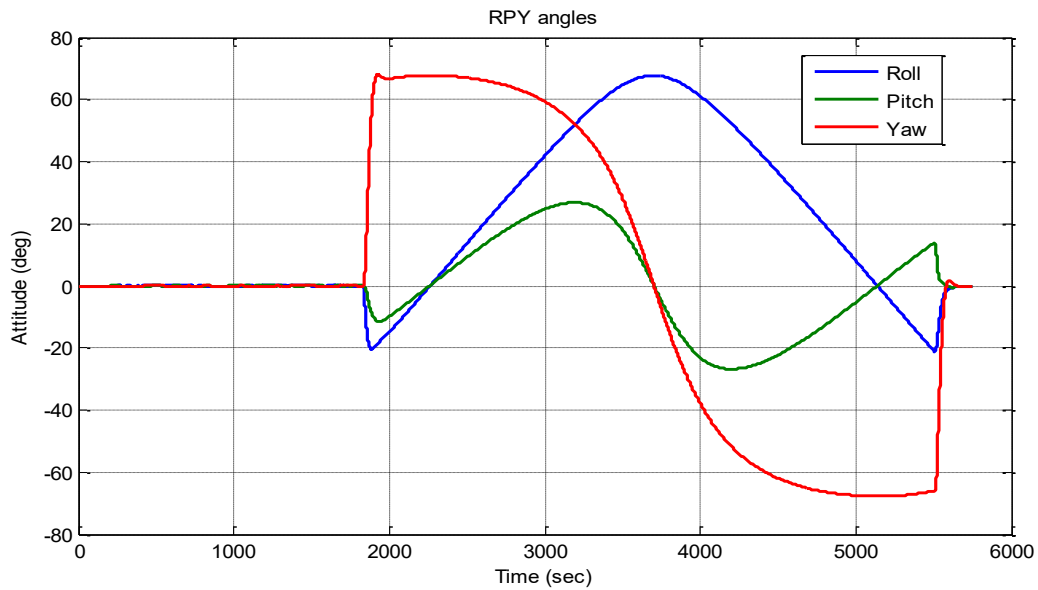


Figure 35: True RPY attitude during Sun tracking 3-axis control.

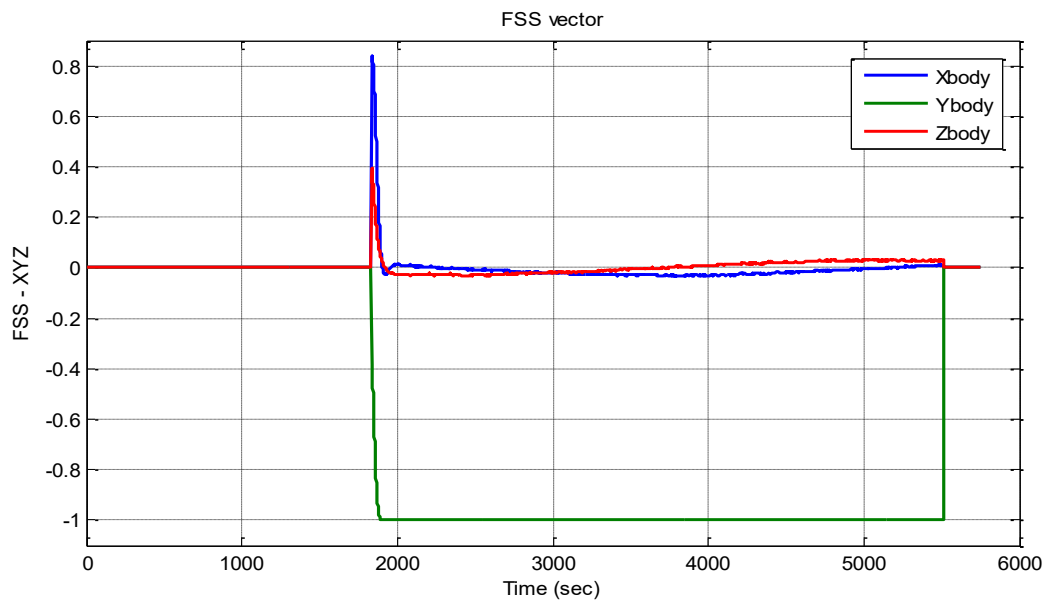


Figure 36: CubeSense Sun vector measurement during Sun tracking control in the sunlit part of an orbit.

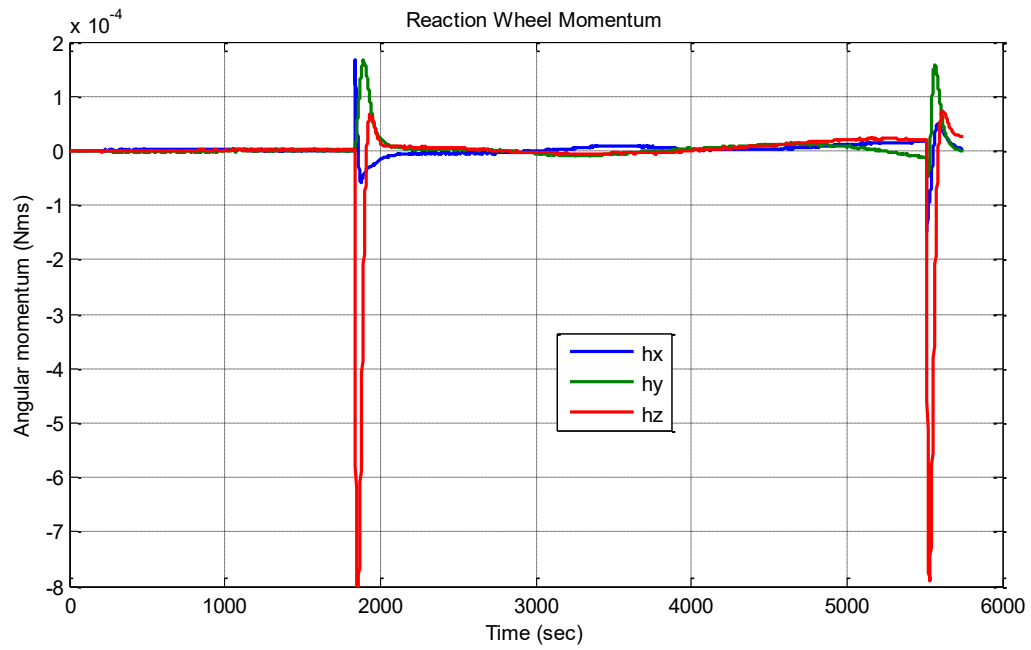


Figure 37: XYZ reaction wheel angular momentum during Sun tracking control.



19. Ground Target Tracking Controllers

When the XYZ reaction wheel 3-axis controller gives satisfactory performance, the earth observation ground tracking controller can be enabled close to a target area on the ground.

Prior to activating the target tracking controller, it is necessary to set the reference target location by specifying the [CmdTargetLatitude](#), [CmdTargetLongitude](#) and [CmdTargetAltitude](#) parameters in the [ReferenceTarget](#) telecommand message to the desired target latitude, longitude and altitude. The ground target must be specified using WGS-84 referenced latitude, longitude and altitude.

There are different control modes available to track a ground target.

- [ConTgtTrack](#): Track the reference ground location by pointing the satellite $+Z_B$ axis to the target,
- [ConTgtSteer](#): Track the reference ground location by pointing the satellite $+Z_B$ axis to the target, but with finer control over steering by using input from external sensors,
- [ConGndTrack](#): Track the reference ground location by pointing the configured SBC axis ([ConfigAdcsSatellite.TgtTrackBodyVecX/Y/Z](#)) to the target.

Figure 38 shows tracking control of a ground target during an overhead pass (at max 90° elevation). The RPY attitude errors are displayed in Figure 39. Figure 40 illustrates the XYZ reaction wheel angular momentum during the manoeuvres towards, during and back; from the target tracking control mode to nadir pointing control mode.

NOTE: In this control mode, no prior calculation is performed to determine if the specified target (latitude, longitude, altitude) is currently visible to the satellite. The attitude will be controlled to point to the target, even if the target is on the other side of the Earth.

Table 38: Target tracking 3-axis control commissioning.

PREREQUISITES	Satellite must be in a stable nadir pointing 3-axis reaction wheel mode with angular rates in the body axes as estimated and measured by the rate sensors and EKF. Orbit parameters must be set from TLEs, and Unix time synchronised.	
Command sequence	COMMAND	PARAMETERS
	PowerState	If not already enabled, Gyr0Power set to PowerOn . Mag0Power or Mag1Power to PowerOn Rwl<X>Power to PowerOn (for wheel that corresponds to Y_B aligned wheel, or all 4 wheels for pyramid) Various sensors powered on as needed
	EstMode (Estimation mode)	EstModeMain set to EstGyroEKF or EstFullEKF
	ConMode (Control mode)	ControlMode set to ConTgtTrack or ConGndTrack
Telemetry logging	TELEMETRY FRAME	PERIOD
	TlmEstimatorMainHighRes (Main estimator high-resolution telemetry)	1s
	TlmSensorRawRwl (Raw RWL sensor telemetry)	1s
	TlmSensorCalRwl (Calibrated RWL sensor telemetry)	1s



PREREQUISITES	Satellite must be in a stable nadir pointing 3-axis reaction wheel mode with angular rates in the body axes as estimated and measured by the rate sensors and EKF. Orbit parameters must be set from TLEs, and Unix time synchronised.	
	TlmModels (Models telemetry)	10s
Duration	1 pass	
Test condition	- Observe the RPY attitude changes for target tracking - Observe reaction wheel performance - Observe the satellite Lat/Lon/Alt position near the target area	

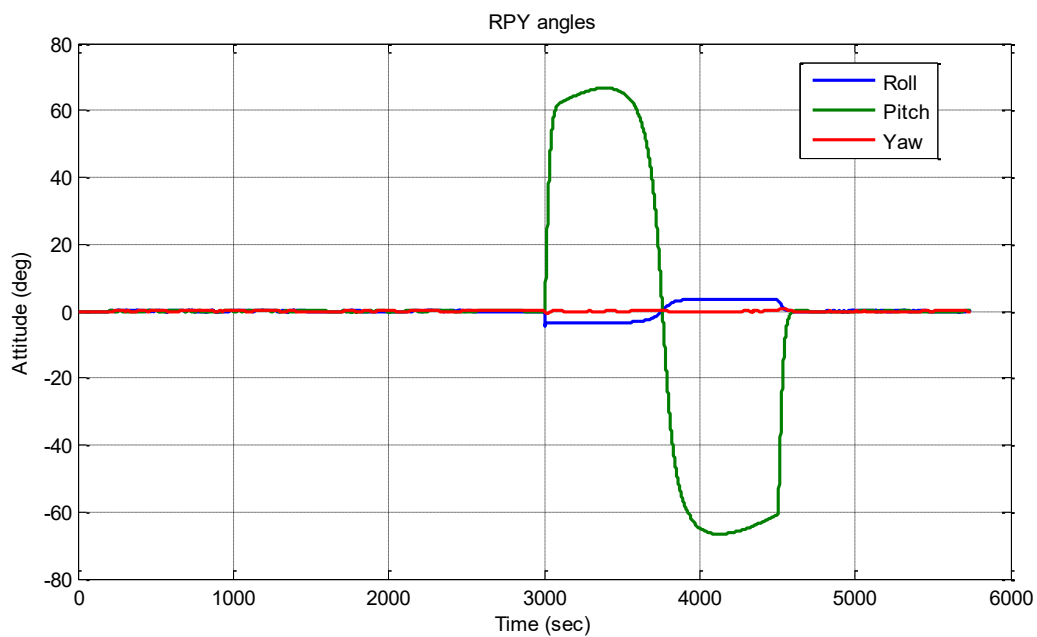


Figure 38: True RPY attitude during ground target tracking reaction wheel control.

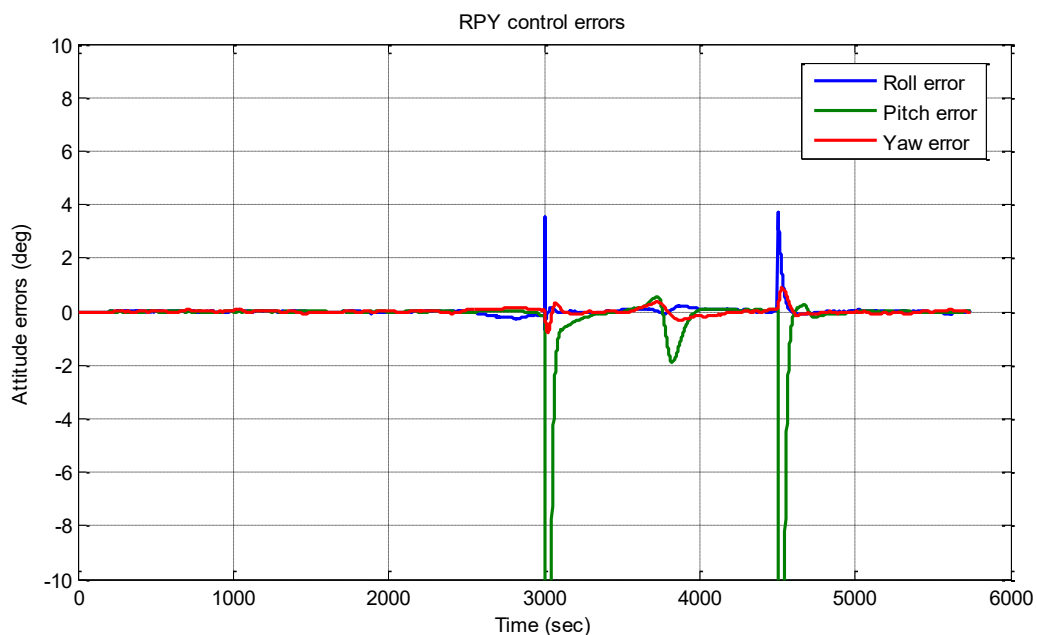




Figure 39: RPY tracking errors during ground target tracking reaction wheel control.

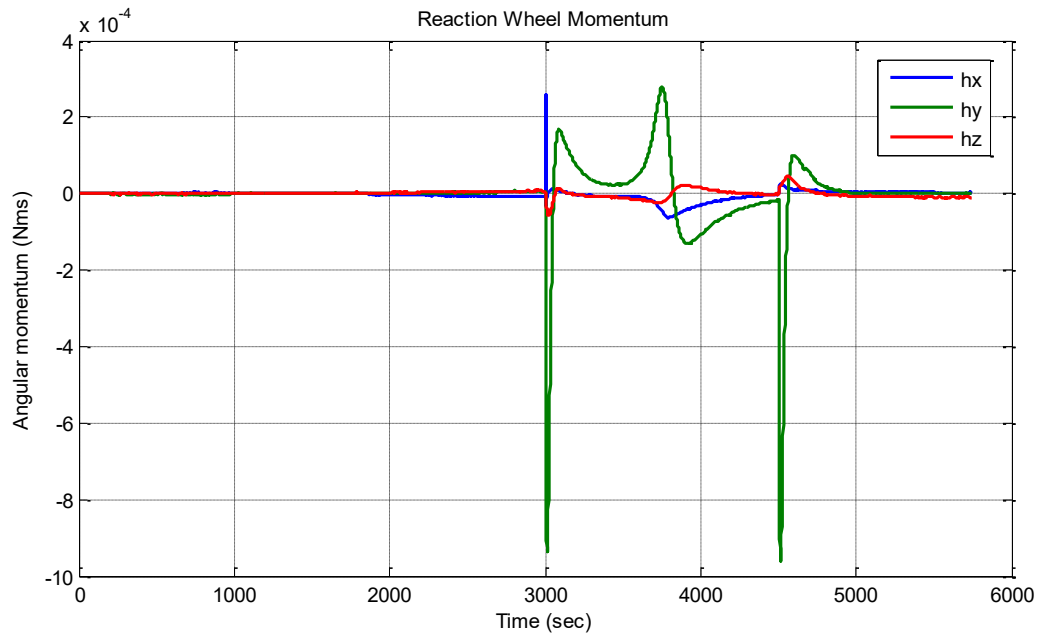


Figure 40: XYZ reaction wheel angular momentum during ground target tracking reaction wheel control.



20. GNSS Receiver Commissioning

Referring to section 2.5, there are different orbit modes available which determine how GNSS data is used by the ADCS.

As an initial step where the GNSS is only commissioned, it is recommended to use the OrbTle orbit mode, so that the CubeADCS relies solely on SGP4 predicted orbital position and velocity. It is thus important to use up to date TLE orbital elements and synchronized Unix time.

The GNSS receiver module must be enabled separately (outside of the CubeADCS).

The GNSS is verified by comparing the measured position and velocity to that of the SGP4 propagator output.

The steps in Table 39 can be followed to verify GNSS operation.

Once the GNSS has been verified, other orbit modes may be selected that make use of GNSS data.

Table 39: GPS receiver commissioning.

PREREQUISITES	Satellite must be in a stable nadir pointing 3-axis reaction wheel mode with angular rates in the body axes as estimated and measured by the rate sensors and EKF. Orbit parameters must be set from TLEs, and Unix time synchronised.	
COMMAND SEQUENCE	COMMAND	PARAMETERS
	OrbMode	OrbitMode set to OrbTle
TELEMETRY LOGGING	TELEMETRY FRAME	PERIOD
	TlmModels (Models Telemetry)	10s
	TlmGnssRaw (raw GNSS sensor telemetry)	10s
DURATION	1 Orbit	
TEST CONDITION	- Observe the GPS status parameters if enough satellites were tracked and the GPS solution status is successful. - Observe the GPS time solutions and compare to Unix time. - Observe the ECEF XYZ position and velocity solutions, compare position to satellite WGS-84 latitude, longitude, altitude SGP4 values.	



21. Fault detection, isolation, and recovery (FDIR)

This section describes the FDIR functionality that is implemented for the CubeADCS.



The full FDIR functionality of the CubeADCS is still under development, and information in this section will be updated in the future to include all FDIR features.

21.1 Fault recovery summary

The existing fault recovery features of the ADCS are summarized in Table 40.

Table 40: Fault recovery summary

FAULT	RECOVERY RATIONALE	TEST CONDITION	ACTION TAKEN	STATUS TLM/EVENT	FURTHER DETAIL
Gyro failure	Resume control by automatically changing to estimation mode that does not use gyro	Estimation mode is EstGyro and no (valid) gyroscope measurements are available	Estimation mode changes to EstMagRkf	GyroEstimError event is logged. EstimationModeChanged event is logged - Gyro valid status reported in calibrated gyro TLM	
Gyro failure	Stop controlling if control mode needs gyro, but continue with estimation by changing to mode that does not use gyro	Estimation mode is EstGyroEKF and Control mode requires angular rate information, and no (valid) gyroscope measurements are available	Estimation mode changes to EstFullEKF Control mode changes to ConNone	GyroEstimError event is logged. EstimationModeChanged event is logged - Gyro valid status reported in calibrated gyro TLM	
Incorrect orbit parameters or GNSS data	Prevent ADCS from generating control commands that will disrupt the attitude	Current estimated position or velocity vector (whether it originates from GNSS or SGP4 propagator) is zero length	Control mode changes to ConNone Estimation mode changes to EstNone	OrbitError event is logged	
Estimation error	Prevent ADCS from controlling if the estimated attitude is incorrect	Sum of absolute values of estimated rate vector components exceed 20 deg/s (EstGyroEKF and	Estimation mode change to EstNone Control mode change to ConNone	EstimRateError event is logged	



		EstFullEkf) or 100 deg/s (EstRkf).			
Payload is pointing too close to the sun	Payload-sun avoidance	Angle between measured sun vector and configured payload SBC vector is less than configured threshold (keep-out angle)	Control mode changes to ConSunDetumble	SunAvoidanceActivated event is logged	21.2
No communication from OBC	Ensure satellite safety when OBC becomes unresponsive	No communication from OBC on particular communications channel for the configured time-out duration	Depends on action selection in FDIR parameters. No action: no action taken Reset: The CubeADCS will reset Safe-Mode: Control mode changes to configured safe-mode control selection. (Estimation mode is decided based on chosen safe-mode control selection). Required sensors and actuators are powered on (if they were not enabled before)	ResetCommsWatchdog or SafeModeCommsWatchdog event logged	21.3
ADCS data integrity issue	Prevent ADCS from generating control commands that will disrupt the attitude	Internal state (position, velocity, estimated quaternion, or angular rates) or reaction wheel speed commands, have a non-number representation in the IEEE754 floating point representation (checks for +/-Inf and NaN conditions)	Control mode changes to ConNone Estimation mode changes to EstNone	AdcsDataIntegrityError event is logged	
Reaction wheel does not follow reference	Stop controlling when the reaction wheels are	Measured and commanded wheel momentum differs by more than 10x	Set RWL error flag (No RWL momentum management takes place)	Rwl<x>ErrorFlag in TlmController	



(commanded) speed	not responding to commands	(pyramid) or 5x (3-axis wheels) maximum torque-increment.			
Reaction wheel has saturated for 60s	Prevent spin-up from magnetic momentum dumping when wheels remain stuck at constant speed.	Commanded reaction wheel angular momentum exceeds maximum angular momentum, for a duration of 60s	Set RWL error flag No RWL momentum management takes place	Rwl<x>ErrorFlag in TlmController	

21.2 Sun Avoidance

This feature is enabled by setting the [SunAvoidanceEn](#) flag to true in the [AdcsConfigController](#) configuration. The [SunKeepOutAng](#) parameter in [AdcsConfigController](#) is the half-cone angle that may not enter the sun, around the configured Sun-avoidance body vector ([SunAvoidBodyVecX/Y/Z](#) in [ConfigAdcsSatellite](#) configuration). If the sun does enter this cone and the enable flag is active, the control mode will automatically change to [ConSunAvoid](#), which will cause the configured sun avoidance body vector to point directly away from the sun.

Following this, after a configurable timeout duration (120s by default), the control will change automatically to [ConSunDetumble](#). The latter mode will reorient the solar panel, configured using the [SunPointBodyVecX/Y/Z](#) vector, towards the sun.

If the sun-avoidance body vector is configured for a sunlight-sensitive payload (telescope), and the sun-pointing vector is configured 90 deg with respect to the sun-avoidance direction, this strategy will ensure firstly that the payload rotates directly away from the sun if the avoidance manoeuvre is triggered, and 2 minutes later it will point the main solar panel towards the sun. The latter manoeuvre is also guaranteed to not point the payload towards the sun, as long as the configured [SunAvoidBodyVecX/Y/Z](#) and [SunPointBodyVecX/Y/Z](#) vectors are perpendicular.

The logic that evaluates if the keep-out region is entered, makes use of the measured sun vector. The FSS measurement will be used primarily, but if it is not available, the calibrated CSS vector measurement is used. If the angle between the measured sun vector (in SBC) and the configured [SunAvoidBodyVecX/Y/Z](#) in [ConfigAdcsSatellite](#) configuration is smaller than the threshold, the transition to [ConSunAvoid](#) will occur.

The sun avoidance activation will not change the estimation mode. [ConSunAvoid](#) and [ConSunDetumble](#) requires that the estimation mode is either [EstGyro](#) or [EstGyroEKF](#). Thus, for the sun avoidance feature to be used, the estimation mode must also be [EstGyro](#) or [EstGyroEKF](#).

21.3 OBC Communications Watchdog

The OBC communications watchdog feature relies on a timeout value that is configurable per communications channel. If the timeout value for a channel is any positive value, the ADCS will register a timeout when no communication is received for the timeout period, and act, also based on a configurable action selection.



Table 41: Action selection for OBC communications timeout

NUMERIC VALUE	NAME	DESCRIPTION
0	No Action	No action
1	Reset	Reset MCU
2	Safe-Mode	Switch to configured safe mode

The timeout and action selection is configurable per communications channel (UART, UART2, I2C, CAN). If the chosen action is to Reset, the CubeADCS will reset. For the “safe-mode” selection, the ADCS will automatically transition to the configured safe control mode.

21.3.1 Safe-mode action

The feature makes use of the already existing safe-mode control selection in [ConfigDefaultModes](#). Only the following selections are valid for the safe-mode control selection:

- [ConBdot](#)
- [ConYspin](#)
- [ConBdot3](#)
- [ConDetumble](#)
- [ConZspin](#)
- [ConSunYspin](#)
- [ConSunZspin](#)
- [ConSunDetumble](#)

The transition to safe-mode requires that at least on MAG and one rate sensor is available for measurement. To satisfy the requirements of a safe-mode, this necessitates that the required nodes are turned on automatically if they are switched off. If the safe mode requires sun measurement, the FSS are also turned on automatically (and likewise for reaction wheels, if safe control mode is [ConSunDetumble](#)).

The [NodeSelection](#) configuration is used in the above situation. All nodes that have bits set in the [NodeSelection](#) are turned on automatically if they are not already powered on.

21.3.2 Events

The [ResetCommsWatchdog](#) event is used to signal a reset due to comms inactivity, while the [SafeModeCommsWatchdog](#) event is used to signal that a safe-mode transition occurred due to comms inactivity.



22. House-keeping actions

The following actions should be performed at every overpass:

22.1 Synchronize time

Time is synchronized by sending a single *Set Unix Time* command to the ADCS.

22.2 ADCS health

The health of the ADCS is assessed by monitoring the error flags in the ADCS State telemetry, and also the current and temperature telemetry frames. Persistent error flags should be addressed, and caution should be taken when current telemetry values deviate too far from the nominal or default values.

22.3 Check for updates and upload TLEs

It is important to use up-to-date two-line elements. TLEs are published at regular intervals, and it is recommended that TLEs are updated on the ADCS whenever new ones are available (this does not have to happen on every pass, but typically once daily).

22.4 House-keeping telemetry ranges

This information will be provided in future versions of this document.